

Path-Integrals in Quantum Field Theory

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Chapter 1

Path-Integrals in Quantum Mechanics: a Breviarium

1.1 The action in Quantum Mechanics

The action functional plays a central role in the formulation of classical mechanics. Of particular relevance is the Hamilton-Jacobi equation for a system with Hamiltonian H and acción S ,

$$H_{clas} \left(x, \frac{\partial S(x, t)}{\partial x} \right) = -\frac{\partial S}{\partial t} \quad (1.1.1)$$

which provides an alternative to the Hamiltonian and Lagrangian formulations, particularly useful to find conserved quantities of a dynamical system even when the equations of motion cannot be solved in close form. In this formulation, the action is taken as an explicit function of coordinates and time. Indeed, one can take the final point of the trajectory as a variable (x) and one can also do the same for the final time (t) so that finally one ends working with $S = S(x, t)$.

In contrast with classical mechanics, neither in the Heisenberg nor in the Schrödinger original formulations of quantum mechanics does the action play a central role.

As it is well known, it was Dirac the one who established in 1926 the connection between commutators in quantum mechanics and Poisson brackets in classical mechanics, clarifying in this way the Heisenberg formulation for quantum systems¹. It is less known that he also found the connection, en the $\hbar \rightarrow 0$ classical limit between Schrödinger wave equation in Quantum mechanics and the Hamilton-Jacobi equation in classical mechanics². But he didn't stop at such finding but also proposed a way to study quantum phenomena by starting

¹Paul A.M. Dirac *The fundamental equations of quantum mechanics*, Proc. Royal. Soc. London **A109** (1926) 642.

²P. A .M. Dirac, *The principles of Quantum Mechanics*, Oxford, 1930.

from classical trajectories, not just those who were extrema of the classical action but also those who were not³.

It was Feynman who provided, 16 years later⁴, a precise prescription to implement Dirac idea. Originally intended to give an alternative formulation of (non-relativistic) Quantum Mechanics it became a particularly powerful tool to study (relativistic) quantum field theories and string theories.

Let us follow Dirac proposal to connect the quantum Schrödinger equation with the classical Hamilton-Jacobi equation. He started from a quantum wave function satisfying the Schrödinger equation in terms of an amplitude A and a phase Φ ,

$$\psi(\vec{x}, t) = A(\vec{x}, t) \exp\left(\frac{i}{\hbar}\Phi(\vec{x}, t)\right) \quad (1.1.2)$$

Here amplitude and phase are real functions that depend smoothly on space-time variables.

Since $\psi(\vec{x}, t)$ obeys Schrödinger equation

$$i\hbar \frac{\partial \psi(\vec{x}, t)}{\partial t} = \hat{H} \psi(\vec{x}, t) \quad (1.1.3)$$

one has

$$i\hbar \frac{\partial A(\vec{x}, t)}{\partial t} - A(\vec{x}, t) \frac{\partial \Phi(\vec{x}, t)}{\partial t} = \exp(-\frac{i}{\hbar}\Phi(\vec{x}, t)) \hat{H} A \exp(\frac{i}{\hbar}\Phi(\vec{x}, t)) \quad (1.1.4)$$

In their coordinate representation operators $\hat{x}$ and $\hat{p}$ satisfy

$$\begin{aligned} \exp(-\frac{i}{\hbar}\Phi) \hat{x} \exp(\frac{i}{\hbar}\Phi) &= \hat{x} \\ \exp(-\frac{i}{\hbar}\Phi) \hat{p} \exp(\frac{i}{\hbar}\Phi) &= \hat{p} + \vec{\nabla}\Phi \end{aligned} \quad (1.1.5)$$

so one can write

$$\exp(-\frac{i}{\hbar}\Phi) \hat{H}(\hat{p}, \hat{x}) \exp(\frac{i}{\hbar}\Phi) = H(\hat{p} + \vec{\nabla}\Phi, \vec{x}) \quad (1.1.6)$$

or

$$i\hbar \frac{\partial A(\vec{x}, t)}{\partial t} - A(\vec{x}, t) \frac{\partial \Phi(\vec{x}, t)}{\partial t} = H(\hat{p} + \vec{\nabla}\Phi, \vec{x}) A \quad (1.1.7)$$

³P. A. M. Dirac, *The Lagrangian in Quantum Mechanics*, Physikalische Zeitschrift der Sowjetunion, **3** (1933) 64.

⁴R. P. Feynman, *The Principle of Least Action in Quantum Mechanics*, Princeton University Ph.D., Thesis, 1942; *Space-Time Approach to Non-Relativistic Quantum Mechanics*, Review of Modern Physics **20** (1948) 367.

As we know, the classical limit of quantum mechanics corresponds to $\hbar \rightarrow 0$ ⁵. This implies that in the classical limit eq.(1.1.7) reduces to

$$-A(x, t) \frac{\partial \Phi(x, t)}{\partial t} = H(\vec{\nabla} \Phi, \vec{x}) A \quad (1.1.8)$$

Note that H in the r.h.s. is just a function of its arguments, not an operator so that factor A in both sides cancel out to give

$$-\frac{\partial \Phi(x, t)}{\partial t} = H(\vec{\nabla} \Phi, \vec{x}) \quad (1.1.9)$$

Now, this result is nothing but the Hamilton-Jacobi equation provided the phase Φ is identified with the classical action S , That is, eq. (1.1.2) can be written in the form

$$\psi(\vec{x}, t) = A \exp\left(\frac{i}{\hbar} S\right) \quad (1.1.10)$$

with S satisfying

$$-\frac{\partial S}{\partial t} = H(\vec{x}, \vec{\nabla} S) \quad (1.1.11)$$

The fact that $\hbar$ units are those of [energy] $\times$ [time] which are also the action units gives another hint to establish Dirac connection (1.1.10).

At this stage, having in mind that at the classical level the particle trajectory corresponds to an extremum of S but that at the quantum level the notion of trajectory is meaningless, Dirac suggested that quantum motion, not just its classical limit, could in general be described by something related to a wave function of the form (1.1.9) provided one makes some sort of to take into account not only the extremum of S but also other trajectories (i.e., other *paths*) which do not correspond to the classical trajectory but should contribute in the quantum case,

$$\psi(\vec{x}, t) \sim \sum_{paths} \exp\left(\frac{i}{\hbar} S\right) \quad (1.1.12)$$

Dirac did not elaborate on this conjecture and it was Feynman who introduced an operative tool giving a precise *recipe* to calculate the sum over paths known today as the path-integral.

⁵Recall the heuristic idea exposed in de Broglie PhD thesis, relating the quantum mechanics wave length λ and the momentum $\vec{p} = m\vec{v}$ of an electron, $\lambda = h/|\vec{p}|$ so that in the limit $\hbar \rightarrow 0$ one has $\lambda \rightarrow 0$ and the electron can be represented by a point like object as in Classical Mechanics. L. de Broglie, *Recherches sur la thorie des quanta*, Thesis, Paris, 1924, Ann. de Physique (10) 3, 22 (1925).

We shall not discuss in this course Feynman proposal of studying the quantum dynamics of a particle of mass m by using the path-integral framework. Instead, we shall directly attack the path-integral approach to quantum field theories. Just for completeness I will just give below the main results in Feynman treatment in quantum mechanics and then pass directly to the quantum field case.

The Feynman construction

For simplicity we consider in what follows a quantum mechanical system with just one degree of freedom so that phase space is 2-dimensional.

We recall that in Schrödinger picture physical states are described by vectors (in a Hilbert space $\mathcal{H}$) evolving in time,

$$|\psi(t)\rangle_S \quad (1.1.13)$$

while operators associated with physical observables are time independent,

$$\hat{O}_S(t) = \hat{O}_S(t_0) = \hat{O}_S \quad (1.1.14)$$

Concerning the Heisenberg picture, a given physical state at different times is described by the same state vector

$$|\psi; t\rangle_H = |\psi; t_0\rangle_H \quad (1.1.15)$$

Two different states in the Heisenberg picture at a given time will be written as follow

$$|\psi_I; t_0\rangle, \quad |\psi_{II}; t_0\rangle \quad (1.1.16)$$

Concerning operators, they do evolve in time

$$\hat{O}_H = \hat{O}_H(t) \quad (1.1.17)$$

Now, what matters in Quantum Mechanics are expectation values and in this respect, one has the identity

$${}_H\langle\psi|\hat{O}_H|\psi\rangle_H = {}_S\langle\psi|\hat{O}_S|\psi\rangle_S \quad (1.1.18)$$

ensuring that, at the physical level, the results that one obtains are picture independent. This can also be understood by noting that operator eigenvalues are picture independent.

Let $|q\rangle$ be an eigenfunction of the position operator $\hat{Q}$. In the Schrödinger picture one has

$$Q_S|q\rangle_S = q|q\rangle_S \quad (1.1.19)$$

Being the Heisenberg operators time dependent, the corresponding eigenfunctions are defined at a given time t . So, one has for $Q_H(t)$ acting on a Heisenberg picture state $|q; t\rangle_H$

$$\hat{Q}_H(t)|q; t\rangle_H = q|q; t\rangle_H \quad (1.1.20)$$

Note that the eigenvector $|q; t\rangle_H$ of $Q_H(t)$ at time t , does not define a state evolving in time as in the Schrödinger picture.

Given the Hamiltonian $\hat{H}$ (remember that for isolated systems the Hamiltonian, generator of time translations, is time independent and hence $\hat{H}_H = \hat{H}_S$) the equation of motion for $\hat{Q}_H$ reads

$$[\hat{H}, \hat{Q}_H(t)] = -i\hbar\dot{\hat{Q}}_H(t) \quad (1.1.21)$$

This translates into an equation of motion for states in the Schrödinger picture

$$i\hbar\frac{\partial}{\partial t}|\psi(t)\rangle_S = \hat{H}|\psi(t)\rangle_S \quad (1.1.22)$$

Finally, the connection between the two pictures is given by the relations

$$\begin{aligned} |q; t\rangle_H &= \exp\left(\frac{i}{\hbar}\hat{H}t\right)|q\rangle_S \\ \hat{Q}_H(t) &= \exp\left(\frac{i}{\hbar}\hat{H}t\right)\hat{Q}_S\exp\left(-\frac{i}{\hbar}\hat{H}t\right) \end{aligned} \quad (1.1.23)$$

Let us consider a quantum system that, by measurement at time $t = t_1$, is in a Heisenberg state $|q_1; t_1\rangle_H$. The probability amplitude for measuring at a later time $t = t_2$ and finding the system in the state $|q_2; t_2\rangle_H$ is given by

$$Z_{1\rightarrow 2} = {}_H\langle q_2; t_2 | q_1; t_1 \rangle_H \quad (1.1.24)$$

Using (1.1.23) we can write this amplitude in the Schrödinger picture,

$$Z_{1\rightarrow 2} = {}_S\langle q_2 | \exp\left(-\frac{i}{\hbar}\hat{H}(t_2 - t_1)\right) | q_1 \rangle_S \quad (1.1.25)$$

Let us consider $t_2 - t_1 = \epsilon$ very small and suppress the subindex S but still working in the Schrödinger picture. We have

$$Z_{1\rightarrow 2} = \langle q_2 | \exp\left(-\frac{i}{\hbar}\hat{H}\epsilon\right) | q_1 \rangle = \langle q_2 | q_1 \rangle - \frac{i}{\hbar}\epsilon\langle q_2 | \hat{H} | q_1 \rangle + O(\epsilon^2) \quad (1.1.26)$$

Now, we have

$$\begin{aligned}
[\hat{q}, \hat{p}] &= \hbar i \\
\hat{p}|p\rangle &= p|p\rangle \\
\langle q|p\rangle &= \frac{1}{\sqrt{2\pi\hbar}} \exp\left(\frac{i}{\hbar}pq\right) \\
\int |q\rangle\langle q|dq &= 1 = \int |p\rangle\langle p|dp
\end{aligned} \tag{1.1.27}$$

so that we can write

$$Z_{1 \rightarrow 2}^\epsilon = \int_{-\infty}^{\infty} \langle q_2|p_1\rangle \langle p_1|q_1\rangle dp_1 - \frac{i}{\hbar}\epsilon \int_{-\infty}^{\infty} \langle q_2|p_1\rangle \langle p_1|\hat{H}|q_1\rangle dp_1 \tag{1.1.28}$$

or

$$\begin{aligned}
Z_{1 \rightarrow 2}^\epsilon &= \int_{-\infty}^{\infty} \exp\left(\frac{i}{\hbar}(q_2 - q_1)p_1\right) \frac{dp_1}{2\pi\hbar} - \\
&\quad \frac{i}{\hbar}\epsilon \int_{-\infty}^{\infty} \exp\left(\frac{i}{\hbar}(q_2 - q_1)p_1\right) H(p_1, q_1) \frac{dp_1}{2\pi\hbar}
\end{aligned} \tag{1.1.29}$$

where we have written

$$H(p_1, q_1) = \sqrt{2\pi\hbar} \exp\left(\frac{i}{\hbar}p_1q_1\right) \langle p_1|\hat{H}|q_1\rangle \tag{1.1.30}$$

One easily sees that $H(p, q)$ is nothing but the classical Hamiltonian. For example, for a particle of mass m in a potential V , the quantum Hamiltonian is $\hat{H} = \hat{p}^2/2m + V(\hat{q})$ and one has for $\hbar$

$$H(p, q) = \sqrt{2\pi\hbar} \exp\left(\frac{i}{\hbar}pq\right) \left(\frac{p^2}{2m} + V(q)\right) \langle p|q\rangle = \left(\frac{p^2}{2m} + V(q)\right) \tag{1.1.31}$$

i.e.

$$\boxed{H(p, q) = \frac{p^2}{2m} + V(q)} \tag{1.1.32}$$

At the order in ϵ that we are working, we can write instead of eq.(1.1.29)

$$Z_{1 \rightarrow 2}^\epsilon = \int_{-\infty}^{\infty} \exp\left(\frac{i}{\hbar}((q_2 - q_1)p_1 - \epsilon H(p_1, q_1))\right) \frac{dp_1}{2\pi\hbar} \tag{1.1.33}$$

Now, for a finite interval (t_i, t_f) we can use the result above as follows. We consider

$$t_0 = t_i < t_1 < t_2 < \dots < t_{N-1} < t_N = t_f \quad (1.1.34)$$

and make

$$t_n - t_{n-1} = \epsilon \quad \forall n \quad (1.1.35)$$

Then we have

$$\begin{aligned} Z_{i \rightarrow f}^N &= \langle q_f | \exp \left(-\frac{i}{\hbar} \hat{H}(t_f - t_i) \right) | q_i \rangle = \\ &\int dq_1 dq_2 \dots dq_{N-1} \langle q_f | \exp \left(-\frac{i}{\hbar} \hat{H}(t_f - t_{N-1}) \right) | q_{N-1} \rangle \times \\ &\langle q_{N-1} | \exp \left(-\frac{i}{\hbar} \hat{H}(t_{N-1} - t_{N-2}) \right) | q_{N-2} \rangle \dots \langle q_1 | \exp \left(-\frac{i}{\hbar} \hat{H}(t_1 - t_i) \right) | q_i \rangle \end{aligned} \quad (1.1.36)$$

or

$$\begin{aligned} Z_{i \rightarrow f}^N &= \int dq_1 dq_2 \dots dq_{N-1} \frac{dp_1}{2\pi\hbar} \frac{dp_2}{2\pi\hbar} \dots \frac{dp_N}{2\pi\hbar} \\ &\exp \left(\frac{i}{\hbar} ((q_f - q_{N-1})p_N + (q_{N-1} - q_{N-2})p_{N-1} + \dots + (q_1 - q_i)p_1) \right) \\ &\exp \left(-\frac{i}{\hbar} \epsilon (h(p_N, q_{N-1}) + h(p_{N-1}, q_{N-2}) + \dots + h(p_1, q_i)) \right) \end{aligned} \quad (1.1.37)$$

We then define

$$Z_{i \rightarrow f} \equiv \lim_{N \rightarrow \infty} \lim_{\epsilon \rightarrow 0} Z_{i \rightarrow f}^N \quad (1.1.38)$$

The argument in the first exponential in (1.1.37) can be written, in the $N \rightarrow \infty$, $\epsilon \rightarrow 0$ limit, in the form

$$\begin{aligned} \lim_{N \rightarrow \infty} \lim_{\epsilon \rightarrow 0} \epsilon \left(\frac{q_f - q_{N-1}}{\epsilon} p_N + \frac{q_{N-1} - q_{N-2}}{\epsilon} p_{N-1} + \dots + \right. \\ \left. \frac{q_1 - q_i}{\epsilon} p_1 \right) = \int_{t_i}^{t_f} \dot{q} p dt \end{aligned} \quad (1.1.39)$$

Concerning the second exponential, its argument can be written, in the same spirit, as $\int h dt$ so that it is natural to write

$$Z_{i \rightarrow f} = \langle q_f t_f | q_i t_i \rangle = \int \mathcal{D}p \mathcal{D}q \exp \left(\frac{i}{\hbar} \int_{t_i}^{t_f} (p \dot{q} - H(p, q)) dt \right) \quad (1.1.40)$$

where $\mathcal{D}p\mathcal{D}q$ is a shorthand notation for

$$\mathcal{D}p\mathcal{D}q = \lim_{N \rightarrow \infty} \lim_{\epsilon \rightarrow 0} \left(\prod_{i=1}^{N-1} dq_i \frac{dp_i}{2\pi\hbar} \right) \frac{dp_N}{2\pi\hbar} \quad (1.1.41)$$

Defined in this way, $\mathcal{D}p\mathcal{D}q$ is what mathematicians call a Liouville measure over all t .

Note that the integrals in (1.1.39) or (1.1.40) have to be done taking care of initial and final conditions q_i, q_f . In contrast, there are no conditions on the p 's which are to be integrated over $(-\infty, \infty)$. Note also that there is one more integral over p than over q .

One should not forget that computing $Z_{i \rightarrow f}$ through eq.(1.1.40),

$$Z_{i \rightarrow f} = \langle q_f t_f | q_i t_i \rangle = \int \mathcal{D}p\mathcal{D}q \exp \left(\frac{i}{\hbar} \int_{t_i}^{t_f} (p\dot{q} - H(p, q)) dt \right) \quad (1.1.42)$$

implies doing an infinite number of integrals over the phase space (p, q) . That is, one is to integrate over all trajectories compatible with $q(t_i) = q_i$ and $q(t_f) = q_f$.

Each integration implied in (1.1.42) can be thought as follows: the argument in the exponential, which we call S is in fact a sum,

$$S = \frac{i}{\hbar} (S_{N,N-1} + S_{N-1,N-2} + \dots + S_{1,i}) \quad (1.1.43)$$

where each term is given by

$$S_{n,n-1} = \epsilon \left(p_n \frac{q_n - q_{n-1}}{\epsilon} - H(p_n, q_n) \right) \quad (1.1.44)$$

Then S involves summing over broken paths as those represented in figure 1. Integrating over all possible values of q 's implies taking into account all possible paths with fixed extrema. Concerning p integrations, we show in figure 2 the corresponding representation.

A more rigorous mathematical setting would imply restricting the functions $q(t)$ contributing to the “path-integral” (1.1.42) to Lipschitzian functions of order 1/2, this meaning that

$$|q(t') - q(t)| < K(t' - t)^{\frac{1}{2}}, \quad t' > t \quad (1.1.45)$$

with K some positive constant.

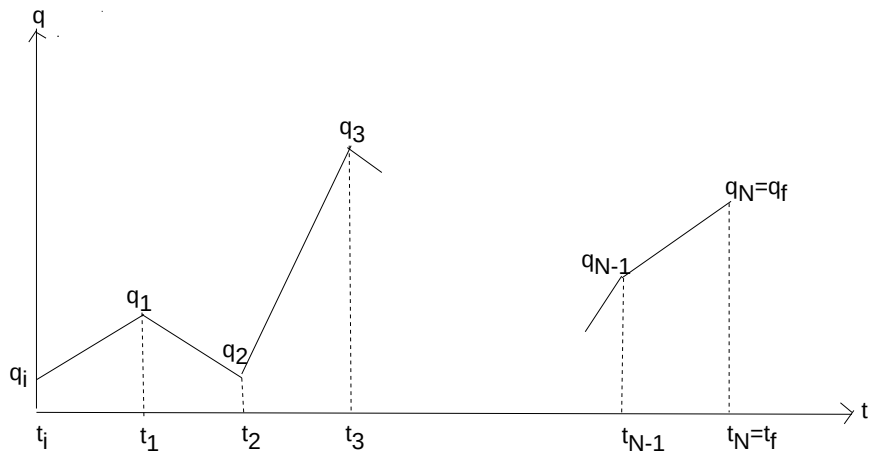


Fig. 1

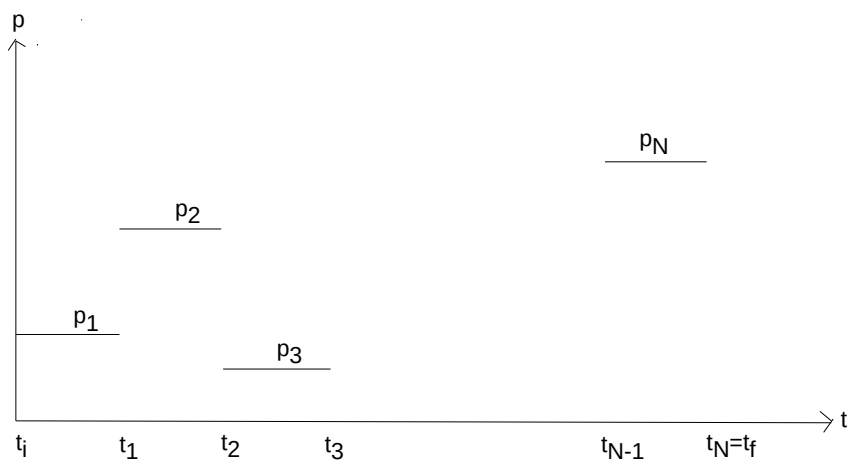


Fig. 2

1.2 The path-integral in terms of the action

Since $H(p, q)$ is just the classical Hamiltonian, we see that the argument in the exponential in (1.1.42) is related to the Lagrangian in the usual way, $L = p\dot{q} - h$. Indeed, consider the case of the quantum Hamiltonian for a particle of mass m in a potential $V(q)$,

$$\hat{H} = \frac{\hat{p}^2}{2m} + V(\hat{q}) \quad (1.2.46)$$

Then, $H(p, q)$ takes the form

$$H(p, q) = \frac{p^2}{2m} + V(q) \quad (1.2.47)$$

and the integrals over p 's can be easily done

$$\begin{aligned} \int \mathcal{D}p \left(\frac{i}{\hbar} \int (p\dot{q} - \frac{p^2}{2m}) dt \right) = \\ \lim_{N \rightarrow \infty} \lim_{\epsilon \rightarrow 0} \prod_{i=1}^N \int \frac{dp_i}{2\pi\hbar} \exp \left(\frac{i}{\hbar} (p_1(q_1 - q_i) + \dots + p_N(q_f - q_{N-1}) - \frac{i\epsilon}{2m\hbar} (p_1^2 + \dots + p_N^2)) \right) \end{aligned} \quad (1.2.48)$$

Let us change from p_i to new variables p'_i

$$p'_i = p_i - \frac{m}{\epsilon} (q_i - q_{i-1}), \quad i = 1, 2, \dots, N \quad (1.2.49)$$

$$dp'_i = dp_i \quad (1.2.50)$$

As we shall often do, we shall write the N change of variables just as

$$\begin{aligned} p' &= p - m\dot{q} \\ \mathcal{D}p' &= \mathcal{D}p \end{aligned} \quad (1.2.51)$$

and then we can treat the whole change of variables as

$$\frac{i}{\hbar} \int_{t_i}^{t_f} (p\dot{q} - \frac{p^2}{2m}) dt = -\frac{i}{\hbar} \int_{t_i}^{t_f} \frac{p'^2}{2m} dt + \frac{i}{\hbar} \int_{t_i}^{t_f} \frac{m\dot{q}^2}{2} dt \quad (1.2.52)$$

Then,

$$\int \mathcal{D}p \exp \left(\frac{i}{\hbar} \int_{t_i}^{t_f} (p\dot{q} - \frac{p^2}{2m}) dt \right) = \exp \left(\frac{i}{\hbar} \int_{t_i}^{t_f} \frac{m\dot{q}^2}{2} dt \right) \times \int \mathcal{D}p' \exp \left(-\frac{i}{\hbar} \int_{t_i}^{t_f} \frac{p'^2}{2m} dt \right)$$

Let us call the path-integral in the r.h.s. of eq.(1.2.53) $\mathcal{N}^{-1}(t_1, t_2)$, a factor that depends on the initial and final time but not on the corresponding conditions for $q(t)$,

$$\mathcal{N}^{-1}(t_f, t_i) = \int \mathcal{D}p' \exp \left(-\frac{i}{\hbar} \int_{t_i}^{t_f} \frac{p'^2}{2m} dt \right) \quad (1.2.53)$$

Then we can write

$$Z_{i \rightarrow f} = \mathcal{N}^{-1}(t_f, t_i) \int \mathcal{D}q \exp \left(\frac{i}{\hbar} \int_{t_i}^{t_f} \left(\frac{m\dot{q}^2}{2} - V(q) \right) dt \right) \quad (1.2.54)$$

or

$$Z_{i \rightarrow f} = \mathcal{N}^{-1}(t_f, t_i) \int \mathcal{D}q \exp \left(\frac{i}{\hbar} \int_{t_i}^{t_f} L(q, \dot{q}) dt \right) \quad (1.2.55)$$

where $L(q, \dot{q})$ is the Lagrangian of the system.

We have then arrived in this particular example to the specific formula suggested by Dirac (compare with eq.(1.1.12))

$$Z_{i \rightarrow f} = \mathcal{N}^{-1}(t_f, t_i) \int \mathcal{D}q \exp \left(\frac{i}{\hbar} S[q; t_i, t_f] \right) \quad (1.2.56)$$

If we now consider in $d = 3$ spatial dimensions and write the corresponding coordinates as $\vec{x} = (x^1, x^2, x^3)$ we then have

$$Z_{i \rightarrow f} = \mathcal{N}^{-1}(t_f, t_i) \int \mathcal{D}\vec{x} \exp \left(\frac{i}{\hbar} S[\vec{x}; t_i, t_f] \right) \quad (1.2.57)$$

with

$$\mathcal{D}\vec{x} = \mathcal{D}x_1 \mathcal{D}x_2 \mathcal{D}x_3 \quad (1.2.58)$$

where the action S is given by

$$S[\vec{x}] = \int_{t_i}^{t_f} d^3x \left(\frac{1}{2} m \dot{\vec{x}}^2 - V(\vec{x}) \right) \quad (1.2.59)$$

and $\mathcal{N}^{-1}(t_f, t_i)$ only depends on the mass of the particle and the initial and final fixed times of trajectories and not on the particular potential V .

The symbol $\int \mathcal{D}\vec{x}$ should be interpreted as a sum over all trajectories (or *paths*) compatible with the initial $\vec{x}(t_i) = \vec{x}_i$ and final conditions $\vec{x}(t_f) = \vec{x}_f$. In order to see the rôle of the classical trajectory among all trajectories included in the integration let us note that in

the $\hbar \rightarrow 0$ (classical) limit the exponential oscillates as $\vec{x}$ varies so as to wash out all contributions except those for which S is fairly constant. This last corresponds to the condition $\delta S / \delta \vec{x} = 0$. That is, in the $\hbar \rightarrow 0$ limit the classical trajectory dominates the path integral.

There is an alternative way of “integrating over all paths” that is simpler than the one described above. Suppose that we have an orthogonal basis $\{\varphi_n\}$ (that satisfies the appropriate boundary conditions) which allows to write any trajectory $x(t)$ in terms of a Bessel-Fourier expansion. Then, we can write

$$x(t) = \sum_n c_n \varphi_n(t) \quad (1.2.60)$$

$$\int_{t_i}^{t_f} \varphi_n(t) \varphi_m(t) dt = \delta_{nm} \quad (1.2.61)$$

Given a certain trajectory $x^I(t)$ one has, through the Bessel-Fourier expansion (1.2.60), a set of coefficients $\{c_n^I\}$. Analogously, for another trajectory $x^{II}(t)$ one has the set $\{c_n^{II}\}$ and so on. Then to integrate over all trajectories with given boundary conditions is to integrate over all possible values of the c_n^I coefficients in a Bessel Fourier expansion. That is

$$\mathcal{D}x(t) \equiv \mathcal{A} \lim_{N \rightarrow \infty} \int_{-\infty}^{\infty} dc_1 \int_{-\infty}^{\infty} dc_2 \dots \int_{-\infty}^{\infty} dc_N = \prod_i \int_{-\infty}^{\infty} dc_i \quad (1.2.62)$$

Here $\mathcal{A}$ is a normalization constant (that we already omitted in the last equality) ensuring that this measure is normalized as the previously defined one.

The path-integral formalism allows to calculate not only transition probabilities like $|Z_{i \rightarrow f}|^2$ but also quantum matrix elements between states $|\vec{x}(t_i)\rangle$ and $|\vec{x}(t_f)\rangle$ of **time-ordered products** of operators $\hat{\mathcal{O}}[x(t)]$. As an example

$$\langle \vec{x}(t_f) | T(\hat{\mathcal{O}}[\vec{x}(t_1)] \hat{\mathcal{O}}[\vec{x}(t_2)]) | \vec{x}(t_i) \rangle = \int \mathcal{D}\vec{x}(t) \exp\left(\frac{i}{\hbar} S[\vec{x}]\right) \mathcal{O}[x(t_1)] \mathcal{O}[x(t_2)] \quad (1.2.63)$$

where T defines a time ordering for product of operators

$$T(\hat{\mathcal{O}}[\vec{x}(t_a)] \hat{\mathcal{O}}[\vec{x}(t_b)]) = \begin{cases} \hat{\mathcal{O}}[\vec{x}(t_a)] \hat{\mathcal{O}}[\vec{x}(t_b)] & \text{if } t_a > t_b \\ \hat{\mathcal{O}}[\vec{x}(t_b)] \hat{\mathcal{O}}[\vec{x}(t_a)] & \text{if } t_b > t_a \end{cases} \quad (1.2.64)$$

It is important to stress that the path-integral recipe implies that quantum mechanics operators are replaced in the integrand by ordinary functionals ordered from the past (at the left) to the future (at the right).

One can summarize this by stating that

- All quantum aspects are taken into account by the path-integral measure; in contrast the integrand is completely classical.
- Concerning calculations involving products of operators, the path-integral leads automatically to time order matrix elements.

Let us end this section by recalling the Dirac-Feynman construction in the form

$$\boxed{\text{Dirac : } \sum_{\text{paths}} \exp\left(\frac{i}{\hbar} S\right) \longrightarrow \text{Feynman : } \int \mathcal{D}x \exp\left(\frac{i}{\hbar} S\right)} \quad (1.2.65)$$

1.3 Adding sources

A practical way of computing the path-integrals defining matrix elements as those above is the following. Consider a “source” $j(t)$ (a function that at the end of the operation will be put to zero) “coupled” to $x(t)$ in a path-integral which we shall call $Z[j(t)]$ defined as follows

$$Z[j(t)] = \int \mathcal{D}x \exp\left(\frac{i}{\hbar} S[x]\right) \exp\left(\frac{i}{\hbar} \int_{t_i}^{t_f} j(t)x(t)dt\right) \quad (1.3.66)$$

The idea is that by differentiation with respect to the source one can make descend from the exponential $x(t)$ (as many as the number of derivatives we make) so that the resulting path-integral corresponds to one giving matrix elements of $\hat{x}$ in the form defined above. But, to see how this works one has to first define the precise meaning of differentiation with respect to a function like $j(t)$. This is what is known as computing a *functional derivative* and is time to give a thumb definition of it.

Quick review of functional derivatives⁶

Consider the following linear functional of $j(t)$, $F[j(t)]$ defined as

$$F[j(t)] = \int_{t_i}^{t_f} K(t, t') j(t') dt' \quad (1.3.67)$$

⁶A nice book where one can find a short discussion on functional derivatives is A. Visconti-Quantum field theory. Volume 1, Pergamon Press, 1969. For a rigorous treatment see the book of one of the founders of functional analysis, Vito Volterra: V. Volterra and J. Peres, 1936

In analogy with ordinary calculus, where one can define partial derivatives of a function $f(x_i)$ through the formula

$$df = \sum_i \frac{\partial f}{\partial y_i} dy^i \quad (1.3.68)$$

one can define the functional derivative of $F[j(t)]$ with respect to $j(t')$ as

$$\frac{\delta F[j(t)]}{\delta j(t')} \equiv K(t, t') \quad (1.3.69)$$

so that in analogy with eq. (1.3.68) one can interpret eq.(1.3.67) as

$$\delta F[j(t)] = \int_{t_i}^{t_f} \frac{\delta F[j(t)]}{\delta j(t')} \delta j(t') dt' \quad (1.3.70)$$

Eqs.(1.3.67),(1.3.69) (or alternatively eq.(1.3.70)) is all one needs in order to compute functional derivatives. It satisfies all derivative properties.

In particular, note that since

$$j(t) = \int_{t_i}^{t_f} \delta(t - t') j(t') dt' \quad (1.3.71)$$

one has the nice formula

$$\frac{\delta j(t)}{\delta j(t')} = \delta(t - t') \quad (1.3.72)$$

which is the equivalent to

$$\frac{\partial y_i}{\partial y_j} = \delta_{ij} \quad (1.3.73)$$

for ordinary derivatives.

Using (1.3.72) one has

$$\frac{\delta}{\delta j(t)} \int_{t_i}^{t_f} j(t') x(t') dt = \int_{t_i}^{t_f} \frac{\delta j(t')}{\delta j(t)} x(t') dt' = \int_{t_i}^{t_f} \delta(t' - t) x(t') dt' = x(t) \quad (1.3.74)$$

so that

$$\frac{\delta Z[j]}{\delta j(t_a)} = \int \mathcal{D}x \exp\left(\frac{i}{\hbar} S[x]\right) \frac{i}{\hbar} x(t_a) \exp\left(\frac{i}{\hbar} \int_{t_i}^{t_f} j(t) x(t) dt\right) \quad (1.3.75)$$

Here and in what follows we avoid the $i \rightarrow f$ subindex in the “generating functional” Z . Hence

$$\left. \frac{\hbar}{i} \frac{\delta Z[j]}{\delta j(t_a)} \right|_{j=0} = \langle x_f | \hat{x}(t_a) | x_i \rangle \quad (1.3.76)$$

More generally

$$\langle x_f | \hat{x}(t_1) \hat{x}(t_2) \dots \hat{x}(t_p) | x_i \rangle = \left(\frac{\hbar}{i} \right)^p \frac{\delta^p Z[j]}{\delta j(t_1) \delta j(t_2) \dots \delta j(t_p)} \Big|_{j=0} \quad (1.3.77)$$

1.4 The Euclidean Action

In computing certain path-integrals, we shall face the problem of “gaussian” integrals but with an imaginary (not a -1) coefficient. So, these integrals are not really gaussian integrals. One way in which this problem can be overcome is by passing from Minkowski time t to Euclidean time τ ,

$$t \rightarrow \tau \equiv it \quad (1.4.78)$$

This can be done by a Wick rotation which is nothing but an analytic continuation. Take for example the Minkowski action for a particle of mass m in a potential V ,

$$\frac{i}{\hbar} S_M = \frac{i}{\hbar} \int dt \left(\frac{m \dot{x}^2}{2} - V(x) \right) \quad (1.4.79)$$

(Here $\dot{f} = df/dt$).

Let us now proceed to analytical continuation (1.4.79) to Euclidean time τ

$$\frac{i}{\hbar} S_M \rightarrow \frac{1}{\hbar} \int d(it) \left(-\frac{m}{2} \left(\frac{dq}{d(it)} \right)^2 - V(q) \right) = -\frac{1}{\hbar} S_E \quad (1.4.80)$$

with the Euclidean action defined as

$$S_E = \int d\tau L_E = \int d\tau \left(\frac{m}{2} \left(\frac{dq}{d\tau} \right)^2 + V(q) \right) \quad (1.4.81)$$

Wick rotation of the basic path-integral then leads to

$$\int \mathcal{D}q \exp \left(\frac{i}{\hbar} S_M \right) \xrightarrow{\text{Wick rotation}} \int \mathcal{D}q \exp \left(-\frac{1}{\hbar} S_E \right) \quad (1.4.82)$$

We see that the Euclidean Lagrangian looks as the Minkowskian energy, a sum of kinetic and potential terms.

Up to here we can summarize the result of passing from Minkowski to Euclidean time by stating that one has to change

- $i/\hbar \rightarrow -1/\hbar$
- $S_M \rightarrow S_E$ or $L_M \rightarrow L_E$
- $d/dt \rightarrow d/d\tau$

In order to write the Euclidean action as the difference of kinetic and potential terms one should define

$$V_E(q) \equiv -V(q) \quad (1.4.83)$$

One can then define an Euclidean Lagrangian

$$L_E = T_E - V_E \quad (1.4.84)$$

and an Euclidean energy density

$$\epsilon_E = T_E + V_E \quad (1.4.85)$$

An intuitive way of understanding the effect of the Wick rotation is the following. The Newton second law reads, in Minkowski space

$$m \frac{d^2 \vec{x}}{dt^2} = -\vec{\nabla} V_M \quad (1.4.86)$$

When passing to Euclidean space one changes $t = -i\tau$ while $\vec{x}$ remains unchanged

$$-m \frac{d^2 \vec{x}}{d\tau^2} = -\vec{\nabla} V_M \quad (1.4.87)$$

With the definition $V_E = -V_M$ Newton's law looks unchanged

$$m \frac{d^2 \vec{x}}{dx_0^2} = -\vec{\nabla} V_E \quad (1.4.88)$$

but then the force $\vec{F}_E = -\vec{\nabla} V_E = \vec{\nabla} V_M = -\vec{F}_M$ points in the opposite direction to that of the Minkowskian one.

The relevance of the Euclidean action goes beyond the fact that it allows to handle more properly gaussian actions. As a chemist discovered in the seventies⁷ it allows to describe tunneling effect using the path-integral and to develop semiclassical approximations. This, as we shall see, has a lot to do with the fact that one can think that, in passing to Euclidean space, the potential is inverted.

⁷K. Freed, *Path Integrals and Semiclassical Tunneling, Wavefunctions, and Energies*, Journal of Chemical Physics **56** (1972) 692.

Coming back to the probability amplitude for a transition from a state $|x_i\rangle$ at $t = t_i$ to a state as $|x_f\rangle$ at $t = t_f$, being the Hamiltonian the time evolution operator we can write

$$\begin{aligned} Z_{M i \rightarrow f} &= \langle x(t_f) | \exp \left(-\frac{i}{\hbar} \hat{H} T \right) | x(t_i) \rangle_S \\ &= \mathcal{N}^{-1}(t_f, t_i) \int \mathcal{D}q \exp \frac{i}{\hbar} S_M[x; t_i, t_f] \end{aligned} \quad (1.4.89)$$

where $T = t_f - t_i$. Now, working with Euclidean time

$$\begin{aligned} Z_{E i \rightarrow f} &= \langle q(\tau_f) | \exp \left(-\frac{1}{\hbar} \hat{H} \mathcal{T} \right) | q(\tau_i) \rangle_S \\ &= \mathcal{N}^{-1}(\tau_f, \tau_i) \int \mathcal{D}q \exp \left(-\frac{1}{\hbar} S_E[q; \tau_f, \tau_i] \right) \end{aligned} \quad (1.4.90)$$

where $\mathcal{T} = \tau_f - \tau_i$. If we now introduce the suggestive notation

$$\beta = \frac{\mathcal{T}}{\hbar} \quad (1.4.91)$$

and consider periodicity in Euclidean time imposing that $x(\tau_i) = x(\tau_f) = \bar{x}$ and then sum over all possible periodic conditions we can write

$$Z_E = \text{Tr}_i \langle \bar{x} | \exp \left(-\beta \hat{H} \right) | \bar{x} \rangle = \int d\bar{x} \mathcal{D}q \exp \left(-\frac{1}{\hbar} S_E[\bar{x}; \beta \hbar] \right)$$

with $\text{Tr}_{\bar{x}}$ meaning that we shall sum over all possible periodic trajectories starting and ending in $\bar{x}$. We see that Z_E can be identified with the Statistical Mechanics quantum partition function of a system with Hamiltonian $\hat{H}$ provided we identify β with

$$\beta = \frac{1}{kT} \quad (1.4.92)$$

with k the Boltzmann's constant and T the temperature. That is why we shall call Z the partition function of the system, even when working in Minkowski space.

We have identified the quantum partition function of a particle with the path-integral over all periodic trajectories. This analogy was presented in a technical way. There should be, however, deeper reasons behind it, connected with the properties of space and time. Although no rigorous explanation exists, arguments coming from gravitation have been invoked by Polyakov to sustain it. Indeed, Polyakov writes on page 4 of his nice book, when he discusses the statistical mechanics classical and quantum partition function in terms of path-integrals⁸:

⁸A.M. Polyakov, *Gauge Fields and Strings*, Harwood Academic Publishers, Worcester, 1987.

... Therefore, the second rule is that in the quantum case the inverse temperature acts as an imaginary time.

Our derivation of these analogies was technical. I feel that there are deep reason for them, connected with the properties of space-time. Although no real explanation exists, I shall give some comments on this below, when disacussing gravity.

Chapter 2

Path-integral formulation for Quantum Field Theories

2.1 From Quantum Mechanics to Quantum field Theory

We have seen that the path-integral defining Z , the partition function that describes the behavior of a quantum mechanical system with one degree of freedom was given by (see eq.(1.1.38))

$$Z_{i \rightarrow f} = \langle q_f t_f | q_i t_i \rangle = \int \mathcal{D}p \mathcal{D}q \exp \left(\frac{i}{\hbar} \int_{t_i}^{t_f} (p \dot{q} - H(p, q)) dt \right) \quad (2.1.1)$$

with the path-integral measure defined as as

$$\mathcal{D}p \mathcal{D}q = \lim_{N \rightarrow \infty} \lim_{\epsilon \rightarrow 0} \left(\prod_{i=1}^{N-1} dq_i \frac{dp_i}{2\pi\hbar} \right) \frac{dp_N}{2\pi\hbar} \quad (2.1.2)$$

with the exponential in the integrand interpreted as

$$S = \frac{i}{\hbar} (S_{N,N-1} + S_{N-1,N-2} + \dots) \quad (2.1.3)$$

$$S_{j,j-1} = \epsilon \left(p_j \frac{q_j - q_{j-1}}{\epsilon} - H(p_j, q_j) \right) \quad (2.1.4)$$

If the system has n degrees of freedom, labeled by an index α ($\alpha = 1, 2, \dots, n$), one has, instead of eqs.(2.1.1)-(2.1.4),

$$\begin{aligned}
Z_{i \rightarrow f} = & \lim_{N \rightarrow \infty} \lim_{\epsilon \rightarrow 0} \prod_{\alpha=1}^n \prod_{i=1}^{N-1} \left(\int dq_i^\alpha \frac{dp_i^\alpha}{2\pi\hbar} \right) \frac{dp_N^\alpha}{2\pi\hbar} \\
& \exp \left(\frac{i}{\hbar} \sum_{(j,\alpha)}^{(N,n)} (p_j^\alpha (q_j^\alpha - q_{j-1}^\alpha) - \epsilon H(p_j^\alpha, q_j^\alpha)) \right)
\end{aligned} \tag{2.1.5}$$

or, compactly

$$Z_{i \rightarrow f} = \langle q_f t_f | q_i t_i \rangle = \int \mathcal{D}p^\alpha \mathcal{D}q^\alpha \exp \left(\frac{i}{\hbar} \int_{t_i}^{t_f} (p^\alpha \dot{q}^\alpha - H(p^\alpha, q^\alpha)) dt \right) \tag{2.1.6}$$

which is nothing but expression (2.1.1) but extended to a system with n degrees of freedom.

What about a system with infinite many degrees of freedom, like a relativistic field theory? For simplicity, we start with a scalar theory in Minkowski (3+1) space-time, with Lagrangian density $\mathcal{L}$ defined as

$$\mathcal{L} = \mathcal{L}(\phi, \partial_\mu \phi) \tag{2.1.7}$$

Here $\partial_\mu = \partial/\partial x^\mu$, $\mu = 0, 1, 2, 3$, $x^0 = t$, $(x^i) = \vec{x}$, $i = 1, 2, 3$. We shall call ‘‘Lagrangian’’ the integral of the Lagrangian density over space (which as stated we took for the moment to be R^3)

$$L \equiv \int d^3x \mathcal{L}(\phi, \partial_\mu \phi) \tag{2.1.8}$$

so that the action is

$$S = \int_{t_i}^{t_f} dt L = \int_{t_i}^{t_f} dt \int d^3x \mathcal{L}(\phi, \partial_\mu \phi) \tag{2.1.9}$$

As a shorthand we shall write

$$S = \int_M d^4x \mathcal{L}(\phi, \partial_\mu \phi) \tag{2.1.10}$$

remembering that the four dimensional integral corresponds to a manifold $M = T \times R^3$ ($T = (t_i, t_f)$). Eventually, we shall make $t_i = -\infty$ and $t_f = \infty$ and then the manifold will be just the usual 4-dimensional Minkowski space.

The Euler-Lagrange equations of motion associated to $\mathcal{L}$ read

$$\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \right) = \frac{\partial \mathcal{L}}{\partial \phi} \tag{2.1.11}$$

Concerning the conjugate momentum Π and the Hamiltonian density H one has

$$\Pi = \frac{\partial \mathcal{L}}{\partial \dot{\phi}} \quad (2.1.12)$$

and

$$H = \Pi \dot{\phi} - \mathcal{L} \quad (2.1.13)$$

We can at this point apply what we have learnt for the quantum-mechanical problem with partition function (2.1.6). To this end, we consider cubical cells in space R^3 of volume ϵ^3 . Then, we define

$$q^\alpha(t) \equiv \frac{1}{\epsilon^3} \int_{V_\alpha} d^3x \phi(\vec{x}, t) \quad (2.1.14)$$

where V_α represents the manifold defined by the α^{th} -cell. Each q^α can be seen as the “mean-value” of ϕ over V_α . We then consider a time partition also with size ϵ so that space-time is now divided in N^4 cubical cells with N the number of cells that will eventually go to infinity¹.

With this, Lagrangian L can be written as

$$L = \int d^3x \mathcal{L} = \sum_\alpha \epsilon^3 \mathcal{L}^\alpha(q^\alpha, q^{\alpha+\vec{d}}, \dot{q}^\alpha) \quad (2.1.15)$$

The “conjugate momentum” to q^α is then

$$p^\alpha(t) = \frac{\partial L}{\partial \dot{q}^\alpha} = \epsilon^3 \frac{\partial \mathcal{L}^\alpha}{\partial \dot{q}^\alpha} = \epsilon^3 \Pi^\alpha(t) \quad (2.1.16)$$

and the Hamiltonian density becomes

$$H = \sum_\alpha \epsilon^3 (\Pi^\alpha \dot{q}^\alpha - \mathcal{L}^\alpha) = \sum_\alpha \epsilon^3 H^\alpha \quad (2.1.17)$$

Now equation (2.1.6) can be used just by identifying Π^α with p^α , noting that the sum over cells, $\sum_\alpha$, becomes in the limit $\epsilon \rightarrow 0$ (and $N \rightarrow \infty$) an integral over the space manifold, $\int d^3x$. Then, instead of formula (2.1.6) we have, for the partition function of the scalar field theory

$$Z = \int \mathcal{D}\Pi \mathcal{D}\phi \exp \left(\frac{i}{\hbar} \int_M d^4x (\Pi(x) \dot{\phi}(x) - H(\Pi, \phi)) \right) \quad (2.1.18)$$

¹If one works taking $c = 1$ units of time and space have the same length dimensions $[l]$ and masses have $[m] = 1/l$ dimensions.

(Here $(x) = (t = x^0, \vec{x})$).

We can take the initial state at $t_i = -\infty$ as the state of least energy for the system, i.e. the “vacuum state”. Again at $t_f = \infty$ we shall consider that the system is again in its vacuum state (unique, for the moment). Then Z can be interpreted as the vacuum to vacuum amplitude for a system with dynamics governed by Lagrangian L and Hamiltonian H . Adding a source J one defines the “generating functional” $Z[j]$,

$$Z[j] = \int \mathcal{D}\Pi \mathcal{D}\phi \exp \left(\frac{i}{\hbar} \int_M d^4x (\Pi(x) \dot{\phi}(x) - H(\Pi, \phi)) \right) \exp \left(\frac{i}{\hbar} \int_M d^4x j(x) \phi(x) \right) \quad (2.1.19)$$

After differentiation with respect to $j(x)$ one can compute vacuum expectation values of products of fields and, from these, physical quantities like scattering amplitudes.

Let us now copy the steps that led us in the case of just one degree of freedom, from the Hamiltonian formulation of Z to the corresponding Lagrangian formulation. Suppose the Hamiltonian for the scalar field theory is quadratic in the momenta

$$H = \frac{1}{2} \Pi^2 + \frac{1}{2} (\partial_i \phi)^2 + V[\phi] \quad (2.1.20)$$

with $\vec{x} = (x^i)$, $(i = 1, 2, 3)$. Then, as in the quantum mechanical case, after a change of variables we can factor out a gaussian integral over momenta and end with a path-integral over “coordinates”, in this case over fields. Indeed, the integral over momenta reads

$$\int \mathcal{D}\Pi \exp \left(-\frac{i}{2\hbar} \int d^4x (\Pi^2 - 2\Pi \dot{\phi}) \right) = \mathcal{N} \exp \left(\frac{i}{2\hbar} \int d^4x \dot{\phi}^2 \right) \quad (2.1.21)$$

with

$$\mathcal{N} = \int \mathcal{D}\Pi \exp \left(-\frac{i}{2\hbar} \int d^4x (\Pi - \dot{\phi})^2 \right) = \int \mathcal{D}\Pi' \exp \left(-\frac{i}{2\hbar} \int d^4x \Pi'^2 \right) \quad (2.1.22)$$

With this, $Z[j]$ reads

$$Z[j] = \int \mathcal{D}\phi \exp \left(\frac{i}{\hbar} \int_M d^4x \left(\frac{1}{2} \partial_\mu \phi \partial^\mu \phi - V[\phi] + j\phi \right) \right) \quad (2.1.23)$$

or

$$Z[j] = \int \mathcal{D}\phi \exp \left(\frac{i}{\hbar} \int_M d^4x (\mathcal{L} + j\phi) \right) \quad (2.1.24)$$

This expression is perfectly covariant. The integration measure $\mathcal{D}\phi = \prod_x d\phi(x)$ which we shall use is not a completely rigorous definition but, as we shall see, one can give it a precise

meaning by comparing results using this path-integral measure with usual perturbation theory in the framework of field theory canonical quantization so that such identity justifies to take such measure as a definition. In particular, let us point that the quantity

$$G^{(n)}(x_1, x_2, \dots, x_n) \equiv \frac{1}{Z[0]} \left(\frac{\hbar}{i} \right)^n \frac{\delta^n Z}{\delta j(x_1) \delta j(x_2) \dots \delta j(x_n)} \Big|_{j=0} \quad (2.1.25)$$

calculated in the path-integral framework is nothing but the n -point Green function obtained in the context of field theory canonical quantization. As stated, this can be seen by comparing, perturbatively, the result obtained from the path-integral expression (2.1.24) with that arising in the canonical approach.

In the case of quantum mechanics, we have also seen that once momentum variables were integrated out, one ends with $Z[0] = Z$ in the form

$$Z = \int Dx(t) \exp \left(\frac{i}{\hbar} S \right) \quad (2.1.26)$$

where S is the classical action and then an alternative definition of the path-integral measure was to write

$$\mathcal{D}x(t) = \prod_n dc_n \quad (2.1.27)$$

where the c_n s are the (real) coefficients of a Bessel-Fourier expansion in terms of an arbitrary basis. Integrating over all possible coefficient ensembles one is also covering all field space.

The same can be done in quantum field theory case: starting from the path-integral partition function

$$Z = \int D\phi(x) \exp \left(\frac{i}{\hbar} S[\phi] \right) \quad (2.1.28)$$

one can take as measure $\mathcal{D}\phi$

$$D\phi \equiv \prod_n dc_n \quad (2.1.29)$$

where c_n are (real) coefficients of the Bessel-Fourier expansion of the (real) scalar field $\phi(x)$,

$$\phi(x) = \sum c_n \varphi_n(x) \quad (2.1.30)$$

in terms of an appropriate ortonormal basis $\{\phi_n(x)\}$. Again, the result should be independent of the basis choice.

2.2 The Scalar $\lambda\phi^3$ QFT: a simple example

Let us consider the so-called “ $\lambda\phi^3$ ” theory, i.e. one in which the potential takes the form

$$V[\phi] = \frac{\lambda}{3!}\phi^3 + \frac{1}{2}m^2\phi^2 \quad (2.2.31)$$

(m will be identified with the mass of the scalar particle associated to the field $\phi(x)$). Signs are chosen so as to guarantee the stability of the solution $\phi = 0$ of the classical equations of motion.

From here on we shall work in Euclidean space where $Z[j]$ reads (after integrating by parts the kinetic energy term for ϕ and neglecting surface terms)

$$Z[j] = \int \mathcal{D}\phi \exp \left(-\frac{1}{\hbar} \int_M d^4x \left(\phi \frac{(-\square + m^2)}{2} \phi + \frac{\lambda\phi^3}{3!} + j\phi \right) \right) \quad (2.2.32)$$

Let us call A the differential operator appearing in the quadratic part of the Lagrangian,

$$A = -\square + m^2 \quad (2.2.33)$$

Consider now the case in which the “coupling constant” λ vanishes, that is, the “free” (i.e. non self-interacting) case. Calling $Z_0[j]$ the corresponding generating functional, we have

$$Z_0[j] = \int \mathcal{D}\phi \exp \left(-\frac{1}{\hbar} \int_M d^4x \left(\phi \frac{A}{2} \phi + j\phi \right) \right) \quad (2.2.34)$$

Now, for the $j = 0$ case the path-integral can be done following a line similar to that of the quantum mechanical example in which we expanded the path-integral variable in Bessel-Fourier series. Indeed, in view of eq.(2.2.32) it is convenient to consider the eigenvalue problem

$$A\varphi_n = \lambda_n\varphi_n \quad (2.2.35)$$

and then write ϕ taking $\{\varphi_n\}$ as the appropriate basis,

$$\phi(x) = \sum_n c_n \varphi_n \quad (2.2.36)$$

That is, the basis for the expansion is precisely the one arising from the A operator so that the first term in the exponential argument will be diagonal. We then have, for the free theory (i.e. $j = 0$) and in the absence of sources (i.e. $j = 0$) that the partition function (which we call $Z_0[j = 0]$) takes the form

$$Z_0[j = 0] = \prod_n \int dc_n \exp \left(-\frac{1}{\hbar} \sum_n \lambda_n c_n^2 \right) = \left(\prod_n \lambda_n \right)^{-1/2} \equiv \det^{-1/2} A \quad (2.2.37)$$

We have ignored in the r.h.s. $\hbar$ factors, π factors, etc, since as we shall see, they cancel out when computing appropriately normalized expectation values.

Now we can also compute $Z_0[j \neq 0]$ by “completing squares” so as eliminate the linear ϕ term from the path-integral. To this end we write

$$Z_0[j] = \int \mathcal{D}\phi \exp \left(-\frac{1}{2\hbar} \int d^4x (\phi - \bar{\phi}) A (\phi - \bar{\phi}) - \frac{1}{2\hbar} \int d^4x C \right) \quad (2.2.38)$$

and find C and $\bar{\phi}$ so that the identity the r.h.s. of this equation does coincide with Z_0 as defined in (2.2.34). The answer is

$$C = - \int d^4x' j(x) A^{-1}(x, x') j(x') \quad (2.2.39)$$

$$\bar{\phi} = - \int d^4x' A^{-1}(x, x') j(x') \quad (2.2.40)$$

where $A^{-1}(x, x')$ is the Green function for the operator A ,

$$(-\square_x + m^2) A^{-1}(x, x') = \delta^4(x - x') \quad (2.2.41)$$

Let us now make a change of the path-integral variables $\phi \rightarrow \phi'$ in the generating functional (2.2.38)

$$\phi' = \phi - \bar{\phi} \quad (2.2.42)$$

which, being a shift has a trivial Jacobian,

$$\mathcal{D}\phi' = \mathcal{D}\phi \quad (2.2.43)$$

so that

$$Z_0[j] = \int \mathcal{D}\phi' \exp \left(-\frac{1}{2\hbar} \int d^4x \phi' A \phi' \right) \exp \left(\frac{1}{2\hbar} \int d^4x d^4x' j(x) A^{-1}(x, x') j(x') \right) \quad (2.2.44)$$

or

$$Z_0[j] = Z_0[j = 0] \exp \left(\frac{1}{2\hbar} \int d^4x d^4x' j(x) A^{-1}(x, x') j(x') \right) \quad (2.2.45)$$

We can use formula (2.2.45) in order to compute the (vacuum) expectation of the product of two fields at different points in the free ($\lambda = 0$) theory

$$\langle \phi(x_1) \phi(x_2) \rangle_0 = \frac{\hbar^2}{Z_0[0]} \frac{\delta^2 Z_0[j]}{\delta j(x_1) \delta j(x_2)} \Big|_{j=0} \quad (2.2.46)$$

One finds

$$\langle \phi(x_1)\phi(x_2) \rangle_0 = \hbar A^{-1}(x_1, x_2) \quad (2.2.47)$$

which is nothing (apart from a $\hbar$ factor which from here on we put to 1) but the (two-points) Green function $G_0^{(2)}(x_1, x_2)$ for the free theory

$$AG_0^{(2)}(x, y) = (-\square_x + m^2)G_0^{(2)}(x, y) = \delta^{(4)}(x - y) \quad (2.2.48)$$

$$G_0^{(2)}(x, y) = G_0^{(2)}(x - y) = \int \frac{d^4k}{(2\pi)^4} \exp(ik(x - y)) \frac{1}{k^2 + m^2} \quad (2.2.49)$$

We can then write the generating functional $Z[j]$ in terms of the partition function $Z_0[j = 0]$ in the form

$$Z_0[j] = \exp \left(-\frac{1}{2} \int d^4x d^4x' j(x) G_0^{(2)}(x, x') j(x') \right) Z_0[j = 0] \quad (2.2.50)$$

The expectation value (2.2) is also known as the “correlation function” between fields ϕ at points x and y in space-time. It can be interpreted physically as the amplitude for propagation of a particle or excitation between y and x .

In the free theory, it is simply the Feynman propagator that arises in canonical field theory quantization (for $n = 2$). It can be connected with the “correlation functions” in Statistical Mechanics that quantify how microscopic variables vary one with respect of another on average across space and time. Because of this connection one usually calls $Z[j = 0]$ the partition function of the system and $Z[j]$ the generating function for Green functions (extending the use to the case of n points correlation functions).

The expectation value (2.2) is in fact the “vacuum expectation value” (v.e.v.) of the product $\phi(x)\phi(y)$ that one computes in the framework of the canonical quantization of field theories. The connection between both approaches is derived in detail in sections 4.2 and 9.2 of Peskin-Schroeder book “An Introduction to Quantum Field Theory”².

2.3 The Wick theorem

This “theorem”, due to a work of Gian Carlo Wick³ (based in previous work by Freeman Dyson⁴), allows to evaluate vacuum expectation values of product of n fields (v.e.v.’s) at

²M. Peskin and D.V. Schroeder, *An Introduction to Quantum Field Theory*, West View Press, 1995.

³G.C. Wick, *Physical Review* **80** (1950) 268

⁴F.J. Dyson, *Physical Review* **75** (1949) 486.

different points x_i ($i = 1, 2, \dots, n$) in terms of the vacuum expectation values of product of just two fields.

From the partition function for the free theory defined by eq.(2.2.50) one easily finds the formula that proves the theorem. For example, for the 4-points correlation function one gets, after differentiation

$$\begin{aligned} \langle \phi(x_1)\phi(x_2)\phi(x_3)\phi(x_4) \rangle_0 &= G_0^{(2)}(x_1, x_2)G_0^{(2)}(x_3, x_4) + \\ &G_0^{(2)}(x_1, x_3)G_0^{(2)}(x_2, x_4) + \\ &G_0^{(2)}(x_1, x_4)G_0^{(2)}(x_2, x_3). \end{aligned} \quad (2.3.51)$$

One can easily prove that in the free case using $Z_0[j]$ as defined in eq.(2.2.45), all v.e.v.'s with an odd number of points vanish since there will always be an odd number of fields ϕ in the path-integrand which will change sign when $\phi \rightarrow -\phi$ while the exponential is an even functional under such change. Concerning the even case with $n > 4$, it just gives analogous results as that for the case $n = 4$.

Bellow we shall discuss how to extend the evaluation of v.e.v.'s when a self-interaction like $\lambda\phi^3$ is included (self-)interacting theory. To this end we shall develop a perturbative scheme so that finally any Green function can be expressed in terms of free two-point Green functions thanks to Wick theorem. But before let us study some formal properties of $Z[j]$.

2.4 Integration by parts and equations of motion

One easily see that

Lemma 1: The path-integral of a total functional derivative is zero. In particular

$$\int \mathcal{D}\phi \frac{\delta}{\delta\phi(y)} \exp \left(-S[\phi] + \int d^4x j(x)\phi(x) \right) = 0 \quad (2.4.52)$$

For a “proof” of this result, one goes back to the definition of the path-integral measure in terms of cells of volume ϵ^4 and uses the equivalent result for normal integrals provided boundary terms vanish.

The following identity follows from eq.(2.4.52)

$$\int \mathcal{D}\phi \left(j(y) - \frac{\delta S}{\delta\phi(y)} \right) \exp \left(-S[\phi] + \int d^4x j(x)\phi(x) \right) = 0 \quad (2.4.53)$$

Now, the derivative with respect to ϕ in the l.h.s. can be replaced by a derivative with respect to j

$$\int \mathcal{D}\phi \left(j(y) - \frac{\delta S}{\delta\phi(y)} \left[\frac{\delta}{\delta j} \right] \right) \exp \left(-S[\phi] + \int d^4x j(x)\phi(x) \right) \quad (2.4.54)$$

and then from (2.4.52) we have

$$\left(j(y) - \frac{\delta S}{\delta \phi(y)} \left[\frac{\delta}{\delta j} \right] \right) Z[j] = 0 \quad (2.4.55)$$

This equation is known as the Schwinger-Dyson equation of motion, the quantum version of the classical equation of motion in the presence of a source. One can derive an infinite set of equations using this master equation. For example, writing for S

$$S[\phi] = \frac{1}{2} \int d^4x \phi A \phi + S_{int}[\phi] \quad (2.4.56)$$

one has

$$\frac{\delta S}{\delta \phi} = A\phi + \frac{\delta S_{int}}{\delta \phi} \quad (2.4.57)$$

In the particular case of the $\lambda\phi^3$ theory,

$$\frac{\delta S}{\delta \phi} = A\phi + \frac{1}{2}\lambda\phi^2 \quad (2.4.58)$$

Then, the Schwinger-Dyson equation reads

$$\left(A \frac{\delta}{\delta j(y)} + \frac{\lambda}{2} \left(\frac{\delta}{\delta j(y)} \right)^2 - j(y) \right) Z[j] = 0 \quad (2.4.59)$$

Now, differentiating with respect to $j(z)$ and then putting $j = 0$ one gets, after some algebra

$$A \frac{\delta^2}{\delta j(y) \delta j(z)} Z[j] \Big|_{j=0} - \delta^4(y-z) Z[0] + \frac{\lambda}{2} \frac{\delta}{\delta j(z)} \left(\frac{\delta}{\delta j(y)} \right)^2 Z \Big|_{j=0} = 0 \quad (2.4.60)$$

Dividing by $Z[0]$ and rearranging we finally have

$$(-\square + m^2) \langle \phi(y) \phi(z) \rangle + \frac{\lambda}{2} \langle \phi^2(y) \phi(z) \rangle = \delta^4(y-z) \quad (2.4.61)$$

to be compared with the classical equation of motion

$$(-\square + m^2) \phi(y) + \frac{\lambda}{2} \phi^2(y) = 0 \quad (2.4.62)$$

Conversely, one can show that the solution of the Schwinger-Dyson equation is precisely the generating functional $Z[j]$. Indeed, writing

$$Z[j] = \exp(-W[j]) \quad (2.4.63)$$

one can think of W as a sort of Laplace transform of S and then if one essays for the solutions $Z_{sol}[j]$ of Schwinger-Dyson equation a Laplace transform one finds that the argument in the exponential should be in fact the action S .

Delta functionals

A sensible path-integral representation for the “ δ -functional” $\delta[F[\phi]]$ is the following

$$\delta[F[\phi]] = \int \mathcal{D}\lambda \exp \left(- \int d^4x F[\phi] \lambda(x) \right) \quad (2.4.64)$$

the natural extension of the definition of the δ -function in terms of Fourier transforms. To see that (2.4.64) gives a sound result let us consider the following path-integral

$$Z_r[j] = \int \mathcal{D}\phi \delta[F[\phi]] \exp \left(-S + \int d^4x j\phi \right) \quad (2.4.65)$$

and see how the δ -functional modifies the integration domain Using the representation (2.4.64) we have

$$Z_r[j] = \int \mathcal{D}\phi \mathcal{D}\lambda \exp \left(-S + \int d^4x (j\phi - \lambda(x)F[\phi]) \right) \quad (2.4.66)$$

But in view of our Lemma 1, we have

$$\int \mathcal{D}\phi \mathcal{D}\lambda \frac{\delta}{\delta\lambda} \exp \left(-S + \int d^4x j\phi \right) = 0 \quad (2.4.67)$$

or

$$\int \mathcal{D}\phi F[\phi] \exp \left(-S + \int d^4x j\phi \right) = 0 \quad (2.4.68)$$

Which can be written compactly as

$$F \left[\frac{\delta}{\delta j(x)} \right] Z_r[j] = 0 \quad (2.4.69)$$

this showing that indeed Z_r corresponds to a restraint path-integral, restrained in the sense that it only takes into account configurations for which $F = 0$.

2.5 Green functions of the interacting theory

We are now ready to study the Green functions of the $\lambda\phi^3$ theory when the interaction is present. Let us begin by evaluating the two-point function $G^{(2)}$,

$$G^{(2)}(u, v) = \frac{1}{Z} \int \mathcal{D}\phi \exp \left(-S_0 - \frac{\lambda}{3!} \int d^4x \phi^3(x) \right) \phi(u) \phi(v) \equiv \frac{\text{num}}{Z} \quad (2.5.70)$$

with Z defined as

$$Z = \int \mathcal{D}\phi \exp \left(-S_0 - \frac{\lambda}{3!} \int d^4x \phi^3(x) \right) \quad (2.5.71)$$

and S_0 the free (quadratic) action.

Consider first the denominator Z . Let us expand the interaction term in powers of λ ,

$$Z = \sum_{n=0}^{\infty} (-1)^n \frac{\lambda^n}{n!(3!)^n} \int d^4x_1 d^4x_2 \dots d^4x_n \langle \phi^3(x_1) \phi^3(x_2) \dots \phi^3(x_n) \rangle_0 \times Z_0 \quad (2.5.72)$$

The v.e.v. in the r.h.s. corresponds to a free theory. As we have seen, only if it corresponds to an even number of fields it will be different from zero. We can then write

$$Z = \sum_{k=0}^{\infty} \frac{\lambda^{2k}}{(2k)!(3!)^{2k}} \int d^4x_1 d^4x_2 \dots d^4x_{2k} \langle \phi^3(x_1) \phi^3(x_2) \dots \phi^3(x_{2k}) \rangle_0 \times Z_0 \quad (2.5.73)$$

Consider the first non-trivial term, namely $k = 1$

$$I^{(2)} = \frac{\lambda^2}{(3!)^2 2!} \int d^4x_1 d^4x_2 \langle \phi(x_1) \phi(x_1) \phi(x_1) \phi(x_2) \phi(x_2) \phi(x_2) \rangle_0 \times Z_0 \quad (2.5.74)$$

Since the v.e.v. corresponds to a free theory, we can apply the Wick theorem to decompose it into products of two-point Green functions for the free theory, which are known. Consider first the case in which the three “contractions” corresponds to fields ϕ at different points x_1 and x_2 . The resulting product gives

$$p_{1,2} = G_0^{(2)}(x_1 - x_2)^3 \quad (2.5.75)$$

How many of these cubic terms are there? Consider the first $\phi(x_1)$. It can be contracted with any of the three ϕ at points x_2 . Once a given contraction is done, the second $\phi(x_1)$ has just two possible $\phi(x_2)$ to contract with. The remaining last $\phi(x_1)$ has just no option an only one possible contraction. So the number of terms identical to $p_{1,2}$ is 3.2. We then have

$$P_{1,2} = 3.2 p_{1,2} = 3! G_0^{(2)}(x_1 - x_2)^3 \quad (2.5.76)$$

But there is also the possibility that two of the $\phi(x_1)$ points contract between themselves. There is for the remaining $\phi(x_1)$ no other possibility than contracting with one of the three $\phi(x_2)$. And the last two $\phi(x_2)$ have to contract among themselves. Let us call $p_{1,1}$ this kind of terms. They take the form

$$p_{1,1} = G_0^{(2)}(0) G_0^{(2)}(x_1 - x_2) G_0^{(2)}(0) \quad (2.5.77)$$

Concerning the number of times this term appears, the first $\phi(x_1)$ has 2 possible choices to contract with another $\phi(x_1)$. The third $\phi(x_1)$ can choose among 3 $\phi(x_2)$'s and then the other two $\phi(x_2)$ have just one possibility. So the factor is again $3 \cdot 2 = 3!$. Using the same notation we have

$$P_{1,1} = 3!G_0^{(2)}(0)G_0^{(2)}(x_1 - x_2)G_0^{(2)}(0) \quad (2.5.78)$$

We then have

$$\begin{aligned} \langle \phi(x_1)\phi(x_1)\phi(x_1)\phi(x_2)\phi(x_2)\phi(x_2) \rangle_0 &= P_{1,2} + P_{1,1} \\ &= 3!G_0^{(2)}(x_1 - x_2) \left(G_0^{(2)}(x_1 - x_2)^2 + G_0^{(2)}(0)^2 \right) \end{aligned} \quad (2.5.79)$$

We then have

$$\begin{aligned} Z &= \left(1 + \frac{\lambda^2}{2 \cdot 3!} \int d^4x_1 d^4x_2 G_0^{(2)}(x_1 - x_2) \left(G_0^{(2)}(x_1 - x_2)^2 + G_0^{(2)}(0)^2 \right) + \right. \\ &\quad \left. \dots \right) Z_0 \end{aligned} \quad (2.5.80)$$

It will be convenient at this point to write the free two-point Green function in terms of its Fourier transform (2.2.49)

$$G_0^{(2)}(z) = \int \frac{d^4k}{(2\pi)^4} \exp(ikz) \frac{1}{k^2 + m^2} \quad (2.5.81)$$

so that calling $z = x_1 - x_2$, we have for the term with $G_0^{(2)}(x_1 - x_2)^3$

$$\begin{aligned} I_3 &= \frac{\lambda^2}{2 \cdot 3!} \int d^4x_1 d^4x_2 \int \frac{d^4k_1}{(2\pi)^4} \int \frac{d^4k_2}{(2\pi)^4} \int \frac{d^4k_3}{(2\pi)^4} \\ &\quad \exp(ik_1z) \exp(ik_2z) \exp(ik_3z) \frac{1}{k_1^2 + m^2} \frac{1}{k_2^2 + m^2} \frac{1}{k_3^2 + m^2} \end{aligned} \quad (2.5.82)$$

Integrating out x_1 we get a delta-function,

$$\int d^4x_1 \exp(i(k_1 + k_2 + k_3)x_1) = (2\pi)^4 \delta^{(4)}(k_1 + k_2 + k_3) \quad (2.5.83)$$

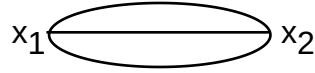
Integrating out x_2 we get the same result so that we have

$$\begin{aligned} I_3 &= \frac{1}{(2\pi)^4} \int d^4k_1 d^4k_2 d^4k_3 \delta^{(4)}(k_1 + k_2 + k_3) \delta^{(4)}(k_1 + k_2 + k_3) \\ &\quad \frac{1}{k_1^2 + m^2} \frac{1}{k_2^2 + m^2} \frac{1}{k_3^2 + m^2} \end{aligned} \quad (2.5.84)$$

Finally, integrating over k_3 ,

$$I_3 = \frac{\delta^{(4)}(0)}{(2\pi)^4} \int d^4k_1 d^4k_2 \frac{1}{k_1^2 + m^2} \frac{1}{k_2^2 + m^2} \frac{1}{(k_1 + k_2)^2 + m^2} \quad (2.5.85)$$

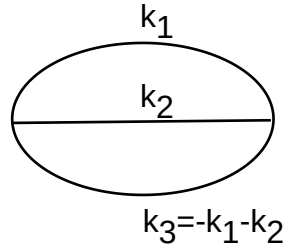
We can represent this result by the following (Feynman) diagram:



In this diagram

1. Each line represents a propagator. That is, a two-point free Green function in coordinate space, $G_0^{(2)}(x_1 - x_2)$ or its Fourier transform in momentum space, $1/(k^2 + m^2)$.
2. Each independent loop represents an integral. (In the diagram above one considers as independent any pair of loops among the 3 possible loops of two lines since always the third one can be expressed in terms of the other two).

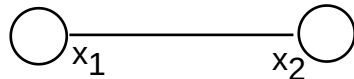
We can be more specific and draw the Feynman diagram, in momentum space indicating the momentum associated with each line:



Concerning $G_0^{(2)}(0)^2 G_0^{(2)}(x_1 - x_2)$, which can be more instructively written as

$$G_0^{(2)}(0) G_0^{(2)}(x_1 - x_2) G_0^{(2)}(0) = G_0^{(2)}(x_1 - x_1) G_0^{(2)}(x_1 - x_2) G_0^{(2)}(x_2 - x_2) \quad (2.5.86)$$

its Feynman diagram can be drawn as follows:



Then, in coordinate space, we can represent diagrammatically Z in the form

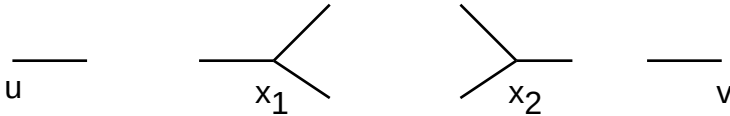
$$1 + \lambda^2 (\text{diagram of a circle with a horizontal line through its center} + \text{diagram of two circles connected by a horizontal line}) + \dots$$

What about the numerator in (2.5.70)? Expanding as we did for Z , we have

$$\begin{aligned} \text{num} = & \left(\langle \phi(u)\phi(v) \rangle_0 + \frac{\lambda^2}{2!(3!)^2} \int d^3x_1 d^3x_2 \langle \phi(u)\phi^3(x_1)\phi^3(x_2)\phi(v) \rangle_0 + \dots \right) \times Z_0 \end{aligned} \quad (2.5.87)$$

Since Z_0 appears as an overall factor both in the numerator and the denominator, we shall omit it from here on.

In order to analyse the number of ways we can do contractions when applying the Wick theorem in the second term in the r.h.s. of (2.5.87), it will be useful to draw the following picture in which each field $\phi(z)$ is represented by a short line with a z in one of its end,



Let us make diagrams as follows. First consider the different ways to join the line that represents $\phi(u)$ with any of the three lines at x_1 representing the three fields at the vertex x_1 , or with any of the three lines at x_2 . We then count a factor 2 and draw one of the two equivalent diagrams, say joining $\phi(u)$ with one $\phi(x_1)$ and $\phi(v)$ with one $\phi(x_2)$. But in fact we have three choices for joining $\phi(u)$ with one leg in x_1 and three other choices for joining $\phi(u)$ with x_2 . We then have a factor $2 \times 3 \times 3$:

$$2.3.3 \quad \left(\begin{array}{c} \text{---} \\ \text{u} \end{array} \text{---} x_1 \begin{array}{c} \diagup \\ \diagdown \end{array} \quad \begin{array}{c} \diagdown \\ \diagup \end{array} x_2 \text{---} v \right)$$

Now, for joining each one of the two remaining legs in x_1 with one in x_2 we have two choices. The overall factor is then 2.3.3.2 or $3!^2$ and the diagram (including the factor counting the number of times it appears) will be then drawn as follows:

$$(3!)^2 \quad \text{u} \text{---} x_1 \text{---} \text{---} x_2 \text{---} v$$

There are other two possible diagrams appearing in the numerator:

$$\begin{array}{cc} \text{---} \text{---} & \text{---} \text{---} \\ \text{u} \text{---} v & \text{u} \text{---} v \end{array}$$

Diagrams like the two above are called *disconnected* diagrams in contrast with the one in the top of this page which is a connected one.

We see that graphically, we can then represent the numerator in the form

$$\text{num} = \text{---} + \frac{\lambda^2}{2!} \text{---} \text{---} + \frac{\lambda^2}{2!(3!)^2} \text{---} [\text{---} + \text{---}]$$

Now we can factorize a propagator in the first and third terms in the r.h.s. and the same can be done with the second and the following term which is not represented in the figure above. Then, the numerator can be rearranged as

$$\text{num} = \text{---} [1 + \frac{\lambda^2}{2!(3!)} (\text{---} + \text{---})] + \frac{\lambda^2}{2!} \text{---} [1 + \dots]$$

We then see that each bracket is nothing but the denominator Z so that we can write:

$$\text{num} = Z (\text{---} + \frac{\lambda^2}{2} \text{---} + \dots)$$

The factor Z in the numerator is then cancelled out by the denominator so that we finally have for $G^{(2)}(u, v)$

$$G^{(2)}(u, v) = u \text{---} v + \frac{\lambda^2}{2} u \text{---} \text{---} v + \dots$$

We see that only connected diagrams contribute to $G^{(2)}(u, v)$. This is true in general and this is the reason why one calls $Z[j]$ the generating functional for connected diagrams.

Some comments

I- Diagrams like



are called tadpole diagrams.



The following article in the Webster 's New Collegiate Dictionary clarifies the figure and the name:

tad.pole 'tad-,pol/n

[ME taddepol, fr. tode toad + polle head]

(15c):

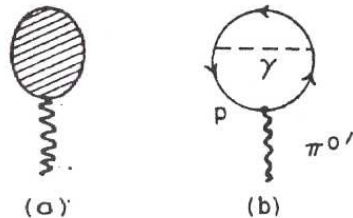
a larval amphibian; specif: a frog or toad larva that has a rounded body with a long tail bordered by fins and external gills soon replaced by internal gills and that undergoes a metamorphosis to the adult.

It was Sidney Coleman who called such a diagram a "tadpole diagram" in one of his papers in collaboration with Sheldon Glashow⁵. They wrote:

If the scalar meson octet does exist, there is the possibility of a class of Feynman diagrams that vanish for every other kind of particle. These are the scalar tadpoles, diagrams with only one external line. In the limit of exact unitary symmetry, all scalar tadpoles vanish; but as we turn on the various symmetry-breaking interactions, the scalar tadpoles acquire nonzero values... Figure 1 shows a typical electromagnetic contribution to the π'_0 tadpole.

⁵S. R. Coleman and S. L. Glashow, "Departures from the eightfold way: Theory of strong interaction symmetry breakdown," Phys. Rev. **134** (1964) B671.

FIG. 1. (a) The general scalar meson tadpole diagram. (b) A typical $\pi^{0'}$ tadpole, due to electromagnetism.



The editor of the journal was not satisfied, but he finally accepted since Coleman proposed in his answer to replace “tadpole” by “spermion”⁶.

Consider for example the tadpole diagram on page 103, with one loop in each extreme. This diagram is problematic: the integral over each loop, which corresponds to $G_0^{(2)}(0)$, is divergent since, according to our Feynman recipe

$$G_0^{(2)}(0) = \int \frac{d^4k}{(2\pi)^2} \frac{1}{k^2 + m^2} \propto \int_0^\infty \frac{k^3 dk}{k^2 + m^2} \quad (2.5.88)$$

Now, this divergence can be easily eliminated by a procedure that is standard in Quantum Field Theory. One adds to the $\lambda\phi^3$ Lagrangian L a “counterterm” which in the present case should be linear in ϕ , so that the resulting Lagrangian $\bar{L}$ reads

$$\bar{L} = L + a\lambda\phi \quad (2.5.89)$$

with a some constant to be chosen. The presence of this new term in the action generates new diagrams. For example, when computing Z there will be a new diagram of order λ^2 coming from the product of just one factor $\lambda\phi^3$ and a linear λa factor in the expansion of the λ terms of the action: Calling T this new contribution

$$T = a\lambda^2 \int d^4x_1 d^4x_2 \langle \phi(x_1) \phi^3(x_2) \rangle_0 = a\lambda^2 G_0^{(0)}(0) \int d^4x_1 d^4x_2 G_0^{(2)}(x_1 - x_2) \quad (2.5.90)$$

The new diagram is, then, also divergent. An appropriate choice of the constant a allows to cancel this divergent contribution with that arising from the tadpole. Diagrammatically one has

⁶ As Coleman and Glashow indicate in footnote 11 of their work, the contribution of tadpole diagrams was first suggested by Julian Schwinger in his renowned paper *A Theory of the Fundamental Interactions*.

$$\text{---}\bigcirc\text{---}\bigcirc + \underset{a}{\bullet}\text{---}\bigcirc = \text{finite}$$

One can arrange a so that the new term just cancels the divergency without adding any finite contribution.

In this way one can eliminate any divergency coming from tadpole diagrams. The procedure is the path-integral version of normal ordering in the operator approach to QFT. The rationale of the approach can be schematized as follows: unless one communicates with God to know wheter L or $\bar{L}$ describes the actual physical system, both Lagrangian are candidates on an equal foot.

Let us summarize the recipe for constructing any Feynman diagram in the $\lambda\phi^3$ theory:

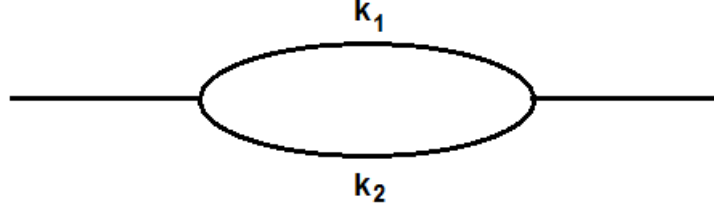
$$k_1\text{---}k_2 = \delta^{(4)}(k_1+k_2) \frac{1}{k_1^2+m^2}$$

$$k_1 \begin{array}{l} \nearrow k_2 \\ \searrow k_3 \end{array} = \lambda \delta^{(4)}(k_1+k_2+k_3)$$

It remains to calculate the combinatorial factor with which each diagram enters. The analysis goes as follows: the n^{th} term in the expansion has an overall factor $(-\lambda)^n/(n!(3!)^n)$. If we call N the number of times a given diagram appears in such a term, the factor f we are looking for is

$$f = N \frac{(-\lambda)^n}{n!(3!)^n} \tag{2.5.91}$$

Given a diagram, there is a number called its order (order of the symmetry group) r . For example, the simple loop diagram has an order $r = 2$:



This number r corresponds to the number of diagrams that are identical when one interchanges lines in a symmetry operation. In the example above, there are 2 identical diagrams, one with the ' k_1 ' line above the ' k_2 ' line and one with the ' k_1 ' line below the ' k_2 ' one. Then, $r = 2$. Now, one can prove that for a $\lambda\phi^3/3!$ interaction term N is related to r as follows

$$N = \frac{n!(3!)^n}{r} \quad (2.5.92)$$

One then has

$$f = \frac{(-\lambda)^n}{r} \quad (2.5.93)$$

One has to add some factors if one is working in momentum space. First, there is a $1/(2\pi)^4$ for each propagator. But each loop integral leads to a $2\pi^4\delta(k_1+k_2+\dots)$ at each vertex where $k_1, k_2, \dots$ join. Finally, there is an overall $2\pi^4\delta(k_1+k_2+\dots)$ which guarantees conservation of four momenta. Then, if we call l the number of lines and V the number of vertices, we have a factor F of the form

$$F = \left(\frac{1}{(2\pi)^4}\right)^l ((2\pi)^4)^V (2\pi)^4 = \left(\frac{1}{(2\pi)^4}\right)^{l-V-1} \quad (2.5.94)$$

The quantity $l - V - 1$ is just the number L of independent loops of the diagram. Then, in momentum space the total factor $\mathcal{F}$ in front of a given diagram is

$$\mathcal{F} = \frac{(-\lambda)^n}{r} \left(\frac{1}{(2\pi)^4}\right)^L \quad (2.5.95)$$

II- Usually, the denominator (Z) in v.e.v.'s is written as

$$\text{den} = \exp \left(\sum_i D_i(\text{vac}) \right) \quad (2.5.96)$$

where $D_i(\text{vac})$ is the contribution of the i^{th} connected vacuum diagram (i.e. a diagram without external legs). The exponential appears because one has a symmetry factor of the form

$$\prod_i \frac{1}{n_i! r_i^{n_i}}$$

for any diagram with n_1 components of the “type” 1, n_2 of the type 2, $\dots$. The numerator is just the denominator times the sum of connected components without vacuum components,

$$\text{num} = \exp \left(\sum_i D_i(\text{vac}) \right) \times \sum \text{Diag}(\text{connected}, \text{external legs}) \quad (2.5.97)$$

or. compactly

$$\text{num} = \exp \left(\sum_i D_i(\text{vac}) \right) \times \sum D(c, e) \quad (2.5.98)$$

so that

$$\frac{\text{num}}{\text{den}} = \sum D(c, e) \quad (2.5.99)$$

III- We have seen that the generating functional for the free theory can be computed to be

$$Z_0[j] = Z_0[0] \exp \left(\frac{1}{2\hbar} \int d^4x_1 d^4x_2 j(x_1) G_0^{(2)}(x_1 - x_2) j(x_2) \right) \quad (2.5.100)$$

with $G_0^{(2)}(x_1 - x_2)$ defined through

$$(-\square + m^2) G_0^{(2)}(x_1 - x_2) = \delta^{(4)}(x_1 - x_2) \quad (2.5.101)$$

Now, when an interaction L_{int} is added, the generating functional cannot be computed exactly,

$$Z[j] = \int \mathcal{D}\phi \exp \left(-\frac{1}{\hbar} S_0 \right) \exp \left(-\frac{1}{\hbar} \int d^4x L_{\text{int}}[\phi] \right) \exp \left(\int d^4x j(x) \phi(x) \right) \quad (2.5.102)$$

The interaction Lagrangian can be written in the form

$$L_{\text{int}} = \exp \left(-\int d^4z j(z) \phi(z) \right) L_{\text{int}} \left[\frac{\delta}{\delta j(x)} \right] \exp \left(\int d^4z j(z) \phi(z) \right) \quad (2.5.103)$$

and the same for any functional of L_{int} . In particular

$$\begin{aligned} \exp\left(-\frac{1}{\hbar} \int d^4x L_{int}[\phi]\right) &= \exp\left(-\int d^4z j(z)\phi(z)\right) \\ &\times \exp\left(-\frac{1}{\hbar} \int d^4x L_{int}\left[\frac{\delta}{\delta j}\right]\right) \exp\left(\int d^4z j(z)\phi(z)\right) \end{aligned} \quad (2.5.104)$$

and then

$$Z[j] = \exp\left(-\frac{1}{\hbar} \int d^4x L_{int}\left[\frac{\delta}{\delta j}\right]\right) Z_0[j] \quad (2.5.105)$$

this formula can be taken as the perturbative (in the sense the exponential has to be expanded in powers of the coupling constant appearing in L_{int}) definition of the path-integral.

Chapter 3

Path-integral Quantization of Gauge Theories

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3.1 The Abelian case

Electrodynamics, a paradigm of gauge field theories, admits a formulation in which the basic objects are *gauge fields* $A_\mu(x)$ ($\mu = 0, 1, 2, \dots, d$ with $d+1$ the dimensionality of space-time). In terms of these gauge fields (*connections* in the mathematical language of fiber bundles¹) one constructs the electromagnetic field tensor $F^{\mu\nu}(x)$ (the *curvature*),

$$F^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\mu \quad (3.1.1)$$

The Maxwell equations (in the absence of sources) are compactly written as

$$\partial_\mu F^{\mu\nu} = 0 \quad (3.1.2)$$

$$\partial_\mu {}^*F^{\mu\nu} = 0 \quad (3.1.3)$$

where we have defined the *dual* of $F^{\mu\nu}$ as

$${}^*F^{\mu\nu} = \frac{1}{2} \varepsilon^{\mu\nu\alpha\beta} F_{\alpha\beta} \quad (3.1.4)$$

The observable in the theory, the electric E^i and magnetic B^i fields ($i = 1, 2, 3$) are then given by

$$E^i \equiv F^{0i} \quad B^i \equiv {}^*F^{0i} \quad (3.1.5)$$

¹A book that introduces rigorously fiber bundles is: S. Kobayashi and K. Numizu, “Foundations of Differential Geometry”, Interscience, New York, 1963; a book written by physicists: “Fiber Bundle Techniques in Gauge Theories”, W. Drechsler and M.E. Mayer, Lectures Notes in Physics, Springer, Berlin, 1977.

Eq.(3.1.2) can be derived, *à la* Euler-Lagrange, from the Maxwell action,

$$S = \frac{1}{4} \int d^4x F^{\mu\nu} F_{\mu\nu} \quad (3.1.6)$$

while eq.(3.1.3) is just a mathematical property (the Bianchi identity) of the curvature defined in (3.1.1).

One calls the theory a gauge theory because under the “gauge” change

$$A_\mu(x) \rightarrow A_\mu(x)^\Lambda = A_\mu(x) + \partial_\mu \Lambda(x) \quad (3.1.7)$$

with Λ an arbitrary real function satisfying $\partial_i \partial_j \Lambda = \partial_j \partial_i \Lambda$, the observables remain unchanged

$$E^{\Lambda^i}(x) = E^i(x) \quad B^{\Lambda^i}(x) = B^i(x) \quad (3.1.8)$$

As a prologue to the study of non-Abelian gauge theories we can identify Λ with an algebra valued function of an (Abelian) $U(1)$ algebra. Indeed, a group element of $U(1)$ can be always written as

$$g(x) = \exp(i\Lambda(x)) \quad (3.1.9)$$

and in that case the gauge transformation (3.1.7) takes a form which looks more complicated but which can be more easily generalized to other (non-Abelian) groups

$$A_\mu(x) \rightarrow A_\mu(x)^g = A_\mu(x) + \frac{1}{i} g^{-1} \partial_\mu g \quad (3.1.10)$$

The advantage of this definition of a gauge transformation is that it can be immediately generalized to the case of non-Abelian gauge groups in which Λ takes values in the Lie algebra of the gauge group.

In order to quantize a gauge theory in the path-integral framework we shall start from the classical action (3.1.6) and identify the basic field variable to integrate out *integral over all fields*.

We shall consider the Lagrangian formulation but the most rigorous formulation should start from the Hamiltonian which requires a more detailed analysis which you can find in the notes of my graduate course on Path-Integrals in QFT and Statistical mechanics, following the treatment of Slavnov-Faddeev book² or any other book on QFT.

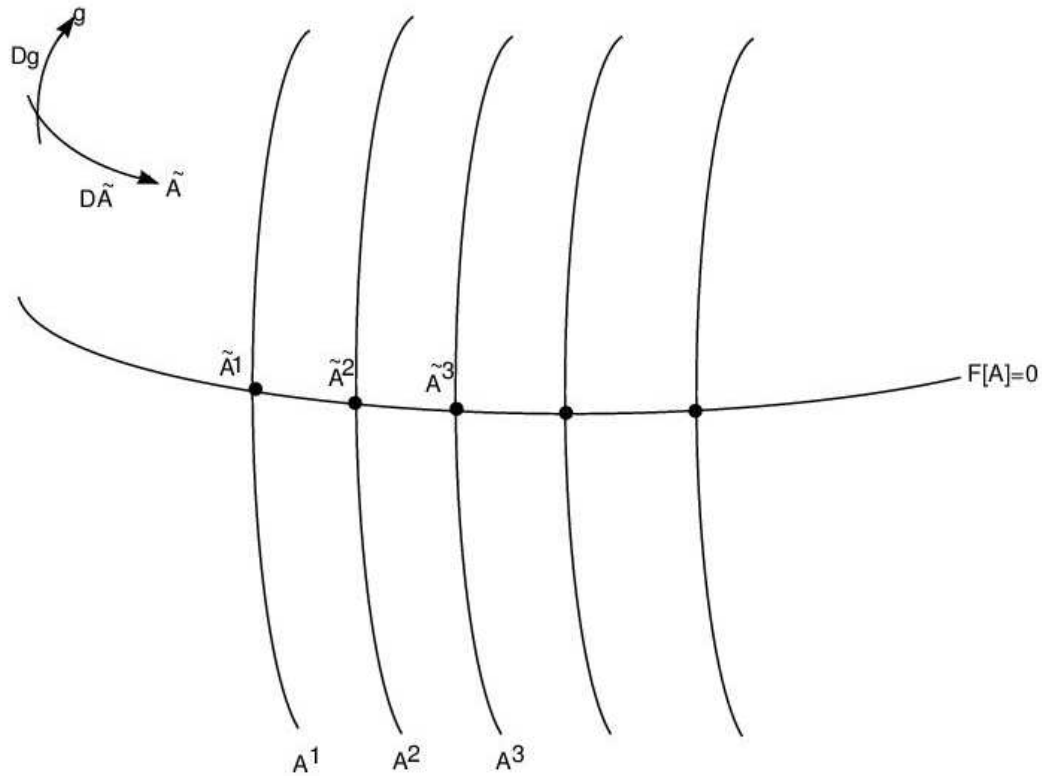
In the Lagrangian formulation, the natural path-integral variables seems to be the gauge fields A_μ . One faces however a problem: there are there d path-integral variables in d space-time dimensions but due to gauge-invariance not all of them are independent in the sense that several choices of A_μ describe the same physical situation, at last at the classical level.

²A. A. Slavnov and L. D. Faddeev. “Gauge Fields: An Introduction To Quantum Theory”, Frontiers in Physics) , Springer, 1993.

If we were to represent graphically the would be domain of path-integration, it should look as in the next figure. There, vertical lines are the orbits of the gauge group. One moves along an orbit by gauge transforming a connection/gauge field. An infinitesimal gauge transformation corresponds to a movement tangent to the orbit. That is, given an infinitesimal parameter ϵ , one has for the gauge transformation,

$$A_\mu^\epsilon(x) = A_\mu(x) + \partial_\mu \epsilon \quad (3.1.11)$$

and then $\partial_\mu \epsilon$ represent a vector tangent to the orbit.



The fact that the space over which one integrates has this structure poses a problem: the existence of zero modes of the “natural” operator whose eigenfunctions are used in the Bessel-Fourier expansion leading to a simple path-integral measure.

In the Lagrangian formulation one can solve the problem by restricting the integration to one representative of the gauge fields for each orbit. That is, one takes the whole family of gauge-transformed of a given configuration as one point in the integration domain. Different

“points” then corresponds to inequivalent gauge fields (i.e., gauge fields that are not the gauge transformed of any other).

The process of selecting one representative is what was called in classical electrodynamics gauge fixing. Indeed, even at the classical level, when studying Maxwell equations it was necessary to fix the gauge. That is, to chose a gauge condition (for example the Lorenz gauge condition³, the Coulomb condition, etc).

A gauge condition can be written as an equation of the form

$$F[A_\mu] = 0 \quad (3.1.12)$$

This is represented in the previous figure as the line crossing the orbits and thus selecting a representative of each gauge equivalent configurations.

Suppose we adopt the usual definition of the Lagrangian version of the partition function

$$Z = \int \mathcal{D}A_\mu \exp \left(-\frac{1}{\hbar} S[A] \right) \quad (3.1.13)$$

In order to make an expansion around a classical configuration $A_\mu^{cl}(x)$ to study small quantum fluctuation one can propose the following change of path-integral variables

$$A_\mu(x) = A_\mu^{cl}(x) + \sqrt{\hbar} a_\mu(x) \quad (3.1.14)$$

Then, in terms of the new variable a_μ one has (up to second order)

$$Z = \exp \left(-\frac{1}{\hbar} S[A_\mu^{cl}] \right) \int \mathcal{D}a_\mu \exp \left(- \int d^4x d^4z a_\mu(x) S_{\mu\nu}^{(2)}(x, z) a_\nu(z) \right) \quad (3.1.15)$$

where

$$S_{\mu\nu}^{(2)}(x, z) = \left. \frac{\delta^2 S}{\delta A_\mu(x) \delta A_\nu(z)} \right|_{A=A^{cl}} \quad (3.1.16)$$

When writing the action in terms of the new variable a_μ , we have supposed that there are no linear terms (i.e., A_μ^{cl} is a solution of the eqs. of motion). For example, one can use $A_\mu^{cl} = 0$.

Denoting by Λ a gauge transformation parameter we write the gauge transformed of A_μ^{cl} as

$$A_\mu^{cl}(x, \Lambda) = A_\mu^{cl}(x) + \partial_\mu \Lambda(x) \quad (3.1.17)$$

³L. Lorenz, “On the Identity of the Vibrations of Light with Electrical Currents” Phil. Mag. 34 (1867) 287 (Ludwig Lorenz, not Hendrik Lorentz!).

Then, if

$$\frac{\delta S}{\delta A_\mu^{cl}(x, \Lambda)} = 0 \quad (3.1.18)$$

$$\frac{\delta}{\delta \Lambda(z)} \frac{\delta S}{\delta A_\mu^{cl}(x, \Lambda)} = \int d^4 y \frac{\delta^2 S}{\delta A_\mu^{cl}(x, \Lambda) \delta A_\nu^{cl}(y, \Lambda)} \frac{\delta A_\nu^{cl}(y, \Lambda)}{\delta \Lambda(z)} = 0 \quad (3.1.19)$$

or

$$\int d^4 y S_{\mu\nu}^{(2)} \frac{\delta A_\nu^{cl}(y, \Lambda)}{\delta \Lambda(z)} = 0 \quad (3.1.20)$$

this meaning that the quadratic form $S_{\mu\nu}^{(2)}$ has a zero mode $a_\mu^{(0)}$ of the form

$$a_\mu^{(0)}(y) = \int d^4 y \frac{\delta A_\nu^{cl}(y, \Lambda)}{\delta \Lambda(z)} \alpha(z) = \int d^4 y \partial_\mu \delta(y - z) \alpha(z) = \partial_\mu \alpha(z) \quad (3.1.21)$$

where $\alpha(z)$ is an arbitrary function. This poses a serious problem since in the zero-mode direction the path-integral variable c_n^0 (the coefficient accompanying the zero mode in the Bessel-Fourier expansion) is no more gaussian leading to a divergent result. d.

To handle this problem note that orthogonality of zero and non zero modes $a_\mu^{(n)}$

$$0 = \int d^4 x a_\mu^{(0)}(x) a_\mu^{(n)}(x) = \int d^4 x \partial_\mu \alpha(x) a_\mu^{(n)}(x) = - \int d^4 x \alpha(x) \partial_\mu a_\mu^{(n)}(x) \quad (3.1.22)$$

Since this should be valid for any α we conclude that the eigenfunctions should be fixed to the Lorenz gauge,

$$\partial_\mu a_\mu^{(n)}(x) = 0 \quad (3.1.23)$$

This result suggests that the “natural” gauge condition to impose on the gauge field A_μ (and consequently on a_μ is in this Abelian case the Lorenz gauge. Indeed, if we impose

$$\partial_\mu a_\mu(x) = 0 \quad (3.1.24)$$

we can expect to eliminate the zero mode from the quadratic integration and trade it for an integration over the gauge parameter Λ . (Evidently, we can chose the *background* configuration A_μ^{cl} to also satisfy the Lorentz condition).

Then starting by the natural gauge condition which in this case reads

$$F[A] = \partial_\mu A_\mu = 0 \quad (3.1.25)$$

we write a resolution of the identity in the form

$$1 = \Delta_{FP}[A] \int \delta(F[A^\Lambda]) \mathcal{D}\Lambda \quad (3.1.26)$$

Here we have called $\Delta_{FP}[A]$ (the Faddeev-Popov determinant) a functional of the gauge fields to be adjusted so that the r.h.s. is 1.

We introduce this “resolution of tyhe identity” in the partition function (3.1.13)

$$Z = \int \mathcal{D}A_\mu \mathcal{D}\Lambda \Delta_{FP}[A] \exp\left(-\frac{1}{\hbar}S[A]\right) \delta(F[A^\Lambda]) \quad (3.1.27)$$

and make a shift in the path-integral variable A_μ , in the form

$$A_\mu = A_\mu^{cl}(x, \Lambda) + \sqrt{\hbar}a_\mu = A_\mu^{cl}(x) + \partial_\mu \Lambda + \sqrt{\hbar} \sum_n' c_n a_\mu^{(n)}(x) \quad (3.1.28)$$

with c_n the Bessel-Fourier coefficients of the expansion of a_μ (without including the zero mode) in terms of the basis $\{a_\mu^{(n)}\}$ of the quadratic operator $S_{\mu\nu}^{(2)}$,

$$\int dy S_{\mu\nu}^{(2)}(x, y) a_\mu^{(n)}(y) = \lambda_n a_\mu^{(n)}(x) \quad (3.1.29)$$

Note that the zero mode although absent from the sum, is still there through the $\partial_\mu \Lambda$ which is precisely the zero mode!. Also, working in the Lorenz gauge one has $\delta(F[A^\Lambda] = \delta(\partial_\mu A_\mu + \square \Lambda) = \delta(\square \Lambda)$

In this form, we pass from the integration measure $\mathcal{D}A_\mu \mathcal{D}\Lambda$ to $\prod_n' dc_n \mathcal{D}\Lambda$ and we have, instead of (3.1.15),

$$\begin{aligned} Z &= \exp\left(-\frac{1}{\hbar}S[A_\mu^{cl}]\right) \int \mathcal{D}\Lambda \prod_n dc_n \Delta_{FP}[A^{cl}, \sqrt{\hbar}c_n] \exp(-c_n^2 \lambda_n) \delta[\square \Lambda] \\ &= \exp\left(-\frac{1}{\hbar}S[A_\mu^{cl}]\right) \frac{\Delta_{FP}[A_\mu^{cl}]}{\det \square} (\det' S^{(2)})^{-1/2} \end{aligned} \quad (3.1.30)$$

In the second line we have kept the term of order 0 in $\hbar$ in the Faddeev-Popov determinant since the rest of it (which depends on $\sqrt{\hbar}c_n'$) corresponds to higher order corrections in $\hbar$.

Let us now evaluate Δ_{FP} . Starting from (3.1.26) we expand up to first order in Λ since higher order terms vanish.

$$1 = \Delta_{FP}[A] \int \delta\left(F[A] + \frac{\delta F}{\delta A_\mu} \partial_\mu \Lambda\right) \mathcal{D}\Lambda \quad (3.1.31)$$

or, using explicitly that F corresponds to the Lorentz condition

$$1 = \Delta_{FP}[A] \int \delta(\partial_\mu A_\mu + \partial_\mu \partial_\mu \Lambda) \mathcal{D}\Lambda = \Delta_{FP}[A] \frac{1}{\det \square} \quad (3.1.32)$$

We then have, in the Lorentz gauge

$$\Delta_{FP}[A] = \det \square \quad (3.1.33)$$

and then the final form of Z up to this order is

$$Z = \exp \left(-\frac{1}{\hbar} S[A_\mu^{cl}] \right) (\det' S^{(2)})^{-1/2} \quad (3.1.34)$$

Note that given the Lorentz condition, we have

$$\partial_\mu A_\mu^\Lambda = \partial_\mu A_\mu + \square \Lambda \quad (3.1.35)$$

3.2 Zero-modes and Gribov ambiguities

Suppose that A_μ satisfies the Lorentz condition. Is it possible that other $\Lambda \neq 0$ exists so that also A_μ^Λ also satisfies this condition? From (3.1.35) we see that this implies

$$\square \Lambda = 0 \quad (3.2.36)$$

Equation (3.2.36) has no well behaved solutions other than the trivial one. In contrast, in the case of non-Abelian gauge theories, V.N. Gribov⁴ explicitly constructed non-trivial examples in which the analogous equation has non-trivial solutions for the case of the Coulomb gauge. Soon after the mathematician I.M. Singer showed rigorously that no continuous choice of exactly one gauge field on each orbit can be made⁵. Thus the Gribov ambiguity for the Coulomb gauge will occur in all other gauges.

This problem is called the Gribov problem: one cannot fix univocally the gauge and there are always *Gribov copies*: there are gauge equivalent configurations all satisfying a given gauge condition. One can see one of the problems implied by this by noting that in (1.1.26) we were able to extract the $1/\square$ precisely because eq.(1.1.33) has just the trivial solution $\Lambda = 0$.

Now, it can be shown that the Gribov problem does not exist for small deformations of an irreducible gauge potential, in the “natural” gauge and this justifies gauge fixing within the framework of perturbation expansion. “Small” means that a certain norm (Sobolev $k = 0$ norm) of the gauge field is small enough to include the closest ambiguity⁶.

⁴V.N. Gribov, . “Quantization of non-Abelian gauge theories”, Nucl. Phys. B. 139 (1978) 1.

⁵I.M. Singer, “Some Remarks on the Gribov Ambiguity”, Commun. Math. Phys. 60 (1978) 7.

⁶M. Daniel and C.M. Viallet, “ The gauge fixing problem around classical solutions of the Yang-Mills theory”, Phys. Lett. 76B (1978) 458.

For a generic gauge condition $F[a] = 0$, the Faddeev Popov determinant is in the Abelian case,

$$\Delta_{FP} = \det \left(\frac{\delta F}{\delta A_\mu} \partial_\mu \right) \quad (3.2.37)$$

Zero modes in non-Abelian gauge theories

Let us first fix conventions for the case of non-Abelian gauge theories with group G and a Lie Algebra $\mathcal{G}$ generated by the set of generators t^a ,

$$[t^a, t^b] = f^{abc} t^c \quad a = 1, 2, \dots \dim \mathcal{G} \quad (3.2.38)$$

$$\text{tr } t^a t^b = \frac{1}{2} \delta^{ab} \quad (3.2.39)$$

$$f^{abc} f^{pbc} = \delta^{ap} C[G] \quad (3.2.40)$$

with $C[G]$ the quadratic Casimir of G .

For non-Abelian gauge groups, the gauge field takes values in the Lie Algebra $\mathcal{G}$,

$$A_\mu(x) = A_\mu^a(x) t^a \quad (3.2.41)$$

A gauge transformation is associated now with an element g of the gauge group G ,

$$A_\mu(x) \rightarrow A_\mu^g = g^{-1} A_\mu g + \frac{1}{i} g^{-1} \partial_\mu g \quad (3.2.42)$$

with

$$g = \exp(i t^a \Lambda^a(x)) \quad (3.2.43)$$

For an infinitesimal gauge transformation one has

$$g = \exp(i \Lambda(x)) \approx 1 + i \Lambda(x) + O(\Lambda^2) \quad (3.2.44)$$

where we have written $\Lambda = \Lambda^a t^a$. In this case one has

$$A_\mu^g \approx (1 - i \Lambda) A_\mu (1 + i \Lambda) + \partial_\mu \Lambda + O(\Lambda^2) \quad (3.2.45)$$

or

$$\delta A_\mu \equiv A_\mu^g - A_\mu = \partial_\mu \Lambda + i[A_\mu, \Lambda] \quad (3.2.46)$$

We write compactly eq.(5.1.9) in the form

$$\delta A_\mu = D_\mu[A] \Lambda \quad (3.2.47)$$

where the covariant derivative $D_\mu[A]$ is defined as

$$D_\mu[A] = \partial_\mu + i[A_\mu, \] \quad (3.2.48)$$

As the normal derivative in the case of Abelian groups, the covariant derivative generates *infinitesimal gauge transformations* tangent to the orbits of the gauge group. One can easily see that under gauge transformation, this derivative changes precisely in a covariant way

$$D_\mu[A] \rightarrow D_\mu[A^g] = g^{-1} D[A_\mu] g \quad (3.2.49)$$

We are now at the point where the curvature $F_{\mu\nu}$ can be defined as an object will also belongs to the Lie Algebra $\mathcal{G}$,

$$F_{\mu\nu} = F_{\mu\nu}^a t^a \quad (3.2.50)$$

To see its origin, let us calculate the commutator of two covariant derivatives acting over some Lie Algebra valued function

$$[D_\mu, D_\nu]f = i[\partial_\mu A_\nu - \partial_\nu A_\mu + i[A_\mu, A_\nu], f] \equiv i[F_{\mu\nu}, f] \quad (3.2.51)$$

The curvature or non-Abelian field strength is here defined as

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + i[A_\mu, A_\nu] \quad (3.2.52)$$

or, in components,

$$F_{\mu\nu}^a = \partial_\mu A_\nu^a - \partial_\nu A_\mu^a + i f^{abc} A_\mu^b A_\nu^c \quad (3.2.53)$$

One easily sees that under a gauge transformation it does not remain invariant (as it happens in the Abelian case) but it changes as the covarian derivative, that is, *covariantly*,

$$F_{\mu\nu}(x) \rightarrow F_{\mu\nu}^g(x) = g^{-1} F_{\mu\nu}(x) g \quad (3.2.54)$$

Let us now write the non-Abelian version of Maxwell Lagrangian. It has been written for the first time by Yang and Mills⁷ and is the “natural” extension of the Maxwell Lagrangian,

$$L = \frac{c}{2} \text{tr} F_{\mu\nu} F^{\mu\nu} \quad (3.2.55)$$

with c a constant to be adjusted later and tr the trace of the t^a generators (for example the Pauli matrices appropriately normalized for $G = SU(2)$). Using (5.1.17) one proves that the Yang-Mills Lagrangian is gauge invariant,

$$L[A^g] = L[A] \quad (3.2.56)$$

⁷ C.N. Yang and R.L. Mills, “ Conservation of Isotopic Spin and Isotopic Gauge Invariance”, Phys. Rev. **96** (1954) 191.

The equations of motion are

$$D_\mu F^{\mu\nu} = \partial_\mu F^{\mu\nu} + i[A_\mu F^{\mu\nu}] = 0 \quad (3.2.57)$$

Concerning the Bianchi identity, in the non-Abelian case the equation for the dual field strength takes the form

$$D_\mu {}^*F^{\mu\nu} = 0 \quad (3.2.58)$$

with the dual of $F_{\mu\nu}$ defined as

$${}^*F_{\mu\nu} = \frac{1}{2}\varepsilon_{\mu\nu\alpha\beta}F^{\alpha\beta} = \varepsilon_{\mu\nu\alpha\beta}(\partial^\alpha A^\beta - A^\alpha A^\beta) \quad (3.2.59)$$

In order to fix an appropriate constant c in (5.1.18) let us consider the shift

$$A_\mu \rightarrow eA_\mu \quad (3.2.60)$$

for which the curvature changes as

$$\partial_\mu A_\nu - \partial_\nu A_\mu + i[A_\mu, A_\nu] \rightarrow e(\partial_\mu A_\nu - \partial_\nu A_\mu + ie[A_\mu, A_\nu]) \equiv eF_{\mu\nu} \quad (3.2.61)$$

We see that, apart for the overall factor which we can factor out in the action integral, the field strength depends on the charge e through the field commutator. If we make this replacement in the Yang-Mills Lagrangian (5.1.18) we have

$$L = \frac{e^2 c}{2} \text{tr} F_{\mu\nu} F^{\mu\nu} \quad (3.2.62)$$

Hence, if we chose $c = 1/e^2$ we finally have

$$L = \frac{1}{2} \text{tr} F_{\mu\nu} F^{\mu\nu} \quad (3.2.63)$$

We are now ready to consider the zero-mode associated to gauge invariance of the $SU(2)$ Yang-Mills theory. There are 3 parameters $\alpha^a(x)$ ($a = 1, 2, 3$) associated with the $SU(2)$ group so that the zero-mode reads

$$\varphi_\mu^{ab(0)} = D_\mu^{ab}[A_{cl}]\alpha^b \quad (3.2.64)$$

for arbitrary α^b .

Again, eliminating gauge zero-modes illuminates the way in which the natural gauge condition should be chosen since, as in the Abelian case, the natural procedure in order to avoid the integrating over the gauge zero mode is, as in the Abelian case, to fix the gauge so

that all non-zero modes are orthogonal to it. Imposing such condition to any nonzero mode $\varphi_\nu^{a(n)}$ implies that

$$\int dx \varphi_\nu^{a(n)}(x) D_\mu^{ab}[A_{cl}] \alpha^b(x) = 0 \quad (3.2.65)$$

or, after integrating by parts

$$\int dx D_\mu^{ab}[A_{cl}] \varphi_\nu^{a(n)}(x) \alpha^b(x) = 0 \quad (3.2.66)$$

so that a gauge choice of the form

$$D_\mu[A_{cl}] A_\mu = 0 \quad (3.2.67)$$

and the corresponding one for each non-zero mode will guarantee their orthogonality with the zero-mode which can then be trivially eliminated using the Faddeev-Popov method. Gauge condition (3.2.67) is the “natural gauge”⁸.

⁸A. M. Polyakov, “Quark Confinement and Topology of Gauge Groups,” Nucl. Phys. B **120** (1977) 429; see also F. A. Schaposnik, “Path-Integral in Quantum Field Theory and Statistical Mechanics”, <https://sites.google.com/site/schaposnik/>.

Chapter 4

Path-integrals for fermionic fields

4.1 Anticommuting objects

Using the path-integral approach we have developed an alternative quantization procedure for scalar fields, gauge fields and, that can be used in the same vein for other bosonic fields. That is, fields that obey Bose-Einstein statistics with canonical commutation relations. Indeed, in the canonical approach to study a quantum field theory, given the Lagrangian say for a scalar field,

$$\mathcal{L} = \int d^3x L[\phi, \partial_\mu \phi] \quad (4.1.1)$$

and the canonical conjugate momentum defined as

$$\pi(\vec{x}, t) = \frac{\delta L}{\delta (\partial_0 \phi(\vec{x}, t))} \quad (4.1.2)$$

one imposes canonical commutation relations *at equal times*,

$$[\phi(\vec{x}, t), \pi(\vec{y}, t)] = \hbar i \delta^{(3)}(\vec{x} - \vec{y}) \quad (4.1.3)$$

$$[\phi(\vec{x}, t), \phi(\vec{y}, t)] = [\pi(\vec{x}, t), \pi(\vec{y}, t)] = 0 \quad (4.1.4)$$

One then inverts (4.1.2) to obtain $\partial_0 \phi$ in terms of π and ϕ and in this way one constructs the Hamiltonian,

$$H = \int d^3x (\pi \partial_0 \phi - L) \quad (4.1.5)$$

One can then follow the standard procedure to quantize the theory.

What about fermions? Within the canonical approach, one has to modify the prescription above in order to take into account Pauli principle related to Fermi-Dirac statistics.

There is in fact a previous problem related to the fact that the Dirac Lagrangian for a (massive) fermion is first order,

$$L = \frac{i}{2}(\bar{\psi}i\not{\partial}\psi - (\partial_\mu\bar{\psi})\gamma^\mu\psi) - m\bar{\psi}\psi \quad (4.1.6)$$

with $\bar{\psi} = \psi^\dagger\gamma^0$. Now, being first order, the canonical momentum is related to (and not independent from) ψ and $\psi^\dagger$ and the Hamiltonian cannot be constructed following the steps described above for scalars.

One can however obtain the Hamiltonian from the energy-momentum tensor,

$$H = \int d^3x T_{00} \quad (4.1.7)$$

and define equal-time *anti-commutation* relations which are consistent with the Fermi-Dirac statistics.

$$\{\psi_\alpha(\vec{x}, t), \psi_\beta^\dagger(\vec{y}, t)\} = \delta^{(3)}(\vec{x} - \vec{y})\gamma_{\alpha\beta}^0 \quad (4.1.8)$$

$$\{\psi_\alpha(\vec{x}, t), \psi_\beta(\vec{y}, t)\} = \{\psi_\alpha^\dagger(\vec{x}, t), \psi_\beta^\dagger(\vec{y}, t)\} = 0 \quad (4.1.9)$$

As in the bosonic case, a Fock space can be constructed for fermions with creation and annihilation operators satisfying anticommutation relations,

$$\{c_n, c_m^\dagger\} = \delta_{nm} \quad (4.1.10)$$

$$\{c_n, c_m\} = \{c_n^\dagger, c_m^\dagger\} = 0 \quad (4.1.11)$$

The quantum theory for fermionic fields can be then developed without major problems.

Concerning the path-integral approach, we have seen that all the quantum aspects for the scalar theory (or in general a bosonic theory) was concentrated in the path-integral measure $\mathcal{D}\phi$, which could be written as

$$\mathcal{D}\phi = \prod_n dc_n \quad (4.1.12)$$

where the c_n 's are the Bessel-Fourier expansion coefficients of the classical scalar field. Note that these c -number coefficients are reminiscent of the creation and annihilation operators in the canonical quantization approach. But are just the so called c -numbers, real numbers in the case of real scalar fields.

If we think in something similar for fermion fields, it is evident that if we propose such a Bessel-Fourier expansion, we shall end with a quantum theory corresponding to commuting fields, since we are doing nothing different from what we did in the bosonic case.

Only if we try for the Bessel-Fourier expansion coefficients *not c-numbers but anticommuting numbers* we have a chance of ending with the correct quantum theory.

Let us see how one can achieve this by starting, for simplicity, with two elements a and $\bar{a}$ satisfying a Grassmann algebra¹

$$\{a, \bar{a}\} = a\bar{a} + \bar{a}a = 0 \quad (4.1.13)$$

$$\{a, a\} = 2a^2 = 0, \quad \{\bar{a}, \bar{a}\} = 2\bar{a}^2 = 0 \quad (4.1.14)$$

. Confront these equations with those defining the algebra for anticommuting operators, eqs.(4.1)-(4.1)):

$$\begin{aligned} \{c_n, c_m^\dagger\} &= \delta_{nm} \\ \{c_n, c_m\} &= \{c_n^\dagger, c_m^\dagger\} = 0 \end{aligned}$$

In this two-dimensional case, any polynomial of a and $\bar{a}$ can have at most $2^2 = 4$ terms,

$$f(a, \bar{a}) = f_0 + f_1 a + f_2 \bar{a} + f_3 a \bar{a} \quad (4.1.15)$$

with the coefficients f_i being c -numbers. We shall call a and $\bar{a}$ the generators of the two-dimensional Grassmann algebra.

To see how one can represent these generators consider the simple case in which the polynomial depends solely on a

$$f(a) = f_0 + f_1 a \quad (4.1.16)$$

We can define the following two operators $\hat{a}$ and $\hat{a}^*$ acting on $f(a)$,

$$\hat{a}f(a) = af(a) \quad (4.1.17)$$

$$\hat{a}^*f(a) = \frac{d}{da}f(a) \quad (4.1.18)$$

with d/da a formal “derivative”. Then

$$\hat{a}f(a) = af_0 \quad (4.1.19)$$

$$\hat{a}^*f(a) = f_1 \quad (4.1.20)$$

One easily sees that

¹H. Grassmann, Die Lineale Ausdehnungslehre Ein neuer Zweig der Mathematik (The Linear Extension Theory – A new Branch of Mathematics), Wiegand, Leipzig, 1884.

$$(\hat{a}\hat{a}^* + \hat{a}^*\hat{a})f(a) = f(a) \quad (4.1.21)$$

$$\hat{a}^2 f(a) = a^2 f_0 = 0, \quad \hat{a}^{*2} f(a) = a^* f_1 = 0 \quad (4.1.22)$$

Comparing this algebra with that defined in eqs.(4.1)-(4.1) for fermionic creation and annihilation operators we see that they formally coincide. We have then been able to obtain a simple representation for an algebra of the kind Fermi-Dirac creation and annihilation operators obey.

In summary, concerning fermions, one should just consider expansions using as coefficients elements of anticommuting Grassmann algebras of the type above defined. It is then reasonable to infer that the path-integral measure should be defined in terms of these coefficients, as we did in the case of boson fields, except that in that case the coefficients were just c numbers.

To this end, we have to define integration over anticommuting numbers, a task that, according to Coleman², “make strong men quail”³. But we are women and men that will not quail so we shall proceed.

In the case of $f(a)$ functions, we need to know the result of the following integral

$$I = \int f(a) da \quad (4.1.23)$$

with $f(a)$ defined as in (4.1.16). In fact, we need just to define the following two integrals

$$\int da \quad \text{and} \quad \int ada \quad (4.1.24)$$

To find a sensible definition, we demand the usual properties

$$\int da[f(a) + g(a)] = \int da f(a) + \int da g(a) \quad (4.1.25)$$

$$\int da f(a+b) = \int da f(a) \quad (4.1.26)$$

The first condition poses no problem. Concerning the second one

$$\int da f(a+b) = \int da[f_0 + f_1(a+b)] = f_0 \int da + f_1 \int da(a+b) \quad (4.1.27)$$

²S. Coleman, “An introduction to spontaneous symmetry breakdown and gauge fields” in *Aspects of symmetry*, Cambridge University press, 1985.

³Hombres fuertes desfallecer

or

$$\int da f(a+b) = f_0 \int da + f_1 \int daa + f_1 \left(\int da \right) b \quad (4.1.28)$$

From eq.(4.1.26) the r.h.s. cannot depend on b . This means that the integral in the parenthesis in the third term should vanish and then

$$\int da = 0 \quad (4.1.29)$$

$$f_1 \int daa = f_0 \int da + f_1 \int daa = f_1 \int daa \quad (4.1.30)$$

Using condition (4.1.29) this equation reduces to an identity independently of the value of $\int daa$. We can then fix this value to one as a simple normalization

$$\int daa = 1 \quad (4.1.31)$$

All this leads to

$$\int da f(a) = f_1 \quad (4.1.32)$$

while

$$\int da f(a) a = f_0 \quad (4.1.33)$$

We summarize in the next formulæ these results emphasizing the coincidence between integrals and what one should expect for “derivatives” with respect to anticommuting objects

$$\begin{aligned} \int da &= 0, & (d/da)1 &= 0 \\ \int daa &= 1, & (d/da)a &= 1 \end{aligned} \quad (4.1.34)$$

In some sense, integrating and differentiating anticommuting objects coincide: the integral and the derivative of a constant vanishes and the integral or derivative of the variable also coincide.

For consistency we should also impose that differentials and variables anticommute

$$\{a, da\} = 0 \quad (4.1.35)$$

and make the choice

$$\int daa = - \int ada = 1 \quad (4.1.36)$$

All these properties are also imposed to $\bar{a}$,

$$\int d\bar{a} = 0, \quad \int d\bar{a}\bar{a} = 1, \quad \{\bar{a}, d\bar{a}\} = 0 \quad (4.1.37)$$

We also impose

$$\{da, d\bar{a}\} = \{a, d\bar{a}\} = \{\bar{a}, da\} = 0 \quad (4.1.38)$$

Consider now as an example the following integral

$$I = \int dad\bar{a} \exp(-\lambda\bar{a}a) \quad (4.1.39)$$

Only the first two term survive in an expansion in powers of λ so that one has

$$\begin{aligned} \int d\bar{a}da \exp(-\lambda\bar{a}a) &= \int d\bar{a}da (1 - \lambda\bar{a}a) = -\lambda \int d\bar{a}da\bar{a}a \\ &= \lambda \int dad\bar{a}\bar{a}a = \lambda \int daa = \lambda \end{aligned} \quad (4.1.40)$$

This result should be contrasted with that obtained with an integral over (commuting) real numbers,

$$\int_{-\infty}^{\infty} dx \exp(-\lambda x^2) = \frac{\sqrt{\pi}}{\sqrt{\lambda}} \quad (4.1.41)$$

Remember that representing a gaussian path-integral for bosonic fields as an infinite product of gaussian integrals over real numbers we got the result of the inverse of the square root of the product of λ 's. In the present case, one can then advance that an infinite product of integrals of the kind (4.1.39) will lead to the product of λ 's without a square root and appearing in the numerator.

To see this in detail, let us extend the number of anticommuting elements of the Grassman algebra to $2N$, satisfying the anticommutation relations

$$\{a_n, a_m\} = \{\bar{a}_n, \bar{a}_m\} = \{a_n, \bar{a}_m\} = 0 \quad (4.1.42)$$

which, as we did in the case of $N = 1$, can be seen to lead to $2N$ operators satisfying

$$\{\hat{a}_n, \hat{a}_m\} = \{\hat{\bar{a}}_n, \hat{\bar{a}}_m\} = 0, \quad \{\hat{a}_n, \hat{\bar{a}}_m\} = \delta_{nm} \quad (4.1.43)$$

Concerning integrals, the rules extend to

$$\int \prod_{n,m=1}^N d\bar{a}_m da_n \left\{ \begin{pmatrix} \bar{a}_1 & \bar{a}_2 & \bar{a}_3 & \cdots & \bar{a}_N & a_1 & a_2 & \cdots & a_N \\ & \bar{a}_2 & \bar{a}_3 & \cdots & \bar{a}_N & a_1 & a_2 & \cdots & a_N \\ & & \bar{a}_3 & \cdots & \bar{a}_N & a_1 & a_2 & \cdots & a_N \\ & & & \ddots & & & & & \\ & & & & \ddots & & & & \\ & & & & & a_1 & a_2 & \cdots & a_N \\ & & & & & & a_2 & \cdots & a_N \\ & & & & & & & \ddots & \\ & & & & & & & & \ddots \\ & & & & & & & & & a_N \end{pmatrix} \right\} = \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} \right\} \quad (4.1.44)$$

Consider now the integral

$$I_N = \int \prod_{n,m=1}^N d\bar{a}_m da_n \exp(-\lambda_1 \bar{a}_1 a_1 - \lambda_2 \bar{a}_2 a_2 - \dots - \lambda_N \bar{a}_N a_N) \quad (4.1.45)$$

If one now expands the exponential, according to eq.(4.1.44) only the term containing N 's $\bar{a}$'s and N 's a 's will give a nonvanishing result and then we have

$$I_N = \prod_n (-1)^N \lambda_n \quad (4.1.46)$$

Note that in this formula there is an ambiguity concerning the sign of the determinant, a point that is very important related to appropriate regularization that we shall not cover here and just consider that we are solely computing the absolute value of the determinant⁴.

From this result, we infer the following one

$$\int \prod_{n,m} d\bar{a}_m da_n \exp \left(- (\bar{a}_1 \ \bar{a}_2 \ \dots \ \bar{a}_N) A \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix} \right) = \det(A) \quad (4.1.47)$$

⁴For example in the case of the Dirac determinant $D[A]$ in the presence of gauge fields, one has $\det D[A] = (-1)^{n_-} |\det D[A]|$ where n_- is the number of negative eigenvalues of $|D_a|$. As n_- is infinite, some regularization is needed. A rigorous method to solve conundrum is to use the so-called the η -invariant regularization of Atiyah, Pato di, and Singer, see M. F. Atiyah, V. Patodi, and I. M. Singer, "Spectral Asymmetry In Riemannian Geometry", Math. Proc. Camb. Phil. Soc. 77 (1975).

with A some diagonalizable matrix.

As advanced, we got a determinant in the numerator (instead of square root of a determinant in the denominator). Also note that under the change

$$a_j \rightarrow \mu a_j \quad (4.1.48)$$

with μ some real number, the differential changes as

$$da_j \rightarrow \frac{1}{\mu} da_j \quad (4.1.49)$$

This follows from the fact that $\int daa = 1$. It should not be a surprise in view of the connection already signaled between the integral and the derivative for anticommuting objects.

As advanced, we are brave women and men that now ready to define a path-integral for fermions. Consider

$$Z = \int \mathcal{D}\bar{\psi} \mathcal{D}\psi \exp(S[\bar{\psi}, \psi]) \quad (4.1.50)$$

with S the Dirac action,

$$S = \int d^4x L \quad (4.1.51)$$

and L the Lagrangian defined in (4.1.6) which we can compactly write

$$L = \bar{\psi}(i\not{\partial} + m)\psi \quad (4.1.52)$$

In view of our experience, it is natural to consider the eigenvalue problem for this free Dirac operator

$$(i\not{\partial} + m)\varphi_n(x) = \lambda_n \varphi_n(x) \quad (4.1.53)$$

$$\int d^4x \varphi_n^\dagger \varphi_m = \delta_{nm} \quad (4.1.54)$$

and expand the fermionic variables in (4.1.50) in terms of the basis provided by this operator,

$$\psi(x) = \sum_n^\infty a_n \varphi_n(x) , \quad \bar{\psi}(x) = \sum_n^\infty \bar{a}_n \varphi_n^\dagger(x) \quad (4.1.55)$$

where the Bessel-Fourier coefficients are **anticommuting Grassmann objects**. Note that we are taking $\bar{\psi}$ and ψ as independent path-integral variables and not as related through the usual operation, $\bar{\psi} = \psi^\dagger \gamma^0$. We then define the path-integral over all fermion fields as an integral over the Grassmann coefficients,

$$\mathcal{D}\bar{\psi} \equiv \prod_n^\infty d\bar{a}_n , \quad \mathcal{D}\psi \equiv \prod_n^\infty da_n \quad (4.1.56)$$

and then the partition function for free fermions can be written in the form,

$$\begin{aligned}
Z_0 &= \prod_n \int d\bar{a}_n da_n \exp \left(- \int d^4x \sum_n \bar{a}_n \varphi_n^\dagger (i\cancel{\partial} + m) \sum_m a_m \varphi_m \right) \\
&= \prod_n \int d\bar{a}_n da_n \exp \left(- \sum_n \lambda_n \bar{a}_n a_n \right) = \mathcal{N} \prod_n \lambda_n \\
&\equiv \det(i\cancel{\partial} + m)
\end{aligned} \tag{4.1.57}$$

where we have absorbed a normalization constant $\mathcal{N}$ and defined the determinant of the Dirac operator as the (unbounded) product of its eigenvalues. One should compare this result for free fermions,

$$Z_0^{fermions} = \det(i\cancel{\partial} + m) \tag{4.1.58}$$

with that arising in the case of a free scalar field

$$Z_0^{bosons} = \det^{-1/2}(\square + m) \tag{4.1.59}$$

If instead of considering free fermions, we study the case in which they are subject to an external electromagnetic field with potential A_μ , so that the resulting gauge invariant Lagrangian is

$$L = \bar{\psi}(i\cancel{\partial} + m + e\cancel{A})\psi \tag{4.1.60}$$

and the path-integral quantization proceeds in the same way. One ends with

$$Z[A] = \det(i\cancel{\partial} + m + e\cancel{A}) \tag{4.1.61}$$

We can now use the identity, valid for matrices but also for operators,

$$\log \det A = \text{tr} \log A \tag{4.1.62}$$

in order to write

$$Z[A] = \exp \left(\text{tr} \log(i\cancel{\partial} + m + e\cancel{A}) \right) \tag{4.1.63}$$

Consider now

$$P[A] = \log \left(\frac{i\cancel{\partial} + m + e\cancel{A}}{i\cancel{\partial} + m} \right) = \log(1 + eG_0\cancel{A}) \tag{4.1.64}$$

with G_0 the free fermion Green function,

$$(i\cancel{\partial} + m)G_0(x - y) = \delta^{(4)}(x - y) \tag{4.1.65}$$

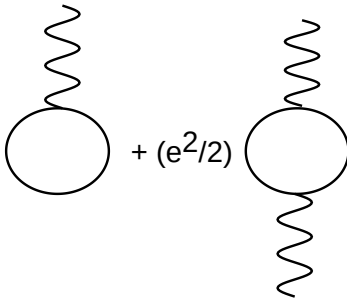
Expanding $P[A]$ one has

$$P[A] = \sum_n \frac{1}{n!} (-e)^n (G_0 \not{A})^n \quad (4.1.66)$$

Then, the normalized version of $Z[A]$ can be identified with

$$Z[A] = \exp(\text{tr} P[A]) \quad (4.1.67)$$

which gives nothing but the vacuum to vacuum amplitude for fermions in an external fields, as can be seen by comparing with the result obtained via canonical quantization,

$$\text{tr } P[A] = 1 - e \text{ (diagram)} + (e^2/2) \text{ (diagram)} + \dots$$


This result was proved for the first time by Matthews and Salam⁵.

⁵P.T. Matthews and A. Salam, Nuovo Cim. **2** (1955) 120.

4.2 BRST Symmetry

BRST à la ancienne

In 1975 Becchi, Rouet and Stora⁶. and, independently Tyutin⁷, discovered that although gauge symmetry has been fixed, the partition function Z has still a symmetry related to gauge invariance but with an anticommuting parameter. This symmetry showed its power when it made possible to prove, in a clear and complete form, the renormalizability of non-Abelian gauge theories. In particular, the proof helped in taking seriously these theories as candidates to describe electroweak and strong interactions.

The way in which BRST invariance was discovered is related to the Faddeev-Popov method but it can be seen that a BRST principle can replace and generalize this method. We shall discuss here the historical approach. Concerning the “modern” approach, we leave for the interested reader an Appendix which will not be discussed during the lectures.

Let us start from formula (5.1.39) for the Euclidean partition function of a non-Abelian gauge theory and, for simplicity, put $\hbar = 1$

$$Z = \mathcal{N} \int \mathcal{D}A_\mu \Delta_{FP}[A] \exp(-S[A]) \delta(F[A]) \quad (4.2.68)$$

We then introduce a path-integral representation for the delta function enforcing the gauge condition.

$$Z = \int \mathcal{D}A_\mu \mathcal{D}b \exp\left(-\frac{1}{4}\text{tr} \int d^4x F_{\mu\nu}^2\right) \Delta_{FP}[A] \exp\left(-\text{tr} \int d^4x F[A]b(x)\right) \quad (4.2.69)$$

Here $b(x) = b^a(x)t^a$ is a Lie-algebra valued auxilliary scalar. Concerning the Faddeev-Popov determinant, it is given by

$$\det \Delta_{FP}[A] = \det D_\mu^{ab}[A] \frac{\delta F^c[A]}{\delta A_\mu^b} = \det \mathcal{M}^{ac}[A] \quad (4.2.70)$$

As we have seen, for the Lorenz and Coulomb gauge one has

$$\begin{aligned} \det \mathcal{M}^{ac}[A] &= \det D_\mu^{ac} \partial_\mu && \text{Lorenz gauge} \\ \det \mathcal{M}^{ac}[A] &= \det D_i^{ac} \partial_i && \text{Coulomb gauge} \end{aligned} \quad (4.2.71)$$

⁶C. Becchi, A. Rouet and R. Stora, Comm. Math. Phys. **42** (1975) 127

⁷I.V. Tyutin, Lebedev preprint N^o 39, 1975 (unpublished.)

We have learnt that determinants of differential operators like $\mathcal{M}$ can be represented as path integrals over anticommuting variables,

$$\det \mathcal{M} = \int \mathcal{D}\bar{c}^a \mathcal{D}c^a \exp \left(- \int d^4x \bar{c}^a \mathcal{M}^{ab} c^b \right) \quad (4.2.72)$$

where $\bar{c}(x) = \bar{c}^a(x)t^a$ and $c(x) = c^a(x)t^a$ are anticommuting Lie algebra valued scalars. Although integration over anticommuting variables was originally introduced to describe fermions, it should be stressed here that $\bar{c}$ and c are a pair of anticommuting scalars (not fermions transforming like 1/2 spinors). Concerning gauge transformations, remember that the Faddeev-Popov determinant is gauge invariant and this, in turn, is a result of the fact that $\mathcal{M}$ transforms covariantly. Then, if one is going to use fields $\bar{c}$ and c to exponentiate $\mathcal{M}$, these fields have to transform in the adjoint representation of the gauge group to get a scalar effective action,

$$\begin{aligned} \bar{c} &\rightarrow \bar{c}^g = g^{-1}(x) \bar{c} g(x) \\ c &\rightarrow c^g = g^{-1}(x) c g(x) \end{aligned} \quad (4.2.73)$$

The fact that $\bar{c}$ and c are not physical fields, in the sense that they will not appear as physical *in* or *out* states when computing v.e.v.'s makes plausible the name of *ghosts* that they have received.

We can then write the path-integral Z in the compact form

$$Z = \int \mathcal{D}A_\mu \mathcal{D}b \mathcal{D}\bar{c} \mathcal{D}c \exp \left(- \int d^4x L_{eff} \right) \quad (4.2.74)$$

where the effective Lagrangian L_{eff} is defined as

$$L_{eff} = \text{tr} \left(\frac{1}{4} F_{\mu\nu}^2 + bF[A] + \bar{c} \mathcal{M} c \right) \quad (4.2.75)$$

Being the gauge fixed to the condition $F[A] = 0$, this effective Lagrangian is not gauge invariant due to the presence of the term containing, precisely, the gauge fixing function $F[A]$. One can see however that L_{eff} has still a remanent invariance, associated to gauge transformations, parametrized by an anticommuting global quantity instead of the commuting local parameter $\epsilon(x)$ associated to general gauge transformations.

The transformation laws that define this new invariance (called BRST invariance) are inferred from the gauge transformations laws of the physical fields, in this case the gauge field. Under an infinitesimal gauge transformation the gauge fields transforms as

$$\delta_{gauge} A_\mu^a = D_\mu^{ab}[A] \epsilon^b(x) \quad (4.2.76)$$

We then define a BRST transformation δ_{BRST} as if it were a gauge transformation but with a parameter $\epsilon^a(x) = c^a(x)\lambda$, with λ an anticommuting constant parameter,

$$\delta_{BRST}A_\mu^a = D_\mu^{ab}[A]c^b(x)\lambda \quad (4.2.77)$$

Since c^a is an anticommuting ghost, the product with λ gives a commuting result for function $\epsilon(x)$.

It is not difficult to define BRST transformation laws for b , $\bar{c}$ and c ensuring that L_{eff} is BRST invariant under the complete set of transformations. Indeed, one can see that if one imposes

$$\begin{aligned} \delta_{BRST}\bar{c}^a &= -b^a\lambda \\ \delta_{BRST}c^a &= -\frac{1}{2}f^{abc}c^bc^c\lambda \\ \delta_{BRST}b^a &= 0 \end{aligned} \quad (4.2.78)$$

one has

$$\delta_{BRST}L_{eff} = 0 \quad (4.2.79)$$

This can be proved by direct calculation. Instead, we shall use an important property associated to δ_{BRST} namely that it is nilpotent

$$\delta_{BRST}^2 = 0 \quad (4.2.80)$$

It is important to stress that this property holds without use of the fact $\lambda^2 = 0$. In fact, from here on, we shall omit parameter λ in the definition of δ_{BRST} . This means that δ_{BRST} so defined is now an anticommuting object that takes a commuting field and transforms it into some anticommuting object.

To prove (4.2.80) one has to see that it is satisfied for each one of the fields. For example, one can easily see that acting on $\bar{c}$ it is nilpotent,

$$\delta_{BRST}^2c = \delta_{BRST}b = 0 \quad (4.2.81)$$

It is a little more involved to consider the case in which δ_{BRST} acts on A_μ ,

$$\begin{aligned} \delta_{BRST}^2A_\mu &= \delta_{BRST}(\partial_\mu c + g[A_\mu, c]) \\ &= -\frac{1}{2}\partial_\mu[c, c] + [\partial_\mu c, c] + [[A_\mu, c], c] - \frac{1}{2}[A_\mu, [c, c]] \end{aligned} \quad (4.2.82)$$

Now, the two first terms in the second line of (4.2.82) cancel each other. Concerning the remaining two, they are zero due to a Jacobi identity for anticommuting objects. We shall

prove this just for the $SU(2)$ case but the proof can be extended to a general gauge group. One has, in components

$$\begin{aligned} \varepsilon^{abc} A_\mu^b c^c \varepsilon^{apq} c^p - \frac{1}{2} \varepsilon^{abc} A_\mu^a \varepsilon^{bpq} c^p c^q &= A_\mu^b c^b c^c - A_\mu^c c^b c^b - \frac{1}{2} A_\mu^b c^b c^c + \\ &\quad \frac{1}{2} A_\mu^b c^c c^b \end{aligned} \quad (4.2.83)$$

Because c^a is anticommuting, the second term in (4.2.83) is zero while the third and forth terms add to cancel the first one. One then has proved that $\delta_{BRST}^2 A_\mu = 0$. We leave as a problem to show that $\delta_{BRST}^2 \bar{c} = \delta_{BRST}^2 c = 0$.

Once nilpotency is proved, it is easy to see that L_{eff} is BRST invariant. Of course the Yang-Mills term in L_{eff} remains invariant under BRST transformation, which is a particular case of a gauge transformation,

$$\delta_{BRST} \text{tr} F_{\mu\nu}^2 = 0 \quad (4.2.84)$$

Concerning the remaining two terms in (4.2.75), let us note that they can be written as the BRST transformed of a given functional. Indeed, consider

$$\delta_{BRST}(\bar{c}F[A]) = (\delta_{BRST} \bar{c})F[A] - \bar{c} \frac{\delta F[A]}{\delta A_\mu} \delta_{BRST} A_\mu \quad (4.2.85)$$

(Here we have taken into account that $\bar{c}$ and δ_{BRST} anticommute). Now, using (4.2.78) we have,

$$- \delta_{BRST}(\bar{c}F[A]) = bF[A] + \bar{c} \frac{\delta F[A]}{\delta A} D_\mu c \quad (4.2.86)$$

We can then write

$$L_{eff} = \frac{1}{4} \text{tr} F_{\mu\nu}^2 - \text{tr} \delta_{BRST}(\bar{c}F[A]) \quad (4.2.87)$$

and then

$$\delta_{BRST} L_{eff} = \frac{1}{4} \text{tr} \delta_{BRST} F_{\mu\nu}^2 - \text{tr} \delta_{BRST}^2(\bar{c}F[A]) = 0 \quad (4.2.88)$$

4.3 The quantum Noether theorem and anomalies

At the classical level the Noether theorem can be presented, for a scalar field as follows:

$$L = L[\phi, \partial_\mu \phi] \quad (4.3.89)$$

and a certain field transformation S

$$\phi(x) \xrightarrow{S} \phi'(x) = \phi(x) + \delta_S \phi(x) \quad (4.3.90)$$

We shall call S a symmetry of the theory if the Euler-Lagrange equations (that is, those that govern dynamics) are not affected by its action. In other words, the equations of motion must remain unaltered under transformation (5.2.118). This implies that the only acceptable change in the Lagrangian under S should be a total derivative which in the relativistic field theory case corresponds to the divergence of a given 4-vector, The Noether theorem⁸ then states that

- For each symmetry S of the theory there exists a locally conserved current j^μ , that is, a current satisfying

$$\partial_\mu j^\mu(x) = 0 \quad (4.3.91)$$

- Consequently, there is a charge Q ,

$$Q = \int d^3x j^0(x) \quad (4.3.92)$$

which is conserved,

$$\frac{dQ}{dt} = 0 \quad (4.3.93)$$

$$j^\mu = \Lambda^\mu - \frac{\partial L}{\partial(\partial_\mu \phi)} \delta_S \phi \quad (4.3.94)$$

$$\partial_\mu j^\mu = 0 \quad (4.3.95)$$

Nota bene: Usually, when $\Lambda \neq 0$ the symmetry is a space-time symmetry like Lorentz symmetry, diffeomorphism symmetry, etc. Internal symmetries correspond in general to $\Lambda = 0$ although there are some exceptions (for example, in the case of Chern-Simons theories an internal symmetry (gauge symmetry) leads to change in the Lagrangian, i.e. $\Lambda \neq 0$).

In the appendix we recall the derivation of the locally conserved current at the classical level. Now we shall discuss the path-integral approach to derive the quantum version of the theorem.

⁸ E. Noether, Nachr. d. Ges. d. Wissensch. zu Göttingen, (1918).

The Noether theorem at the quantum level.

Let us consider the case of a global symmetry, i.e. a symmetry for which the parameter of the transformation is constant. For example in the case of a complex scalar field $\phi = \phi_1 + i\phi_2$ with the a quartic potential,

$$L = \frac{1}{2} \partial_\mu \phi^* \partial^\mu \phi - V[|\phi|] \quad (4.3.96)$$

which is symmetric under phase transformation of the form

$$\phi \rightarrow \exp(i\epsilon) \phi \quad (4.3.97)$$

with ϵ constant. This is a global $U(1)$ symmetry.

Since Lagrangian (4.3.96) is invariant under the transformation (4.3.97), $\delta_\epsilon L = 0$ (i.e. there is no surface term) we can compute the classical Noether current from (5.2.131) putting $\Lambda = 0$. Taking ϕ^* and ϕ as the two independent components of the complex field one gets.

$$j_\epsilon^\mu = -\frac{1}{2} \partial_\mu \phi \delta \phi^* - \frac{1}{2} \partial_\mu \phi^* \delta \phi \quad (4.3.98)$$

or, using

$$\delta \phi = i\epsilon \phi, \quad \delta \phi^* = -i\epsilon \phi^* \quad (4.3.99)$$

$$j^\mu = \frac{i}{2} (\phi^* \partial^\mu \phi - \phi \partial^\mu \phi^*) \quad (4.3.100)$$

(Here we have written $j_\epsilon^\mu = \epsilon j^\mu$, ignoring the constant phase which plays no role in the definition of the conserved charge).

This is a current which is locally conserved at the classical level,

$$\partial_\mu j^\mu = 0 \quad (4.3.101)$$

How can it be tested whether this equation remains valid at the quantum level? Within the path integral approach, we can expect to have, if the symmetry is still present, an identity for the v.e.v.

$$\langle \partial_\mu j^\mu \rangle = 0 \quad (4.3.102)$$

We shall present a simple method to compute the l.h.s. in (4.3.102) and determine whether it vanishes (sometimes it is dubbed "the Noether method", although the path-integral was not yet invented in the times of Emmy Noether). Given the partition function for the theory

$$Z = \int \mathcal{D}\phi^* \mathcal{D}\phi \exp(-\int d^4x L) \quad (4.3.103)$$

one promotes the global infinitesimal symmetry transformations (4.3.97)-(4.3.99) into a local infinitesimal change of variables,

$$\phi \rightarrow \phi' = \phi + i\epsilon(x)\phi, \quad \phi^* \rightarrow \phi'^* = \phi^* - i\epsilon(x)\phi^* \quad (4.3.104)$$

Note that the Lagrangian is no more invariant under this transformation: there will be ϵ derivatives that break the invariance except for constant ϵ . That is, the change in the Lagrangian should come from terms containing derivatives of $\epsilon(x)$ (which are absent in the global case). One can easily see that those terms are given by (for simplicity we take $\epsilon(x)$ small and keep first order in derivatives)

$$\delta_{\partial_\mu \epsilon(x)} L = i \frac{\partial L}{\partial(\partial_\mu \phi)} (\partial_\mu \epsilon) \phi - i \frac{\partial L}{\partial(\partial_\mu \phi^*)} (\partial_\mu \epsilon) \phi^* = (\partial_\mu \epsilon(x)) j^\mu(x) \quad (4.3.105)$$

Note that the global current j^μ factors out from the $\epsilon(x)$ gradient.

Another source of change in Z may come from the path-integral measure. In principle we cannot be sure that transformation (4.3.104) leaves invariant the path integral measure, which is precisely where quantum effects are taken into account. We shall call $J[\epsilon]$ the possible Jacobian which should be analyzed once the character of the transformation is taken into account,

$$\mathcal{D}\phi'^* \mathcal{D}\phi' = J[\epsilon] \mathcal{D}\phi^* \mathcal{D}\phi \quad (4.3.106)$$

Note that the change (4.3.104) is just a scale change for the scalar field, $\phi(x) \rightarrow (1 + \epsilon(x))\phi(x)$, and then the Jacobian will solely depend on ϵ (and not on ϕ).

Let us now write the path-integral defining Z in terms of the new variables

$$Z = \int \mathcal{D}\phi'^* \mathcal{D}\phi' J[\epsilon] \exp \left(- \int d^4x L - \int d^4x (\partial_\mu \epsilon(x)) j^\mu(x) \right) \quad (4.3.107)$$

or, integrating by parts the second term in the exponential,

$$Z = \int \mathcal{D}\phi'^* \mathcal{D}\phi' J[\epsilon] \exp \left(- \int d^4x L + \int d^4x \epsilon(x) \partial_\mu j^\mu(x) \right) \quad (4.3.108)$$

The path-integral in the r.h.s. cannot depend on the arbitrary parameter $\epsilon(x)$ introduced as a change of variables. Or, in other words, Z cannot depend on $\epsilon(x)$. We can then write

$$0 = \frac{1}{Z} \frac{\delta Z}{\delta \epsilon(x)} \Big|_{\epsilon=0} = \left(\langle \partial_\mu j^\mu(x) \rangle + \frac{\delta \log J[\epsilon]}{\delta \epsilon(x)} \right) \Big|_{\epsilon=0} \quad (4.3.109)$$

That is

$$\langle \partial_\mu j^\mu(x) \rangle = - \frac{\delta \log J[\epsilon]}{\delta \epsilon(x)} \Big|_{\epsilon=0} \quad (4.3.110)$$

Eq.(4.3.110) clearly indicates the condition under which a classical global symmetry survives quantization: the Jacobian associated to the local change of path-integral variables related to the transformation should be trivial, $J[\epsilon(x)] = 1$, so that

$$\langle \partial_\mu j^\mu(x) \rangle = 0 \quad (4.3.111)$$

This is completely consistent with the path-integral approach to quantization: the path integral measure is the place where quantum aspects are taken into account and then a quantum violation of a classical symmetry, an anomaly⁹, should manifest as a non-trivial Jacobian associated with the symmetry transformation.

What about the case of symmetry transformations which are already local, like for example gauge transformations?. A local change of variables will leave the Lagrangian invariant and hence no current term is generated in the transformed action so as to follow the Noether method as above. There is a way out to this which we shall exemplify with the scalar field example.

In order to have a complex scalar field theory with a local (gauge) invariance one should replace the derivatives in (4.3.96) by covariant derivatives including a gauge field connection (one then adds also an invariant kinetic Lagrangian for the gauge field). It is only when both fields are transformed that the Lagrangian remains invariant. The trick to obtain a formula like (4.3.110) or (4.3.111) is to proceed within the path-integral approach making a change of variables only for the matter fields (i.e, changing ϕ^* and ϕ but not the gauge field connection. This will produce a change in the Lagrangian which is not compensated by any change in the gauge field part and the change will be precisely of the form calculated above. One then proceeds as before, getting an answer for the v.e.v. of the matter field current divergence.

4.4 Anomalies in fermionic models

Let us consider for simplicity the case of fermions coupled to a $U(1)$ gauge field A_μ in Euclidean space. One can easily see that the Dirac Lagrangian

$$L_F = \bar{\psi}(i\cancel{D} + eA)\psi = \bar{\psi}\cancel{D}[A]\psi \quad (4.4.112)$$

is invariant under global chiral transformations. Indeed, given the γ_5 matrix

$$\gamma_5 = i\gamma_0\gamma_1\gamma_2\gamma_3 \quad (4.4.113)$$

⁹Note the the subtle difference between “anomalous”, a deviation from what is expected and “abnormal” from Latin *abnormis* monstrous

its satisfies the very important property

$$\gamma_\mu \gamma_5 = -\gamma_5 \gamma_\mu \quad (4.4.114)$$

which guarantees that

$$\gamma_5 \not{D}[A] = -\not{D}[A] \gamma_5 \quad (4.4.115)$$

Consider a global chiral (or γ_5) transformation with parameter α

$$\begin{aligned} \psi &\rightarrow \psi' = \exp(\gamma_5 \alpha) \psi \\ \bar{\psi} &\rightarrow \bar{\psi}' = \bar{\psi} \exp(\gamma_5 \alpha) \end{aligned} \quad (4.4.116)$$

Note that, as explained before, $\bar{\psi}$ and ψ are independent path-integral variables and that is the reason why the sign of the exponential arguments coincides and the Lagrangian remains invariant under such change,

$$\begin{aligned} L \rightarrow L' &= \bar{\psi}' \exp(\gamma_5 \alpha) \not{D}[A] \exp(\gamma_5 \alpha) \psi \\ &= \bar{\psi} \exp(\gamma_5 \alpha) \exp(-\gamma_5 \alpha) \not{D}[A] \psi = L \end{aligned} \quad (4.4.117)$$

Here we have used, extending (4.3.109) and using that α is constant

$$\not{D}[A] \exp(\gamma_5 \alpha) = \exp(-\gamma_5 \alpha) \not{D}[A] \quad (4.4.118)$$

We then have

$$\delta_5 L = 0 \quad (4.4.119)$$

and then there should be a Noether conserved current which can be obtained from the formula

$$j_5^\mu = \frac{\partial L}{\partial(\partial_\mu \psi)} \delta_5 \psi \quad (4.4.120)$$

with

$$\delta_5 \psi = \psi' - \psi \approx \alpha \gamma_5 \psi \quad (4.4.121)$$

and one gets

$$j_5^\mu = i \bar{\psi} \gamma^\mu \gamma_5 \psi \quad (4.4.122)$$

which transforms as an axial quadrivector under Lorentz transformations.

We then have, together with the j_μ current associated to the electric charge of the fermions coupled to the $U(1)$ gauge field, an axial current which is classically conserved

$$\partial_\mu j^\mu = 0, \quad j^\mu = i \bar{\psi} \gamma^\mu \psi \quad (4.4.123)$$

$$\partial_\mu j_5^\mu = 0, \quad j_5^\mu = i \bar{\psi} \gamma^\mu \gamma_5 \psi \quad (4.4.124)$$

Together with the electric charge, there is an axial charge which is conserved,

$$\frac{dQ_5}{dt} = 0, \quad Q_5 = \int d^3x j_5^0(t, x) \quad (4.4.125)$$

If we write a four component spinor in the form

$$\psi = \begin{pmatrix} \psi^1 \\ \psi^2 \\ \psi^3 \\ \psi^4 \end{pmatrix} \equiv \begin{pmatrix} \psi^+ \\ \psi^- \end{pmatrix} \quad (4.4.126)$$

and use the chiral representation for Dirac matrices which corresponds to a γ_5 matrix in the form

$$\gamma_5 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \equiv \begin{pmatrix} I_{2 \times 2} & 0 \\ 0 & -I_{2 \times 2} \end{pmatrix} \quad (4.4.127)$$

we see that the chiral charge can be written as

$$Q_5 = \int d^3x (|\psi^+|^2 - |\psi^-|^2) \quad (4.4.128)$$

to be compared with the electric charge

$$Q = \int d^3x (|\psi^+|^2 + |\psi^-|^2) \quad (4.4.129)$$

Note that we have omitted in the definition of the current j_μ a factor e which of course that does not alter the charge Q conservation and also the factor i in the axial charge.

Of course, a mass term for the fermion breaks chiral invariance. Indeed, consider a mass term of the form

$$L_{mass} = im\bar{\psi}\psi \quad (4.4.130)$$

(The "i" in front of the mass arises in Euclidean space so as to ensure a correct pole in the propagator). Then, under an infinitesimal chiral change this term changes as

$$\delta_5 L_{mass} = im\alpha\bar{\psi}\gamma_5\psi \quad (4.4.131)$$

We can still define the axial current in the form $j_5^\mu = i\bar{\psi}\gamma^\mu\gamma_5\psi$ but, instead of eq.(4.4.124), the following equation for the chiral current holds,

$$\partial_\mu j_5^\mu = 2im\bar{\psi}\gamma_5\psi \quad (4.4.132)$$

If $m \approx 0$, one can consider that the axial current is a *partially conserved* current.

The axial symmetry, as the gauge symmetry, is a $U(1)$ symmetry (with a complexification $j = \gamma_5$ generator in Euclidean space, instead of the usual i). In fact, if we look for its action on each Weyl component of the 4 component spinor one has (in Minkowski space for convenience of the argument)

$$\begin{aligned}\psi^+ &\xrightarrow{\text{chiral } t.} \exp(i\alpha)\psi^+ \\ \psi^- &\xrightarrow{\text{chiral } t.} \exp(-i\alpha)\psi^-\end{aligned}\tag{4.4.133}$$

to be compared with gauge transformations

$$\begin{aligned}\psi^+ &\xrightarrow{\text{gauge } t.} \exp(i\alpha)\psi^+ \\ \psi^- &\xrightarrow{\text{gauge } t.} \exp(i\alpha)\psi^-\end{aligned}\tag{4.4.134}$$

It is then natural to define two charges $Q_\pm$,

$$Q_\pm = \frac{1}{2}(Q \pm Q_5)\tag{4.4.135}$$

so that Q_+ changes the right and Q_- the left parts of the fermion spinor. In this context, one says that the massless fermion Lagrangian has a $U(1)_L \times U(1)_R$ symmetry.

We define the generating functional $Z[A]$ for fermions in a gauge field background as

$$Z[A] = \int \mathcal{D}\bar{\psi} \mathcal{D}\psi \exp \left(\int d^4x \bar{\psi} (i\not{D} + e\not{A}) \psi \right) = \det \mathcal{D}[A]\tag{4.4.136}$$

The gauge field background can be thought of as an external source to obtain current correlation function through differentiation (and eventually putting the background to zero). For example

$$\begin{aligned}\langle j^\mu(x) \rangle &= \frac{1}{Z} \frac{\delta Z}{\delta A_\mu(x)} \Big|_{A=0} \\ \langle j^\mu(x) j^\nu(y) \rangle &= \frac{1}{Z} \frac{\delta^2 Z}{\delta A_\mu(x) \delta A_\nu(y)} \Big|_{A=0}\end{aligned}\tag{4.4.137}$$

Evidently, the Lagrangian in (4.4.136) is invariant under $U(1)$ gauge transformations,

$$\begin{aligned}\psi \rightarrow \psi' &= \exp(i\alpha(x))\psi \\ \bar{\psi} \rightarrow \bar{\psi}' &= \bar{\psi} \exp(-i\alpha(x))\end{aligned}\tag{4.4.138}$$

so that at the classical level we have

$$\partial_\mu j^\mu = 0, \quad j_\mu = i\bar{\psi}\gamma^\mu\psi \quad (4.4.139)$$

In order to study the symmetry at the quantum level we promote, following the Noether method, the symmetries to local change of variables. We first consider gauge transformations and then define the change of fermion variables

$$\begin{aligned} \psi &\rightarrow \psi' = \exp(i\alpha(x))\psi \\ \bar{\psi} &\rightarrow \bar{\psi}' = \bar{\psi}\exp(-i\alpha(x)) \end{aligned} \quad (4.4.140)$$

(Note that the background field is not affected by the transformation: it is not a “complete” gauge transformation but just a change of fermion variables. This change produces a change in the Lagrangian, which is proportional to $\partial_\mu\alpha$,

$$\delta_{\alpha(x)}L = i\bar{\psi}\gamma^\mu\psi\partial_\mu\alpha = j^\mu\partial_\mu\alpha \quad (4.4.141)$$

At the path integral level, the change should be also implemented in the path-integral measure and can in principle be accompanied by a Jacobian

$$\mathcal{D}\bar{\psi}\mathcal{D}\psi \rightarrow \mathcal{D}\bar{\psi}'\mathcal{D}\psi' = J^{-1}[\alpha]\mathcal{D}\bar{\psi}\mathcal{D}\psi \quad (4.4.142)$$

Then, when we perform the change of variables in Z we end with

$$\begin{aligned} Z[A] &= J[\alpha] \int \mathcal{D}\bar{\psi}'\mathcal{D}\psi' \exp\left(-\int d^4x (\bar{\psi}'(i\not{\partial} + e\not{A}) + j^\mu\partial_\mu\alpha(x))\psi'\right) \\ &= J[\alpha] \int \mathcal{D}\bar{\psi}'\mathcal{D}\psi' \exp\left(-\int d^4x (\bar{\psi}'(i\not{\partial} + e\not{A}) - \alpha(x)\partial_\mu j^\mu)\psi'\right) \end{aligned} \quad (4.4.143)$$

As usual, we now use the fact that Z is independent of the arbitrary function $\alpha(x)$ so that, differentiating $\log Z$ with respect to α and then putting $\alpha = 0$ we obtain

$$\langle\partial_\mu j^\mu\rangle = -\left.\frac{\delta\log J[\alpha]}{\delta\alpha}\right|_{\alpha=0} \quad (4.4.144)$$

Now, $Z[A]$ in (4.4.143) can also be written, after the change of variables as

$$\begin{aligned} Z[A] &= J[\alpha] \int \mathcal{D}\bar{\psi}'\mathcal{D}\psi' \exp\left(-\int d^4x \bar{\psi}'(i\not{\partial} + e\not{A} + \gamma^\mu\partial_\mu\alpha)\psi'\right) \\ &= J[\alpha] \int \mathcal{D}\bar{\psi}'\mathcal{D}\psi' \exp\left(-\int d^4x \bar{\psi}'(i\not{\partial} + e\not{A}^\alpha)\psi'\right) \\ &= J[\alpha] \det(i\not{\partial} + e\not{A}^{\alpha(x)}) \end{aligned} \quad (4.4.145)$$

where we have written

$$A_\mu^{\alpha(x)} = A_\mu + \frac{1}{e} \partial_\mu \alpha(x) \quad (4.4.146)$$

But we have also $Z[A] = \det(i\cancel{D} + e\cancel{A})$ and then we can write

$$\det(i\cancel{D} + e\cancel{A}) = J[\alpha] \det(i\cancel{D} + e\cancel{A}^{\alpha(x)}) \quad (4.4.147)$$

so that the Jacobian can be seen as the ratio of two determinants: that of the Dirac operator in a background and the one for the gauge transformed background.

Naively, one would say that since

$$i\cancel{D} + e\cancel{A}^{\alpha(x)} = \exp(-i\alpha)(i\cancel{D} + e\cancel{A}) \exp(i\alpha) \quad (4.4.148)$$

and determinants are endowed with the cyclic property, one has

$$\begin{aligned} \det(i\cancel{D} + e\cancel{A}^{\alpha(x)}) &= \det(\exp(-i\alpha)(i\cancel{D} + e\cancel{A}) \exp(i\alpha)) \\ &= \det(\exp(i\alpha) \exp(-i\alpha)(i\cancel{D} + e\cancel{A})) \\ &= \det(i\cancel{D} + e\cancel{A}) \end{aligned} \quad (4.4.149)$$

this giving

$$J[\alpha] = 1 \quad (4.4.150)$$

But the point is the following: the Dirac operator being unbounded, neither $Z[A]$ nor the identity (4.4.147) are well-defined. A regularization prescription for computing the determinant should be adopted. It may happen that a given regularization can not preserve the cyclic property. This would mean that gauge invariance is spoiled. Only when one finds a gauge-invariant regularization one is sure that $J[\alpha] = 1$ or, equivalently, that the path-integral measure can be regularized in a gauge invariant way. As we shall see, when gauge fields are coupled to Dirac fermions this is always possible (In contrast, for Weyl fermions this is not, in general, the case).

Now, suppose that a gauge invariant regularization is found and used to regularize the Jacobian in such a way that $J = 1$. Consider then a chiral transformation instead of a gauge one. Repeating all the precedent steps, one starts from a chiral change of variables

$$\begin{aligned} \psi \rightarrow \psi' &= \exp(\gamma_5 \alpha(x)) \psi \\ \bar{\psi} \rightarrow \bar{\psi}' &= \bar{\psi} \exp(\gamma_5 \alpha(x)) \end{aligned} \quad (4.4.151)$$

This change produces a change in the Lagrangian, which again is proportional to $\partial_\mu \alpha$ (the Lagrangian is invariant under chiral *global* transformations,

$$\delta_{\gamma_5 \alpha(x)} L = \bar{\psi} \gamma^\mu \gamma_5 \psi \partial_\mu \alpha = j_5^\mu \partial_\mu \alpha \quad (4.4.152)$$

At the path integral level, we define the Jacobian

$$\mathcal{D}\bar{\psi}\mathcal{D}\psi \rightarrow \mathcal{D}\bar{\psi}'\mathcal{D}\psi' = J_5^{-1}[\alpha]\mathcal{D}\bar{\psi}\mathcal{D}\psi \quad (4.4.153)$$

so that when we perform the change of variables in Z we end with

$$\begin{aligned} Z[A] &= J_5[\alpha] \int \mathcal{D}\bar{\psi}'\mathcal{D}\psi' \exp \left(- \int d^4x \left(\bar{\psi}'(i\cancel{\partial} + e\cancel{A}) + j_5^\mu \partial_\mu \alpha(x) \right) \psi' \right) \\ &= J_5[\alpha] \int \mathcal{D}\bar{\psi}'\mathcal{D}\psi' \exp \left(- \int d^4x \left(\bar{\psi}'(i\cancel{\partial} + e\cancel{A}) - \alpha(x) \partial_\mu j_5^\mu \right) \psi' \right) \end{aligned} \quad (4.4.154)$$

Again, we use the fact that Z is independent of the arbitrary function $\alpha(x)$ so that, differentiating $\log Z$ with respect to α and then putting $\alpha = 0$ we obtain

$$\langle \partial_\mu j_5^\mu \rangle = - \left. \frac{\delta \log J_5[\alpha]}{\delta \alpha} \right|_{\alpha=0} \quad (4.4.155)$$

An analogous treatment as that followed for gauge transformations leads for $J_5[\alpha]$ to

$$\det(i\cancel{\partial} + e\cancel{A}) = J[\alpha] \det \left(\exp(\gamma_5 \alpha(x)) (i\cancel{\partial} + e\cancel{A}) \exp(\gamma_5 \alpha(x)) \right) \quad (4.4.156)$$

or,

$$J_5[\alpha] = \frac{\det(i\cancel{\partial} + e\cancel{A})}{\det \left(\exp(\gamma_5 \alpha(x)) (i\cancel{\partial} + e\cancel{A}) \exp(\gamma_5 \alpha(x)) \right)} \quad (4.4.157)$$

One again needs to regularize determinants. If one uses (as one should if gauge invariance is to be respected) the regularization leading to $J = 1$, then, in principle, there is no reason to expect that also $J_5 = 1$. We shall show below how one effectively computes this Jacobian, finding that it is not trivial.

Note that the denominator is similar to the previous case when gauge transformations were considered, except that the exponentials have the same sign. Even if the cyclic property of the determinant was to be respected, these exponentials do not cancel but add. A curious fact takes place for the case in which $\alpha(x) = \alpha = \text{constant}$. Since γ_5 anticommutes with the Dirac operator, one can pass one of the exponentials through it by changing its sign and cancel the other one. Then, one would obtain $J_5 = 1$ in this way while the cyclic property would give in principle a different result since, when it is used, the arguments in the exponentials add. The answer to this apparent contradiction will be clear after computation of the chiral Jacobian.

Fujikawa original derivation started from ex.(4.4.153) for an infinitesimal change of variables¹⁰ and we shall describe it below. One can also give an alternative derivation, based on the ratio of determinants previously discussed, which is more appropriate for computing Jacobians (not necessarily chiral ones) for finite changes of variables¹¹.

Fujikawa's idea was to take seriously the definition of the path integral measure in terms of the (anti-commuting) Bessel-Fourier coefficients in the expansion of fermions,

$$\begin{aligned}\psi(x) &= \sum_n c_n \varphi_n(x) \\ \bar{\psi}(x) &= \sum_n \bar{c}_n \varphi_n^\dagger(x)\end{aligned}\tag{4.4.158}$$

Here c_n and $\bar{c}_n$ are Grassman valued objects and $\{\varphi_n(x)\}$ an adequate basis of Dirac spinors,

$$\begin{aligned}\int \varphi_n^\dagger(x) \varphi_m(x) d^4x &= \delta_{nm} \\ \sum_n \varphi_n(x) \otimes \varphi_n^\dagger(y) &= I_{4 \times 4} \delta^{(4)}(x - y)\end{aligned}\tag{4.4.159}$$

Note that in the expansion of $\bar{\psi}$ we use $\varphi^\dagger$ instead of $\bar{\varphi}$. We shall comment on this election later.

The path-integral measure is then defined as

$$\mathcal{D}\bar{\psi} \mathcal{D}\psi = \prod_n d\bar{c}_n dc_n\tag{4.4.160}$$

Now, in order to compute the Jacobian of the chiral transformation, we have to consider a Bessel-Fourier expansion for the chirally transformed fermion fields,

$$\begin{aligned}\psi'(x) &= \sum_n c'_n \varphi_n(x) \\ \bar{\psi}'(x) &= \sum_n \bar{c}'_n \varphi_n^\dagger(x)\end{aligned}\tag{4.4.161}$$

and then interpret the Jacobian as that arising from the change from $\mathcal{D}\bar{\psi} \mathcal{D}\psi$ to $\mathcal{D}\bar{\psi}' \mathcal{D}\psi'$,

$$\prod_n d\bar{c}_n dc_n = J_5[\delta\alpha] \prod_n d\bar{c}'_n dc'_n\tag{4.4.162}$$

¹⁰K. Fujikawa, Phys. Rev. Lett. **42** (1979) 1195; Phys. Rev. **D21** (1980) 2848.

¹¹R.E. Gamboa Saraví, M.A. Muschietti, F.A. Schaposnik and J. E. Solomin, Ann. of Phys. (N.Y.) **157** (1984) 360.

Now, from (4.4.151) we have

$$\psi'(x) = \exp(\gamma_5 \delta \alpha(x)) \psi(x) \quad (4.4.163)$$

so that we can easily connect c'_n with c_n ,

$$\sum_n c'_n \varphi_n(x) = \exp(\gamma_5 \delta \alpha(x)) \sum_n c_n \varphi_n(x) \quad (4.4.164)$$

Multiplying by $\varphi_m^\dagger(x)$ and integrating one gets, after using ortonormality in the l.h.s.

$$c'_m = \int d^4x \varphi_m^\dagger(x) \exp(\gamma_5 \delta \alpha(x)) \sum_n c_n \varphi_n(x) = \sum_n C_{mn} c_n \quad (4.4.165)$$

where

$$C_{mn} = \int d^4x \varphi_m^\dagger(x) \exp(\gamma_5 \delta \alpha(x)) \varphi_n(x) \quad (4.4.166)$$

The same result is obtained for $\bar{\psi}$,

$$\bar{c}'_m = \sum_n C_{mn} \bar{c}_n \quad (4.4.167)$$

so that one can write for the Jacobian,

$$J_5^{-1}[\delta \alpha] = \det^{-2} C_{mn} \quad (4.4.168)$$

where the negative power in the l.h.s. is due to the definition (4.4.142) while that in the r.h.s. arises because c'_n and $\bar{c}'_n$ are Grassman objects and, as we have seen, Jacobians in that case appear with negative powers in comparison with the commuting case.

Now we shall calculate C_{mn} . To this end, we use the fact that we are considering an infinitesimal transformation and write (4.4.166) in the form

$$C_{mn} \approx \int d^4x \varphi_m^\dagger(x) (1 + \gamma_5 \delta \alpha(x)) \varphi_n(x) = \delta_{mn} + \int d^4x \varphi_m^\dagger(x) \gamma_5 \varphi_n(x) \delta \alpha(x) \quad (4.4.169)$$

Then, we can write the C_{mn} determinant in the form

$$\det C = \begin{vmatrix} 1 + \int d^4x a_{11} \delta \alpha(x) & \int d^4x a_{12} \delta \alpha(x) & \dots & \int d^4x a_{1n} \delta \alpha(x) \\ \int d^4x a_{21} \delta \alpha(x) & 1 + \int d^4x a_{22} \delta \alpha(x) & \dots & \int d^4x a_{2n} \delta \alpha(x) \\ \dots & \dots & \dots & \dots \\ \int d^4x a_{n1} \delta \alpha(x) & \int d^4x a_{n2} \delta \alpha(x) & \dots & 1 + \int d^4x a_{nn} \delta \alpha(x) \end{vmatrix} \quad (4.4.170)$$

where

$$a_{nm} = \varphi_n^\dagger(x) \gamma_5 \varphi_m(x) \quad (4.4.171)$$

Then, to first order in $\delta\alpha$ we have

$$\det C = 1 + \sum_n \int d^4x a_{nn} \delta\alpha(x) = \exp \left(\sum_n \int d^4x a_{nn} \delta\alpha(x) \right) \quad (4.4.172)$$

so that we finally have

$$J_5[\delta\alpha] = \exp \left(2 \int d^4x \sum_n \varphi_n^\dagger(x) \gamma_5 \varphi_n(x) \delta\alpha(x) \right) \quad (4.4.173)$$

Defining

$$\mathcal{A} = 2 \sum_n \varphi_n^\dagger(x) \gamma_5 \varphi_n(x) \quad (4.4.174)$$

and using

$$\langle \partial_\mu j^\mu \rangle = - \left. \frac{\delta \log J[\alpha]}{\delta \alpha} \right|_{\alpha=0} \quad (4.4.175)$$

we finally have for the anomaly equation

$$\langle \partial_\mu j_5^\mu \rangle = -\mathcal{A} \quad (4.4.176)$$

Our next task is to compute $\mathcal{A}$ to determine whether the Jacobian is trivial ($\mathcal{A} = 0$) or not. A first naive manipulation shows that $\mathcal{A}$ is ill-defined. Indeed, one has, after using simple Dirac matrices properties

$$\begin{aligned} \mathcal{A} &= 2 \sum_n \varphi_n^\dagger(x) \gamma_5 \varphi_n(x) = 2 \text{tr} \left(\gamma_5 \sum_n \varphi_n(x) \otimes \varphi_n^\dagger(x) \right) \\ &= 2 \text{tr} (\gamma_5 \delta^{(4)}(0)) = 0 \times \infty \quad !!! \end{aligned} \quad (4.4.177)$$

We then see that $\mathcal{A}$ needs some regularization so that the question on the triviality of the Jacobian changes now to that concerning the search of a consistent regularization for $\mathcal{A}$ (which, after rendered finite, could vanish or not).

Since we know that ambiguities manifested in (4.4.177) are related to the fact that the Dirac operator eigenvalues grow with no bound giving divergent determinants, let us propose, following Fujikawa, to introduce a regularization cutting off the contribution of large eigenvalues.

To this end, we shall use the following trivial identity

$$1 = \lim_{M^2 \rightarrow \infty} \exp \left(-\frac{\lambda_n^2}{M^2} \right) \quad (4.4.178)$$

Here λ_n are real eigenvalues which grow with increasing n as rapidly as those associated with the Dirac operator $\mathcal{D}[A]$ (which are real since $\mathcal{D}$ is hermitian). But, in principle, they can be different from $\mathcal{D}[A]$ eigenvalues (for example, one can choose them as the eigenvalues of the free Dirac operator, $i\mathcal{D}$).

Inserting identity (4.4.178) in the expression for $\mathcal{A}$, we have

$$\mathcal{A} = 2 \sum_n \varphi_n^\dagger(x) \gamma_5 \left(\lim_{M^2 \rightarrow \infty} \exp \left(-\frac{\lambda_n^2}{M^2} \right) \right) \varphi_n(x) \quad (4.4.179)$$

We still call $\mathcal{A}$ the l.h.s. in (4.4.179) because we have just inserted the identity and nothing has been changed. The next is the crucial step: we define the regularized anomaly $\mathcal{A}_{reg}$ through the interchange between the sum (responsible for the infinite in (4.4.177)) and the limit:

$$\mathcal{A}_{reg} = 2 \lim_{M^2 \rightarrow \infty} \sum_n \varphi_n^\dagger(x) \gamma_5 \exp \left(-\frac{\lambda_n^2}{M^2} \right) \varphi_n(x) \quad (4.4.180)$$

Moreover, in order to be able to manipulate naively the basis spinors, we can make a point-splitting so that

$$\mathcal{A}_{reg} = 2 \lim_{y \rightarrow x} \lim_{M^2 \rightarrow \infty} \sum_n \varphi_n^\dagger(y) \gamma_5 \exp \left(-\frac{\lambda_n^2}{M^2} \right) \varphi_n(x) \quad (4.4.181)$$

Operations that were ill defined when performed in (4.4.177) are now safe,

$$\begin{aligned} \mathcal{A}_{reg} &= 2 \lim_{y \rightarrow x} \lim_{M^2 \rightarrow \infty} \sum_n \varphi_n^\dagger(y) \gamma_5 \exp \left(-\frac{\lambda_n^2}{M^2} \right) \varphi_n(x) \\ &= 2 \lim_{y \rightarrow x} \lim_{M^2 \rightarrow \infty} \text{tr} \left(\gamma_5 \sum_n \exp \left(-\frac{\lambda_n^2}{M^2} \right) \varphi_n(x) \otimes \varphi_n^\dagger(y) \right) \end{aligned} \quad (4.4.182)$$

Now, a second choice: evidently, the results should be independent of the choice of basis $\{\varphi_n\}$ and eigenvalues λ_n (provided they grow with n as rapidly as those of $\mathcal{D}$). Then, we can exploit this freedom to simplify calculations. Indeed, if we choose the basis as that provided by $\mathcal{D}$ and the eigenvalues λ_n as those precisely associated to $\mathcal{D}$,

$$\mathcal{D}[A] \varphi_n(x) = \lambda_n \varphi_n(x) \quad (4.4.183)$$

We can then write

$$\begin{aligned}
\mathcal{A}_{reg} &= 2 \lim_{y \rightarrow x} \lim_{M^2 \rightarrow \infty} \text{tr} \left(\gamma_5 \sum_n \exp \left(-\frac{\mathcal{D}_x^2[A]}{M^2} \right) \varphi_n(x) \otimes \varphi_n^\dagger(y) \right) \\
&= 2 \lim_{y \rightarrow x} \lim_{M^2 \rightarrow \infty} \text{tr} \left(\gamma_5 \exp \left(-\frac{\mathcal{D}_x^2[A]}{M^2} \right) \sum_n \varphi_n(x) \otimes \varphi_n^\dagger(y) \right) \\
&= 2 \lim_{y \rightarrow x} \lim_{M^2 \rightarrow \infty} \text{tr} \left(\gamma_5 \exp \left(-\frac{\mathcal{D}_x^2[A]}{M^2} \right) \delta^{(4)}(x - y) \right)
\end{aligned} \tag{4.4.184}$$

At this point, we can use the Fourier transform representation for the delta function,

$$\mathcal{A}_{reg} = 2 \lim_{y \rightarrow x} \lim_{M^2 \rightarrow \infty} \text{tr} \left(\gamma_5 \exp \left(-\frac{\mathcal{D}_x^2[A]}{M^2} \right) \frac{1}{(2\pi)^4} \int d^4k \exp(ik(x - y)) \right) \tag{4.4.185}$$

Acting with the Dirac operator one can easily reduce (4.4.185) to

$$\begin{aligned}
\mathcal{A}_{reg} &= 2 \lim_{y \rightarrow x} \lim_{M^2 \rightarrow \infty} \text{tr} \left(\gamma_5 \frac{1}{(2\pi)^4} \int d^4k \exp(ik(x - y)) \right. \\
&\quad \times \exp \left(-\frac{1}{M^2} (\mathcal{D}_x[A] - \not{k})^2 \right) \Big)
\end{aligned} \tag{4.4.186}$$

One can safely take at this stage the limit $y \rightarrow x$,

$$\mathcal{A}_{reg} = 2 \lim_{M^2 \rightarrow \infty} \text{tr} \left(\gamma_5 \frac{1}{(2\pi)^4} \int d^4k \exp \left(-\frac{1}{M^2} (\mathcal{D}_x[A] - \not{k})^2 \right) \right) \tag{4.4.187}$$

It is convenient to change the integration variables

$$p^\mu = \frac{k^\mu}{M} \tag{4.4.188}$$

and work a little the Dirac matrices in the exponential so as to obtain

$$\mathcal{A}_{reg} = 2 \lim_{M^2 \rightarrow \infty} \text{tr} \left(\gamma_5 \frac{M^4}{(2\pi)^4} \int d^4p \exp \left(-\frac{\mathcal{D}_x}{M^2} + \frac{\mathcal{D}_x \not{p} + \not{p} \mathcal{D}_x}{M} - \not{p}^2 \right) \right) \tag{4.4.189}$$

or

$$\begin{aligned}
\mathcal{A}_{reg} &= 2 \lim_{M^2 \rightarrow \infty} \text{tr} \left(\gamma_5 \frac{M^4}{(2\pi)^4} \exp \left(-\frac{\mathcal{D}_x}{M^2} \right) \int d^4p \exp(-p^2) \right. \\
&\quad \times \exp \left(\frac{\mathcal{D}_x \not{p} + \not{p} \mathcal{D}_x}{M} \right) \Big)
\end{aligned} \tag{4.4.190}$$

We can expand the exponentials in $1/M$ powers and just keep terms up to $1/M^4$ in view of the M^4 factor in front of the integral. Terms in the expansion containing odd powers of p vanish after integration because of parity properties. Concerning even powers, after multiplying by γ_5 and computing the Dirac trace terms containing γ_5 and less than four γ matrices vanish when the Dirac trace is evaluated. The answer is then

$$\mathcal{A}_{reg} = -\frac{e^2}{2} F_{\mu\nu} F_{\alpha\beta} \lim_{M^2 \rightarrow \infty} (\text{tr} \gamma_5 [\gamma^\mu, \gamma^\nu] [\gamma^\alpha, \gamma^\beta]) \int d^4 p \exp(-p^2) \quad (4.4.191)$$

where we have used

$$\begin{aligned} \not{D}^2 &= D_\mu D_\nu \gamma^\mu \gamma^\nu \\ &= D_\mu D_\nu \left(\frac{\gamma^\mu \gamma^\nu + \gamma^\nu \gamma^\mu}{2} + \frac{\gamma^\mu \gamma^\nu - \gamma^\nu \gamma^\mu}{2} \right) \\ &= D_\mu D_\nu (g^{\mu\nu} + [\gamma^\mu, \gamma^\nu]) = D_\mu^2 + \frac{ie}{2} F_{\mu\nu} [\gamma^\mu, \gamma^\nu] \end{aligned} \quad (4.4.192)$$

The product of commutators in (4.4.191) is proportional to $\varepsilon^{\mu\nu\alpha\beta} \gamma_5$ so that after computing the trace and the gaussian integral one ends with

$$\mathcal{A}_{reg} = -\frac{e^2}{32\pi^2} \varepsilon^{\mu\nu\alpha\beta} F_{\mu\nu} F_{\alpha\beta} \quad (4.4.193)$$

and the anomaly equation finally reads

$$\langle \partial_\mu j_5^\mu \rangle = \frac{e^2}{16\pi^2} \varepsilon^{\mu\nu\alpha\beta} F_{\mu\nu} F_{\alpha\beta} \quad (4.4.194)$$

Note 1: If instead of using $\exp(-\not{D}^2/M^2)$ as regulator one uses $\exp(-\not{\partial}^2/M^2)$ the regulation also works. In fact, the derivation proceeds exactly as before since $\not{\partial}^2 = \not{D}[A=0]$. This means that instead of the result (4.4.193) for $\mathcal{A}_{reg}$, one would get $\bar{\mathcal{A}}_{reg} = 0$. Of course, this would mean that the basis and eigenvalues one should use are those of the free Dirac operator. Evidently these eigenvalues are not gauge invariant so there is no guarantee that this alternative regularization preserves gauge-invariance. And, in fact, if following the Noether method one controls conservation of the gauge current, one finds an anomaly which, appart from a sign, coincides precisely with the one found for the chiral current with the gauge invariant regularization. Mathematically, both regularizations are equally correct.. Physical requiremmts will decide which one should be selected. If gauge invariance is to be mantained at any cost, the gauge invariant regularization must be adopted and a chiral anomaly will be the prize to pay.

Note 2: In theories with Weyl fermions the Dirac operator takes the form

$$D[A] = i\not{\partial} + (1 - \gamma_5)\not{A} \quad (4.4.195)$$

This is not an hermitian operator,

$$D^\dagger[A] = i\not{\partial} + \not{A}(1 - \gamma_5) = i\not{\partial} + (1 + \gamma_5)\not{A} \quad (4.4.196)$$

Then, one cannot ensure that eigenvalues of $D^2[A]$ are positive-definite and this operator cannot be used as regulator in the same form as done above. One can solve this problem in two ways: one possibility is to analytically continue the operator

$$D^s[A] = i\not{\partial} + (1 - s\gamma_5)\not{A} \quad (4.4.197)$$

with s a complex variable such that $s = -s^*$ so that $D^{s\dagger}[A] = D^s[A]$. One then works with this hermitian operator and at the end of the calculations one puts $s = 1$. A second possibility is to work with a regulator R defined as

$$R = \frac{1}{2}(D[A]D^\dagger[A] + D^\dagger[A]D[A]) \quad (4.4.198)$$

Interestingly enough, the answer obtained for the anomaly in the two cases is different. In the first case one obtains the covariant anomaly, the name referring to the fact the resulting $\mathcal{A}$ is gauge covariant (in fact, it is proportional to F^*F) as in the Dirac fermion case. Using R as regulator, one finds that $\mathcal{A}$ is not gauge covariant. However, this last anomaly gives a partition function $Z[A]$ that behaves correctly under gauge transformations for A . Moreover, the difference in applying two consecutive gauge transformations to $Z[A]$ in different order close the correct gauge group algebra (Wess-Zumino consistency equations). One calls this anomaly the consistent anomaly. The physical situation will decide which answer to chose and hence which one of the two regularization prescriptions one has to adopt.

Chapter 5

Appendix

5.1 The path-integral quantization of non-Abelian gauge theories

Consider a gauge group G with a Lie Algebra $\mathcal{G}$ generated by the set of generators t^a ,

$$[t^a, t^b] = f^{abc} t^c \quad a = 1, 2, \dots \dim \mathcal{G} \quad (5.1.1)$$

$$\text{tr } t^a t^b = \frac{1}{2} \delta^{ab} \quad (5.1.2)$$

$$f^{abc} f^{pbc} = \delta^{ap} C[G] \quad (5.1.3)$$

with $C[G]$ the quadratic Casimir of G .

For non-Abelian gauge groups, the gauge field takes values in the Lie Algebra $\mathcal{G}$,

$$A_\mu(x) = A_\mu^a(x) t^a \quad (5.1.4)$$

A gauge transformation is associated now with an element g of the gauge group G ,

$$A_\mu(x) \rightarrow A_\mu^g = g^{-1} A_\mu g + \frac{1}{i} g^{-1} \partial_\mu g \quad (5.1.5)$$

with

$$g = \exp(i t^a \Lambda^a(x)) \quad (5.1.6)$$

For an infinitesimal gauge transformation one has

$$g = \exp(i \Lambda(x)) \approx 1 + i \Lambda(x) + O(\Lambda^2) \quad (5.1.7)$$

where we have written $\Lambda = \Lambda^a t^a$. In this case one has

$$A_\mu^g \approx (1 - i\Lambda)A_\mu(1 + i\Lambda) + \partial_\mu\Lambda + O(\Lambda^2) \quad (5.1.8)$$

or

$$\delta A_\mu \equiv A_\mu^g - A_\mu = \partial_\mu\Lambda + i[A_\mu, \Lambda] \quad (5.1.9)$$

We write compactly eq.(5.1.9) in the form

$$\delta A_\mu = D_\mu[A]\Lambda \quad (5.1.10)$$

where the covariant derivative $D_\mu[A]$ is defined as

$$D_\mu[A] = \partial_\mu + i[A_\mu, \] \quad (5.1.11)$$

As the normal derivative in the case of Abelian groups, the covariant derivative generates *infinitesimal gauge transformations* tangent to the orbits of the gauge group. One can easily see that under gauge transformation, the covariant derivative changes according to

$$D_\mu[A] \rightarrow D_\mu[A^g] = g^{-1}D[A_\mu]g \quad (5.1.12)$$

We are now at the point where the curvature $F_{\mu\nu}$ can be defined as an object will also belongs to the Lie Algebra $\mathcal{G}$,

$$F_{\mu\nu} = F_{\mu\nu}^a t^a \quad (5.1.13)$$

To see its origin, let us calculate the commutator of two covariant derivatives acting over some Lie Algebra valued function

$$[D_\mu, D_\nu]f = i[\partial_\mu A_\nu - \partial_\nu A_\mu + i[A_\mu, A_\nu], f] \equiv i[F_{\mu\nu}, f] \quad (5.1.14)$$

The curvature or non-Abelian field strength is here defined as

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + i[A_\mu, A_\nu] \quad (5.1.15)$$

or, in components,

$$F_{\mu\nu}^a = \partial_\mu A_\nu^a - \partial_\nu A_\mu^a + if^{abc}A_\mu^b A_\nu^c \quad (5.1.16)$$

One easily sees that under a gauge transformation it does not remain invariant (as it happens in the Abelian case) but it changes as the covariant derivative, that is, *covariantly*,

$$F_{\mu\nu}(x) \rightarrow F_{\mu\nu}^g(x) = g^{-1}F_{\mu\nu}(x)g \quad (5.1.17)$$

Let us now write the non-Abelian version of Maxwell Lagrangian. It has been written for the first time by Yang and Mills¹ and is the “natural” extension of the Maxwell Lagrangian,

$$L = \frac{a}{2} \text{tr} F_{\mu\nu} F^{\mu\nu} \quad (5.1.18)$$

with a a constant to be adjusted later and tr the trace of the t^a generators (for example the Pauli matrices appropriately normalized for $G = SU(2)$). Using (5.1.17) one proves that the Yang-Mills Lagrangian is gauge invariant,

$$L[A^g] = L[A] \quad (5.1.19)$$

The equations of motion are

$$D_\mu F^{\mu\nu} = \partial_\mu F^{\mu\nu} + i[A_\mu F^{\mu\nu}] = 0 \quad (5.1.20)$$

Concerning the Bianchi identity, in the non-Abelian case the equation for the dual field strength takes the form

$$D_\mu {}^*F^{\mu\nu} = 0 \quad (5.1.21)$$

with the dual of $F_{\mu\nu}$ defined as

$${}^*F_{\mu\nu} = \frac{1}{2} \varepsilon_{\mu\nu\alpha\beta} F^{\alpha\beta} = \varepsilon_{\mu\nu\alpha\beta} (\partial^\alpha A^\beta - A^\alpha A^\beta) \quad (5.1.22)$$

Coupling constants

In the $U(1)$ case (i.e., electrodynamics) there is no coupling constant in the gauge-field Lagrangian. Abelian gauge fields do not interact among themselves. In electrodynamics, an Abelian gauge theory, the coupling constant appears when we couple gauge fields (photons) to matter fields (electrons). Only the matter fields are charged while photons are neutral.

In the non-Abelian case, also the gauge field is charged. This can be clearly seen if we make the shift

$$A_\mu \rightarrow e A_\mu \quad (5.1.23)$$

The curvature changes as

$$\partial_\mu A_\nu - \partial_\nu A_\mu + i[A_\mu, A_\nu] \rightarrow e(\partial_\mu A_\nu - \partial_\nu A_\mu + ie[A_\mu, A_\nu]) \equiv e F_{\mu\nu} \quad (5.1.24)$$

¹ C.N. Yang and R.L. Mills, Physical Review **96** (1954) 191.

We see that, apart for the overall factor which we can factor out in the action integral, the field strength depends on the charge e through the field commutator. If we make this replacement in the Yang-Mills Lagrangian (5.1.18) we have

$$L = \frac{e^2 a}{2} \text{tr} F_{\mu\nu} F^{\mu\nu} \quad (5.1.25)$$

Hence, if we chose $a = 1/e^2$ we finally have

$$L = \frac{1}{2} \text{tr} F_{\mu\nu} F^{\mu\nu} \quad (5.1.26)$$

with

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + ie[A_\mu, A_\nu] \quad (5.1.27)$$

and in this form it is clear that in the non-Abelian case, the gauge field is self-interacting with a coupling constant e : the cubic and quartic self-interacting terms in the Lagrangian are

$$L^{cubic} = -2ie^3 \text{tr} \partial_\mu A_\nu [A_\mu, A_\nu] \quad (5.1.28)$$

$$L^{quartic} = \frac{e^4}{2} \text{tr} [A_\mu, A_\nu] [A_\mu, A_\nu] \quad (5.1.29)$$

So, e is really the gauge-field charge.

Quantization

Except for the fact one has to care of the non-commutative character of fields, quantization of non-Abelian gauge theories poses the same problems as those of electrodynamics and hence their solution follows the same Faddeev-Popov recipe. One starts from the partition function

$$Z = \int \mathcal{D}A_\mu \exp \left(-\frac{1}{\hbar} S[A] \right) \quad (5.1.30)$$

where now

$$\mathcal{D}A_\mu = \prod_x \prod_{\mu=0}^d \prod_{a=1}^{\dim \mathcal{G}} dA_\mu^a(x) \quad (5.1.31)$$

and the Yang-Mills action is given by

$$S = \frac{\text{tr}}{2} \int d^d x F_{\mu\nu} F^{\mu\nu} = \frac{1}{4} \int d^d x F_{\mu\nu}^a F^{a\mu\nu} \quad (5.1.32)$$

Evidently, the path-integral measure should be defined in a gauge-invariant way.

$$\mathcal{D}A_\mu = \mathcal{D}A_\mu^g \quad (5.1.33)$$

where

$$A_\mu^g = g^{-1}A_\mu g - \frac{i}{e}g^{-1}\partial_\mu g \quad (5.1.34)$$

This is very important since, otherwise, the quantum theory will not be gauge invariant.

Again, one has to avoid the problem of overcounting gauge configuration which are gauge copies. To this end, one inserts as in the Abelian case a resolution of the identity of the form

$$1 = \Delta_{FP}[A] \int \mathcal{D}g \delta(F[A^g]) \quad (5.1.35)$$

where $\mathcal{D}g$ is a Haar measure

$$\mathcal{D}g = \prod_x dg(x) \quad (5.1.36)$$

so that at each point x , $dg(x)$ is a measure over the group volume. One then gets for Z ,

$$Z = \int \mathcal{D}A_\mu \mathcal{D}g \Delta_{FP}[A] \exp\left(-\frac{1}{\hbar}S[A]\right) \delta(F[A^g]) \quad (5.1.37)$$

One now proceeds exactly as in the Abelian case, either by expanding up to quadratic terms and showing that the zero mode is eliminated or one can follow the more heuristic approach that will be adopted in what follows: one proves that $\Delta_{FP}[A] = \Delta_{FP}[A^g]$ and since $S[A] = S[A^g]$ and $\mathcal{D}A_\mu = \mathcal{D}A_\mu^g$ one writes, instead of (5.1.37),

$$Z = \int \mathcal{D}A_\mu \mathcal{D}g \Delta_{FP}[A^g] \exp\left(-\frac{1}{\hbar}S[A^g]\right) \delta(F[A^g]) \quad (5.1.38)$$

or, renaming $A_\mu^g \rightarrow A_\mu$,

$$\begin{aligned} Z &= \int \mathcal{D}g \int \mathcal{D}A_\mu \Delta_{FP}[A] \exp\left(-\frac{1}{\hbar}S[A]\right) \delta(F[A]) \\ &= V \int \mathcal{D}A_\mu \Delta_{FP}[A] \exp\left(-\frac{1}{\hbar}S[A]\right) \delta(F[A]) \end{aligned} \quad (5.1.39)$$

where V is the (infinite) gauge group volume which has been factored out leaving the path-integral over a representative of each physical configuration, selected by the gauge condition $F[A] = 0$ which is imposed by the delta function.

Concerning the Faddeev-Popov determinant, it can be computed as in the Abelian case, by noting that, if there were no Gribov copies, the condition $F[A^g] = 0$ has just a solution, say that for $g = I$ (one can always rename gauge group elements so that this is true). Since the path-integral where the FP determinant has to be employed is constrained by $\delta(F[A^g])$, this means that the integral over g is constrained to gauge group elements close to the identity. One can then consider in the integrand in the r.h.s. of eq.(5.1.35)

$$g \approx 1 + i\epsilon + O(\epsilon^2) , \quad \mathcal{D}g = \mathcal{D}\epsilon \quad (5.1.40)$$

so that

$$1 = \Delta_{FP}[A] \int \mathcal{D}\epsilon \delta\left(F[A] + \frac{\delta F}{\delta A_\mu} D_\mu \epsilon\right) = \frac{\Delta_{FP}}{\det\left(\frac{\delta F}{\delta A_\mu} D_\mu\right)} \quad (5.1.41)$$

and hence

$$\Delta_{FP} = \det\left(\frac{\delta F}{\delta A_\mu} D_\mu\right) \quad (5.1.42)$$

We see that even in the Lorentz gauge, which in this case implies $\dim \mathcal{G}$ conditions,

$$\partial_\mu A_\mu^a = 0 \quad (5.1.43)$$

the Faddeev-Popov determinant depends on the gauge field,

$$\Delta_{FP} = \det(\partial_\mu D_\mu[A]) = \det(\square + ie\partial_\mu[A_\mu, \]) = \det(\square + ie[A_\mu, \partial_\mu]) \quad (5.1.44)$$

One can see that the Lorentz gauge is not the natural gauge in the non-Abelian case. Indeed, extrema of the Yang-Mills action satisfy

$$\frac{\delta S}{\delta A_\mu^{cl}(x, \Lambda)} = 0 \quad (5.1.45)$$

where we have explicitly indicated that $A_\mu^{cl}(x, \Lambda)$, solution of (5.1.45), depends, before gauge-fixing, of a gauge parameter Λ . Then, as usual, one has

$$\frac{\delta}{\delta \Lambda} \left(\frac{\delta S}{\delta A_\mu^{cl}(x, \Lambda)} \right) \delta \Lambda = \int d^4 y \frac{\delta^2 S}{\delta A_\mu^{cl}(x, \Lambda) \delta A_\nu^{cl}(y, \Lambda)} D_\nu[A^{cl}] \Lambda = 0 \quad (5.1.46)$$

this showing that $D_\mu[A^{cl}] \Lambda$ is a zero mode of the natural operator

Then, when expanding in eigenfunctions of the quadratic operator appearing in (5.1.46), orthogonality between zero and non-zero modes implies

$$\int d^4 x a_\nu^{(n)} D_\nu[A^{cl}] \Lambda = 0 \quad (5.1.47)$$

this in term giving

$$D_\nu[A^{cl}]a_\nu^{(n)}(x) = 0 \quad (5.1.48)$$

Then, in order to eliminate the zero mode from the path-integration, it is natural to select, as gauge condition,

$$D_\mu[A^{cl}]A_\mu(x) = 0 \quad (5.1.49)$$

The corresponding Faddeev-Popov determinant reads,

$$\Delta_{FP} = \det (D_\mu[A^{cl}]D_\mu[A]) \quad (5.1.50)$$

The generating functional

Let us now add a source and construct the generating functional for the Yang-Mills theory,

$$Z[j] = \int \mathcal{D}g \int \mathcal{D}A_\mu \Delta_{FP}[A] \exp \left(-\frac{1}{\hbar} S[A] - \frac{1}{\hbar} \int d^4x j_\mu(x) A_\mu(x) \right) \delta(F[A]) \quad (5.1.51)$$

We want to maintain gauge-invariance at this level but then, since the source term is not in general invariant,

$$\delta_\epsilon \int d^4x j_\mu(x) A_\mu(x) = \int d^4x j_\mu(x) \delta_\epsilon A_\mu(x) = \int d^4x j_\mu(x) D_\mu \epsilon \quad (5.1.52)$$

we have to impose the condition

$$D_\mu[A]j_\mu(x) = 0 \quad (5.1.53)$$

so as to ensure that, by integration by parts, the r.h.s. in (5.1.53) vanishes. Apart from the fact that the Yang-Mills action is not quadratic and hence we cannot perform the path-integral exactly, we have the problem of the Faddeev-Popov determinant which in principle depends on A_μ and also the delta function. We postpone the problem of Δ_{FP} and attack the problem of the delta function.

A possible functional representation for the delta function is

$$\delta(F[A]) = \lim_{\alpha \rightarrow 0} \frac{1}{\sqrt{2\pi\hbar\alpha}} \exp \left(-\frac{1}{\hbar\alpha} \text{tr} \int d^4x F[A]^2 \right) \quad (5.1.54)$$

Hence, one can work at fixed α using this representation and then, at the end of the calculations, take the limit of vanishing α to effectively restrict the path-integral to the gauge condition $F[A] = 0$. In the case $F[A] = \partial_\mu A_\mu$, the quadratic part of the effective Lagrangian is

$$L^{(2)} = \frac{1}{4} (\partial_\mu A_\nu^a - \partial_\nu A_\mu^a)^2 + \frac{1}{2\alpha} \partial_\mu A_\mu^a \partial_\nu A_\nu^a \quad (5.1.55)$$

In the corresponding quadratic action we can integrate by parts so that the quadratic Lagrangian can be written as

$$L^{(2)} = \frac{1}{2} A_\mu^a \left(-\square \delta_{\mu\nu} + \left(1 - \frac{1}{\alpha} \right) \partial_\mu \partial_\nu \right) A_\nu^a \quad (5.1.56)$$

Concerning the cubic and quartic terms (1.2.52) and (1.2.53), i.e., the self-interacting terms, they can be written as

$$L^{int} = e f^{abc} (\partial_\mu A_\nu^a) A_\mu^b A_\nu^c + e^2 (A_\mu^a A_\mu^a A_\mu^b A_\mu^b - A_\mu^a A_\nu^a A_\mu^b A_\nu^b) \quad (5.1.57)$$

Having in mind a perturbative expansion, we start by writing $Z_0[j]$, the generating functional without including interaction terms,

$$Z_0[j] = \int \mathcal{D}A_\mu \Delta_{FP} \exp \left(-\frac{1}{2\hbar} \int d^4x (A_\mu^a M_{\mu\nu}^{ab} A_\nu^b + j_\mu^a A_\mu^a) \right) \quad (5.1.58)$$

where

$$M_{\mu\nu}^{ab} = \left(-\square \delta_{\mu\nu} + \left(1 - \frac{1}{\alpha} \right) \partial_\mu \partial_\nu \right) \delta^{ab} \quad (5.1.59)$$

Let us suppose for the moment that the Faddeev-Popov determinant does not contribute to the quadratic part of the theory so that it can be ignored. Then, the path-integral can be performed and the answer is

$$Z_0[j] = (\det M)^{-1/2} \exp \left(\frac{1}{4\hbar} \int d^4x d^4y j_\mu^a(x) (M^{-1})_{\mu\nu}^{ab}(x, y) j_\nu^b(y) \right) \quad (5.1.60)$$

(Note that since we have included just the inverse of a square root of $\det \mathcal{M}$ coming from the A integration, we have to remember that in fact there are $N = 4 \times \dim \mathcal{G}$ integrations over the gauge field components so that the determinant should include this N apart from being an operator determinant with index x, y). Concerning with $(M^{-1})_{\mu\nu}^{ab}$, it is the Green function of the operator $M_{\mu\nu}^{ab}$,

$$M_{\mu\nu}^{ab} (M^{-1})_{\nu\rho}^{bc}(x, y) = \delta^{(4)}(x - y) \delta_{\mu\rho} \delta^{ac} \quad (5.1.61)$$

Defining the Fourier transform of $(M^{-1})_{\mu\nu}^{ab}$ through

$$(M^{-1})_{\mu\nu}^{ab}(x) = \int \frac{d^4k}{(2\pi)^4} (\bar{M}^{-1})_{\mu\nu}^{ab}(k) \exp(ik_\mu x_\mu) \quad (5.1.62)$$

one easily finds

$$\left(-k^2\delta_{\mu\nu} + \left(1 - \frac{1}{\alpha}\right)k_\mu k_\nu\right) (\bar{M}^{-1})_{\nu\rho}^{ac}(k) = \delta_{\mu\rho}\delta^{ac} \quad (5.1.63)$$

If one now proposes

$$(\bar{M}^{-1})_{\nu\rho}^{ac}(k) = -\delta^{ac} (A(k^2)\delta_{\nu\rho} + B(k^2)k_\nu k_\rho) \quad (5.1.64)$$

one finds

$$A(k^2) = \frac{1}{k^2}, \quad B(k^2) = \frac{\alpha - 1}{k^4} \quad (5.1.65)$$

so that the gauge field propagator reads

$$(\bar{M}^{-1})_{\nu\rho}^{bc}(k) = -\frac{1}{k^2}\delta^{bc} \left(\delta_{\nu\rho} - (1 - \alpha)\frac{k_\nu k_\rho}{k^2}\right) \quad (5.1.66)$$

We can take now the limit $\alpha \rightarrow 0$ which in fact corresponds to the Lorentz gauge. The corresponding propagator is

$$(\bar{M}^{-1})_{\nu\rho}^{ac}(k) = -\frac{1}{k^2}\delta^{ac} \left(\delta_{\nu\rho} - \frac{k_\nu k_\rho}{k^2}\right) \quad (5.1.67)$$

One can easily verify the identity

$$k_\nu (\bar{M}^{-1})_{\nu\rho}^{ac}(k) = 0 \quad (5.1.68)$$

which is nothing but the Lorentz gauge condition for the propagator. (Remember that the propagator is the Fourier transform of the v.e.v. $\langle A_\nu^a(x) A_\rho^c(y) \rangle_0$).

It is interesting to note that in the absence of a gauge fixing one should have from the beginning $\alpha = \infty$. In that case eq.(5.1.63) would read

$$(-k^2\delta_{\mu\nu} + k_\mu k_\nu) (\bar{M}^{-1})_{\nu\rho}^{ac}(k) = \delta_{\mu\rho}\delta^{ac} \quad (5.1.69)$$

One easily convinces oneself by repeating the steps (5.1.66)-5.1.65) that, in that case, $(\bar{M}^{-1})_{\nu\rho}^{bc}(k)$ does not exist. This is another manifestation of the necessity of fixing the gauge.

There is a family of gauge fixing which is very useful for perturbative calculations and which can be implemented as follows. Since the generating function is gauge invariant, it cannot depend on the gauge condition one chooses. Suppose then that, instead of considering the condition $F[A] = 0$ one chooses

$$F[A] = c(x) \quad (5.1.70)$$

then, Z_0 should read

$$Z_0[j] = V \int \mathcal{D}A_\mu \Delta_{FP}[A] \exp \left(-\frac{1}{\hbar} S[A] - \int d^4x j_\mu(x) A_\mu(x) \right) \delta(F[A] - c) \quad (5.1.71)$$

Since Z_0 cannot depend on c , we can integrate trivially over c both sides of this identity with an arbitrary weight that we choose to be

$$\int \mathcal{D}c \exp \left(-\frac{1}{\hbar\alpha} \int d^4x c(x)^2 \right) \quad (5.1.72)$$

with α now arbitrary. In the l.h.s. of (5.1.72) this corresponds to a constant factor that we absorb in the definition. In the r.h.s., the delta function allows to integrate out c so that finally one gets

$$Z_0[j] = V \int \mathcal{D}A_\mu \Delta_{FP}[A] \exp \left(-\frac{1}{\hbar} S[A] - \int d^4x j_\mu(x) A_\mu(x) \right) \exp \left(-\frac{1}{\alpha} \int d^4x F[A]^2 \right) \quad (5.1.73)$$

Comparing (5.1.73) with the result we derived for the Lorentz gauge, we see that all the derivation goes the same except that now α is arbitrary compared to the $\alpha = 0$ case for the Lorentz gauge. Then, the propagator is still given by (5.1.66) but with α any positive real number. In particular, the choice $\alpha = 1$ greatly simplifies its form when $F[A] = \partial_\mu A_\mu$,

$$(\bar{M}^{-1})_{\nu\rho}^{bc}(k) = -\frac{1}{k^2} \delta^{bc} \delta_{\nu\rho} \quad (5.1.74)$$

This gauge is known as the Feynman gauge.

Instantons Instantons is the name given by 't Hooft (see ref. [?]) to the localized finite action solutions of the classical equations of motion of Yang-Mills theory in Euclidean space that were found by Belavin, Polyakov, Schwartz and Tyupkin [?]. These authors originally called their solution a "pseudoparticle" Today the name instanton applies to the classical solution to the equations of motion with localized finite Euclidean action of any non-linear field theory².

²Note that any soliton (localized static finite energy solution) in $d+1$ Minkowski space-time can be seen as an instanton (localized finite action solution) in d Euclidean space-time.

$SU(2)$ Euclidean Yang-Mills configurations and topology

Let us start by writing the $SU(2)$ Yang-Mills Euclidean action

$$S = -\frac{1}{2g^2} \int d^4x \text{tr} F_{\mu\nu} F_{\mu\nu} \quad (5.1.75)$$

Here g is the coupling constant (factored out with an appropriate redefinition of the gauge field $A_\mu(x)$) taking values in the Lie algebra of $SU(2)$,

$$A_\mu(x) = \frac{\sigma^a}{2i} A_\mu^a, \quad a = 1, 2, 3. \quad (5.1.76)$$

with σ^a the Pauli matrices and $\mu = 0, 1, 2, 3$,

$$\sigma^1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma^2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma^3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad (5.1.77)$$

Defined as in (5.1.76) $A_\mu(x)$ are anti-Hermitian matrices. Correspondingly, the field strength is a matrix-valued tensor

$$F_{\mu\nu} = \frac{\sigma^a}{2i} F_{\mu\nu}^a \quad (5.1.78)$$

which in terms of A_μ reads

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu] \quad (5.1.79)$$

which, using the Lie algebra

$$\left[\frac{\sigma^a}{2}, \frac{\sigma^b}{2} \right] = i\epsilon^{abc} \frac{\sigma^c}{2} \quad (5.1.80)$$

becomes in components

$$F_{\mu\nu}^a = \partial_\mu A_\nu^a - \partial_\nu A_\mu^a + \epsilon^{abc} A_\mu^b A_\nu^c \quad (5.1.81)$$

Under an $SU(2)$ (local) gauge transformation A_μ changes as a connection

$$A_\mu(x) \rightarrow A^U_\mu(x) = U(x) A_\mu U^{-1}(x) + U(x) \partial_\mu U^{-1}(x) \quad (5.1.82)$$

For infinitesimal gauge transformations

$$U(x) \approx 1 + i\epsilon(x) \quad (5.1.83)$$

the change reads

$$\delta A_\mu = A^U_\mu - A_\mu = -i D_\mu[A] \epsilon(x) \quad (5.1.84)$$

with $D_\mu[A]$ the covariant derivative

$$D_\mu[A] = \partial_\mu + [A_\mu, \] \quad (5.1.85)$$

Concerning the field strength, it changes covariantly

$$F_{\mu\nu}(x) = U(x)F_{\mu\nu}(x)U^{-1}(x) \quad (5.1.86)$$

The equation of motion resulting from the Euclidean action (5.1.75) reads

$$D_\mu[A]F_{\mu\nu} = 0 \quad (5.1.87)$$

In order to determine the appropriate boundary conditions let us note that finite action configurations require that $F_{\mu\nu}$ must vanish on the boundary of Euclidean four-space, i.e.,

$$\lim_{r \rightarrow \infty} F_{\mu\nu}(x) = 0, \quad r = |x| = \sqrt{x_0^2 + x_1^2 + x_2^2 + x_3^2} \quad (5.1.88)$$

this implying that at infinity A_μ must be a pure gauge

$$\lim_{r \rightarrow \infty} A_\mu(x) = U(x)\partial_\mu U^{-1}(x) \quad (5.1.89)$$

(In fact $F_{\mu\nu}$ must fall to zero faster than $1/r^2$ as $r \rightarrow \infty$.)

The surface at infinity S_∞^3 may be parametrized by three parameters $\alpha_i, i = 1, 2, 3$ (e.g. the 3 polar angles in four dimensional space). Then, working in a gauge in which $A_r = 0$ one can think of $U(x)$ in (5.1.89) at infinity as establishing an application between the the spatial sphere at infinity, S_∞^3 , and the gauge group manifold which can be identified with S^3 ,

$$U(x)|_{r \rightarrow \infty} : S_\infty^3 \rightarrow S^3 \quad (5.1.90)$$

Then the possible U elements can be classified according to the homotopy classes to which they belong, characterized by integer numbers

$$\pi_3(S^3) = \mathbb{Z} \quad (5.1.91)$$

That is, the finite action gauge field configurations can be classified by integers which can be calculated by the so called Pontryagin (or Chern-Pointryagin) index Q which can be written in terms of the field-strength (curvature) $F_{\mu\nu}$ and its dual $*F_{\mu\nu}$

$$Q = -\frac{1}{16\pi^2} \text{tr} \int d^4x F_{\mu\nu} *F_{\mu\nu} = n \in \mathbb{Z} \quad (5.1.92)$$

$$*F_{\mu\nu} = \frac{1}{2}\varepsilon_{\mu\nu\alpha\beta}F_{\alpha\beta} \quad (5.1.93)$$

One can easily prove (see problem) that

$$F_{\mu\nu}*F_{\mu\nu} = 2\partial_\mu j_\mu \quad (5.1.94)$$

where

$$j_\mu = -2\text{tr} \left(\varepsilon_{\mu\nu\alpha\beta} \partial_\nu \partial_\alpha A_\beta + \frac{2}{3} A_\nu A_\alpha A_\beta \right) \quad (5.1.95)$$

Note that j_0 can be identified with the Chern-Simons Lagrangian after reducing Euclidean space-time from $d = 4$ to $d = 3$ dimensions.

Using eq.(5.1.94) one can write the Pontryagin index as an integral over S_∞^3

$$Q = \int_{S_\infty^3} ds_\mu j_\mu \quad (5.1.96)$$

which can be written

$$Q = -\frac{1}{24\pi^2} \text{tr} \int_{S_\infty^3} ds_\mu \varepsilon_{\mu\nu\alpha\beta} U(\partial_\nu U)U^{-1}(\partial_\alpha U)U^{-1}(\partial_\beta U)U^{-1} \quad (5.1.97)$$

Now since the group manifold is also the 3-sphere, its measure $d\mu(U)$ can be also written in terms of three angles ξ ,

$$d\mu(U) = J d\xi_1 d\xi_2 d\xi_3 \quad (5.1.98)$$

The integral in (5.1.97) then counts how many times the group has been wrapped around the spatial S_∞^3 sphere and this number can be seen to be an integer (see a simple proof of this in Rajaraman book³).

The BPST instanton solution

A *chetvyorka* of russian physicists (Alexander Belavin, Alexander Polyakov, Albert Schwarz and Yu. S. Tyupkin) found the first solution to the Yang-Mills equations of motion in Euclidean space⁴ which they called pseudoparticle (the name instanton was afterwards imposed after 't Hooft). They started from the trivial inequality

$$-\text{tr} \int d^4x (F_{\mu\nu} \pm *F_{\mu\nu})^2 \geq 0 \quad (5.1.99)$$

³R. Rajaraman, *Solitons and Instantons*, Elsevier, Amsterdam, 1996.

⁴A. A. Belavin, A. M. Polyakov, A. S. Schwartz and Y. .S. Tyupkin, "Pseudoparticle Solutions of the Yang-Mills Equations," Phys. Lett. B **59** (1975) 85.

(The minus sign in the l.h.s. is due to the fact that with our conventions for generators $(\sigma^a/(2i))A_\mu^a$ is anti-hermitian.)

Eq.(5.1.99) can be written as

$$-\text{tr} \int d^4x F_{\mu\nu} F_{\mu\nu} \geq \mp \text{tr} \int d^4x F_{\mu\nu}^* F_{\mu\nu} \quad (5.1.100)$$

or

$$S \geq \frac{8\pi^2}{g^2} |Q| \quad (5.1.101)$$

where S is the Euclidean Yang-Mills action (5.1.75) and Q the Chern-Pontryagin index (5.1.92).

Obviously the field configuration for which the bound for the action is attained should correspond to a solution to the equations of motion. In view of eq.(5.1.99), such configuration should satisfy the first order equation

$$F_{\mu\nu} = \pm^* F_{\mu\nu} \quad (5.1.102)$$

thus self-dual and self-antidual configurations extremise the action and hence solve the Euclidean Yang-Mills second-order equations of motion (5.1.87).

Belavin et al [?] offered a very simple derivation of the $Q = 1$ (one-instanton) solution which they showed takes the form

$$A_\mu = \frac{|x|^2}{|x|^2 + \lambda^2} U_1(x) \frac{\partial U_1^{-1}(x)}{\partial x_\mu} \quad (5.1.103)$$

with $|x|^2 = x^{02} + \vec{x}^2$ and

$$U_1 = \frac{x_0 + i\vec{x} \cdot \vec{\sigma}}{|x|} \quad (5.1.104)$$

with $\vec{\sigma}$ the Pauli matrices. The scale λ arises from the scale invariance of the theory and determines the instanton size. Apart from this parameter, one can introduce a 4-vector a_μ associated to the instanton center by making the shift $x^\mu \rightarrow x^\mu - a^\mu$, associated with translation invariance of the Yang-Mills Lagrangian.

Belavin et al ansatz is related to the $O(4)$ symmetry that can be introduced by extending the original $SU(2)$ gauge group to $SU(2) \times SU(2) \approx O(4)$. Working with that group they considered the more general spherically symmetric ansatz in $d = 4$ Euclidean space time leading to the $Q = 1$ instanton solution.

In order to find instanton solutions with Pontryagin index $Q = n > 1$ 't Hooft introduced an η tensor that maps antisymmetric representations of $SO(4)$ onto vectors of one of its own subgroups of $SO(3)$ [?]

$$\eta_{a\mu\nu} = \begin{cases} \varepsilon_{a\mu\nu} & \text{if } a, \mu, \nu = 1, 2, 3; \\ -\delta_{a\nu} & \text{if } \mu = 4; \\ \delta_{a\mu} & \text{if } \nu = 4; \\ 0 & \text{if } \mu = \nu = 4. \end{cases} \quad (5.1.105)$$

in terms of this η tensor 't Hooft ansatz reads

$$A_\mu^a = i\eta_{a\mu\nu}\partial_\nu \log \phi(x) \quad (5.1.106)$$

Inserting this ansatz into the selfduality equations (5.1.102) and after some work one gets

$$\frac{1}{\phi(x)}\square\phi(x) = 0 \quad (5.1.107)$$

where $\square\phi$ is the $d = 4$ Laplacian $\square = \partial_\mu\partial_\mu$ (remember taht we are working in Euclidean space).

When ϕ is non-singular eq.(5.1.107) reduces to $\square\phi(x) = 0$ which, in order to have finite action configurations, permits only the trivial solution $\phi = \text{constant}$ so that $A_\mu = 0$. There are however singular $\phi(x)$ solutions leading to non-singular gauge fields. For example consider the case

$$\phi^{(1)} = \frac{1}{x^2} \quad (5.1.108)$$

which for $x \neq 0$ satisfies

$$\frac{1}{\phi^{(1)}}\square\phi^{(1)} = \frac{1}{\phi^{(1)}}\partial_\mu\left(-2\frac{x_\mu}{|x|^4}\right) = 0 \quad (5.1.109)$$

For $x = 0$ one can easily see using Gauss theorem that that

$$\int d^4x \square \frac{1}{|x|^2} = \int dS_\mu \partial_\mu \left(\frac{1}{|x|^2} \right) = -4\pi \quad (5.1.110)$$

showing that $1/|x|^2$ is the 4-dimensional Laplacian Green function. Hence one has

$$\frac{1}{\phi^{(1)}}\square\phi^{(1)} = -4\pi|x|^2\delta^{(4)} = 0 \quad (5.1.111)$$

The precedent discussion shows that $\phi^{(1)}$ solves the equation (5.1.107) and, more generally one has that

$$\phi^{(n)}(x) = 1 + \sum_{i=1}^n \frac{\lambda_i^2}{|x_\mu - a_\mu^i|^2} \quad (5.1.112)$$

this leading to an instanton solution with Chern-Pontryagin index $Q = n$.

In fact, when computing the n -instanton solution from $\phi^{(n)}$ in the form (5.1.112) one finds that the solution $A_\mu^{(n)}$ is singular at points $x_\mu = a_\mu^i$. However these singularities can be removed to infinity by singular gauge transformations, one for each singularity. Such transformations are of the form (5.1.104)

$$U^i(x) = \frac{(x_0 - a_0^i) + i\vec{\sigma} \cdot (\vec{x} - \vec{a}^i)}{|x - a^i|} \quad (5.1.113)$$

and the final form of the regular 't Hooft n -instanton solution reads

$$A_\mu^{(n)} = 2i\eta_{\mu\nu a} \frac{\sum_i^n \frac{\lambda_i^2 (x_\mu - a_{i\mu})}{|x - a_i|^4}}{1 + \sum_i^n \frac{\lambda_i^2}{|x - a_i|^2}} \quad (5.1.114)$$

The associated field strength reads

$$F_{\mu\nu}^a = 2\eta_{a\mu\nu} \sum_{i=1}^n \frac{\lambda_i^2}{(|x - a_i|^2 + \lambda_i^2)^2} \quad (5.1.115)$$

This solution corresponds to an instanton with n units of topological charges, each one located at points a_μ^i in $d = 4$ Euclidean space-time.

The n instanton 't Hooft solution presented in eq.(5.1.114) exhibit a number of parameters that in the context of soliton quantization should be treated as collective coordinates.

The parameters associated to instanton configurations include

1. $4n$ coordinates $a_{i\mu}$ specifying the centers around which each one of the n topological densities are localized. They are associated to translation invariance of the theory.
2. n scale parameters λ_i measuring the size of each region in which the topological density is localized. They are associated to scale invariance of the theory
3. 3 global $SU(2)$ gauge transformations that originally were ignored on physical grounds

One should also take into account the fact that Yang-Mills theory exhibits a complete conformal invariance (not just the scale invariance manifested through the presence of λ_i). As Jackiw and Rebbi have proven⁵ one has to take into account invariance under the whole

⁵ R. Jackiw and C. Rebbi, "Degrees of Freedom in Pseudoparticle Systems," Phys. Lett. B **67** (1977) 189.

conformal group which includes not only translations and dilatations but also the 6 $O(4)$ rotations and the 4 special conformal transformations

$$x_\mu \rightarrow \frac{x_\mu - c_\mu x^2}{1 - 2c_\mu x_\mu + x^2 c^2} \quad (5.1.116)$$

Looking at the solution obtained from 't Hooft ansatz permits to identified $5n + 3$ parameters. When quantizing Yang-Mills theory around an n -instanton those parameters will become collective coordinates. It should be stressed that the correct number of parameters is not the one revealed by the ansatz but, as proven by Atiyah and Ward⁶ the number of parameters is $8n - 3$. If global rotations are counted, the number becomes $8n$ for $SU(2)$ and $4nN$ for $SU(N)$.

5.2 Local and global symmetries and their conserved charges

The Noether theorem at the classical level

Let us briefly review the Noether theorem for a classical system. For simplicity we consider a scalar field with Lagrangian

$$L = L[\phi, \partial_\mu \phi] \quad (5.2.117)$$

and a certain field transformation S

$$\phi(x) \xrightarrow{S} \phi'(x) = \phi(x) + \delta_S \phi(x) \quad (5.2.118)$$

We shall call S a symmetry of the theory if the Euler-Lagrange equations (that is, those that govern dynamics) are not affected by its action. In other words, the equations of motion must remain unaltered under transformation (5.2.118). This implies that the only acceptable change in the Lagrangian under S should be a total derivative which in the relativistic field theory case corresponds to the divergence of a given 4-vector,

$$\delta_S L = \partial_\mu \Lambda^\mu[\phi, \partial_\mu \phi] \quad (5.2.119)$$

Eq.(5.2.119) should hold independently of the eqs. of motion. Indeed, if this is the case, the action change can be written in the form

$$S' = S + \delta_S S \quad (5.2.120)$$

⁶M. F. Atiyah and R. S. Ward, "Instantons and Algebraic Geometry," Commun. Math. Phys. **55** (1977) 117.

$$\delta_S S[\phi, \partial_\mu \phi] = \int_M d^4x \partial_\mu \Lambda^\mu[\phi, \partial_\mu \phi] = \int_{\partial M} \Lambda^\mu dS_\mu \quad (5.2.121)$$

where ∂M is the border of the manifold M on which dynamics is defined. Now, in general, Euler-Lagrange equations are derived by varying fields and their derivatives under the condition

$$\delta\phi|_{\partial M} = 0 \quad (5.2.122)$$

so that the field eqs. derived from S' will not differ from those obtained from S since, in view of (5.2.122), the change of the action on the border should be zero.

The Noether theorem⁷ then states that

- *For each symmetry S of the theory there exists a locally conserved current j^μ , that is, a current satisfying*

$$\partial_\mu j^\mu(x) = 0 \quad (5.2.123)$$

- *Consequently, there is a charge Q ,*

$$Q = \int d^3x j^0(x) \quad (5.2.124)$$

which is conserved,

$$\frac{dQ}{dt} = 0 \quad (5.2.125)$$

The proof of this theorem is constructive. One considers the change in the Lagrangian,

$$\delta_S L = \frac{\partial L}{\partial \phi} \delta_S \phi(x) + \frac{\partial L}{\partial(\partial_\mu \phi)} \delta_S \partial_\mu \phi + O(\delta\phi^2) \quad (5.2.126)$$

or

$$\partial_\mu \Lambda^\mu = \frac{\partial L}{\partial \phi} \delta_S \phi(x) + \frac{\partial L}{\partial(\partial_\mu \phi)} \partial_\mu \delta_S \phi + O(\delta\phi^2) \quad (5.2.127)$$

Here, we have assumed that $\delta \partial_\mu \phi = \partial_\mu \delta \phi(x)$.

At this point we do use the Euler-Lagrange equations

$$\frac{\delta L}{\delta \phi} = \partial_\mu \frac{\partial L}{\partial(\partial_\mu \phi)} \quad (5.2.128)$$

⁷ E. Noether, Nachr. d. Ges. d. Wissensch. zu Göttingen, (1918).

so that one has

$$\partial_\mu \frac{\partial L}{\partial(\partial_\mu \phi)} \delta_S \phi(x) + \frac{\partial L}{\partial(\partial_\mu \phi)} \partial_\mu \delta_S \phi(x) = \partial_\mu \Lambda^\mu \quad (5.2.129)$$

or

$$\partial_\mu \left(\frac{\partial L}{\partial(\partial_\mu \phi)} \delta_S \phi(x) \right) = \partial_\mu \Lambda^\mu \quad (5.2.130)$$

We have then found the locally conserved current j^μ associated with the symmetry S

$$j^\mu = \Lambda^\mu - \frac{\partial L}{\partial(\partial_\mu \phi)} \delta_S \phi \quad (5.2.131)$$

$$\partial_\mu j^\mu = 0 \quad (5.2.132)$$

Nota bene: Usually, when $\Lambda \neq 0$ the symmetry is a space-time symmetry like Lorentz symmetry, diffeomorphism symmetry, etc. Internal symmetries correspond in general to $\Lambda = 0$ although there are some exceptions (for example, in the case of Chern-Simons theories an internal symmetry (gauge symmetry) leads to change in the Lagrangian, i.e. $\Lambda \neq 0$).

5.3 The BRST method

Having learnt that the gauge fixed effective Lagrangian results from the addition to the original Lagrangian of term which can be written as the BRST transformed of some functional, we can think in the extension of the BRST method in the following sense:

Given a Lagrangian with a certain symmetry (not necessarily gauge symmetry), we will need to fix this symmetry in order not to overcount in the path-integral equivalent configurations. Otherwise, one should get a divergent non physical answer. To do this one proceeds as follows

- Given the (infinitesimal) symmetry transformation law for the original field A , with parameter ϵ , one defines a new transformation law δ_{BRST} copied from it but replacing ϵ by an anticommuting field called a ghost, having the same tensorial character as the parameter of the original symmetry.
- Having introduced the ghost field (which we call $c(x)$), one introduces an “anti-ghost” field $\bar{c}(x)$ and an auxiliary field $b(x)$ and completes the transformation law so as to ensure that $\delta_{BRST}^2 = 0$.
- The effective (gauge fixed) Lagrangian L_{eff} is constructed from the original symmetric one L by adding the BRST transformed of an arbitrary functional depending in principle on all fields,

$$L_{eff}[A, b, \bar{c}, c] = L[A] + \delta_{BRST}\mathcal{F}[A, b, \bar{c}, c] \quad (5.3.133)$$

- The properly gauge fixed partition function is defined as

$$Z = \int \mathcal{D}A \mathcal{D}b \mathcal{D}\bar{c} \mathcal{D}c \exp\left(- \int d^4x L_{eff}[A, b, \bar{c}, c]\right) \quad (5.3.134)$$

Note that this prescription leads to results quite more general than those obtained in the original Faddeev-Popov scheme. In particular, if one choses appropriately $\mathcal{F}$, one can obtain effective Lagrangians that depend on ghost through polynomials of degree higher than 2. For example, if one takes

$$\mathcal{F} = b\bar{c}c\bar{c}F[A] \quad (5.3.135)$$

one gets terms quartic in the ghosts that will never produce, by integration, a Faddeev-Popov determinant but that, nevertheless, fix the gauge correctly. This is exploited for example, in quantization of theories containing gravity.

How are we shure that this general prescription is as correct as that we have proved, based in using $\mathcal{F} = \bar{c}F[A]$? To see this we will prove (under certain assumptions) that the v.e.v. of the BRST transformed of an arbitrary functional (a BRST exact term) satisfies the important property

$$\langle \delta_{BRST}G \rangle = 0 \quad (5.3.136)$$

From this, one can show that the v.e.v.'s of a “physical operator” calculated with Lagrangians differing in a BRST exact term coincide.

To prove (5.3.136) , consider the quantity

$$\langle G \rangle = \int \mathcal{D}\Phi \exp(-S_{eff}[\Phi]) G[\Phi] \quad (5.3.137)$$

where Φ represents all fields: original “phsical” fields, ghost fields, auxiliary fields.

Consider the change of variables

$$\Phi \rightarrow \Phi' = \Phi + \lambda \delta_{BRST}\Phi \quad (5.3.138)$$

By assumption, the action is invariant under BRST transformations and then remains unchanged.

If we also assume that the integration measure is BRST invariant, then the only change in the integrand comes from the change in G ,

$$\langle G \rangle = \int \mathcal{D}\Phi' \exp(-S_{eff}[\Phi']) (G[\Phi'] - \lambda \delta_{BRST}G[\Phi']) \quad (5.3.139)$$

or

$$\langle G \rangle = \langle G \rangle - \lambda \langle \delta_{BRST} G \rangle \quad (5.3.140)$$

from which (5.3.136) follows.

Now, it is easy to prove that

$$\int \mathcal{D}\Phi \exp(-S_{eff}[\Phi]) = \int \mathcal{D}\Phi \exp(-S_{eff}[\Phi] + \delta_{BRST} H) \quad (5.3.141)$$

One just expands in powers of $\delta_{BRST} H$ the r.h.s. and uses (5.3.136) to prove that all terms except the first one vanish. This completes the proof of our assertion concerning the fact one can use an arbitrary BRST exact form to fix the symmetry.

BRST symmetry and Noether charge

Up to here we have seen that the gauge symmetry can be fixed by adding to the original gauge-invariant Lagrangian an arbitrary BRST exact term $\delta_{BRST} H[\Phi]$

$$Z = \int \mathcal{D}\Phi \exp(-S[\Phi] + \delta_{BRST} H[\Phi]) \quad (5.3.142)$$

where we indicate the set of all fields (the original A_μ gauge field, the auxiliary field implementing the gauge condition and the ghost and anti-ghost fields) as Φ . Physical quantities do not depend on the particular choice of H since we have also seen that

$$\langle \delta_{BRST} G[\Phi] \rangle = 0 \quad (5.3.143)$$

for arbitrary $G[\Phi]$.

Since the (nilpotent) BRST transformation δ_{BRST} is a symmetry of the action and also of the integration measure, there should be a charge Q_{BRST} conserved both at the classical and at the quantum level. This charge is Grassman odd and should generate BRST transformations,

$$\begin{aligned} \lambda \delta_{BRST} \Phi &= \phi^\lambda - \Phi = \exp(-i\lambda Q_{BRST}) \Phi \exp(i\lambda Q_{BRST}) - \Phi \\ &= -i\lambda [Q_{BRST}, \Phi] \end{aligned} \quad (5.3.144)$$

The nilpotency condition $\delta_{BRST}^2 = 0$ reads in terms of Q_{BRST}

$$\lambda' \lambda \delta_{BRST}^2 = [\lambda' Q_{BRST}, [\lambda Q_{BRST}, \Phi]] = 0 \quad (5.3.145)$$

or

$$\lambda' \lambda \delta_{BRST}^2 = \lambda' \lambda Q_{BRST}^2 \Phi - (\lambda' \lambda + \lambda \lambda') Q_{BRST} \Phi Q_{BRST} + \Phi Q_{BRST}^2 \lambda \lambda' = 0 \quad (5.3.146)$$

Now, since $\lambda \lambda' + \lambda' \lambda = 0$ we see that the condition

$$Q_{BRST}^2 = 0 \quad (5.3.147)$$

ensures nilpotency at the BRST charge level.

In terms of the BRST charge, eq.(5.3.143) reads

$$\langle \alpha | [Q_{BRST}, G] | \alpha \rangle = 0 \quad (5.3.148)$$

In order for this to hold for all BRST changes it is necessary that

$$Q_{BRST} | \alpha \rangle = 0 \quad (5.3.149)$$

for any physical state $| \alpha \rangle = | \text{phys} \rangle$. That is, *the physical states are the kernel of the nilpotent operator Q_{BRST}* . Or, in other words,

$$\exp(i \lambda Q_{BRST}) | \text{phys} \rangle = | \text{phys} \rangle \quad (5.3.150)$$

That is *Physical states are BRST invariant*.

The Noether charge for Yang-Mills theory

Given the effective gauge fixed Yang-Mills Lagrangian

$$\begin{aligned} L_{eff} &= L_{YM} + \delta_{BRST} \mathcal{F} = L_{YM} + \delta_{BRST} (\text{tr} \bar{c} F[A]) \\ &= L_{YM} + \text{tr}(bF) + \text{tr}(\bar{c} \mathcal{M} c) \end{aligned} \quad (5.3.151)$$

which is invariant under the BRST transformation

$$\Phi \rightarrow \Phi + \lambda \delta_{BRST} \Phi \quad (5.3.152)$$

with global parameter λ , we can construct the associated Noether charge by promoting (5.3.152) to a local transformation,

$$\Phi \rightarrow \Phi' = \Phi + \lambda(x) \delta_{BRST} \Phi \quad (5.3.153)$$

Although the Yang-Mills Lagrangian L_{YM} is still invariant under (5.3.153) since the transformation corresponds to a particular gauge transformation, the complete effective Lagrangian is not. Consider for simplicity the Lorentz gauge condition,

$$F[A] = \partial_\mu A_\mu \quad (5.3.154)$$

giving

$$\mathcal{M} = \partial_\mu D_\mu \quad (5.3.155)$$

Then, the fact that $\lambda = \lambda(x)$ implies that a term linear in $\partial_\mu \lambda$ is generated when the Faddeev-Popov term is variated,

$$\delta L_{FP} = L'_{FP} - L_{FP} = \text{tr}(\bar{c} \partial_\mu D_\mu c)' - \text{tr}(\bar{c} \partial_\mu D_\mu c) \quad (5.3.156)$$

or, integrating by parts at the action level,

$$\delta L_{FP} = -\text{tr}(\partial_\mu \bar{c} D_\mu c)' + \text{tr}(\partial_\mu \bar{c} D_\mu c) \quad (5.3.157)$$

Explicitly, the change of variables we are performing is given by

$$\begin{aligned} A'_\mu &= A_\mu + D_\mu(\lambda(x)c) \\ \bar{c}' &= \bar{c} + \lambda(x)b \\ c' &= c + \frac{1}{2}\lambda(x)[c, c] \\ b' &= b \end{aligned} \quad (5.3.158)$$

Then, we have for δL_{FP}

$$\delta L_{FP} = -b(\partial_\mu \lambda)D_\mu c - \frac{1}{2}\partial_\mu \bar{c}(\partial_\mu \lambda)[c, c] \quad (5.3.159)$$

Concerning the term implementing the gauge condition,

$$\delta L_{GF} = (b\partial_\mu A_\mu)' - (b\partial_\mu A_\mu) = b\partial_\mu \partial_\mu(\lambda c) - b\partial_\mu(\lambda \partial_\mu c) = -\partial_\mu b(\partial_\mu \lambda)c \quad (5.3.160)$$

Putting this together, we have

$$\delta_{\lambda(x)} \mathcal{F} = -(\partial_\mu \lambda) \left(bD_\mu c + \frac{1}{2}(\partial_\mu \bar{c})[c, c] + (\partial_\mu b)c \right) \quad (5.3.161)$$

Performing this change at the path-integral level and assuming that the integration measure, which was invariant under global BRST changes is also invariant for the local case, we have

$$Z = \int \mathcal{D}\phi \exp \left(-S_{eff} + \int d^4x (\partial_\mu \lambda) \left(bD_\mu c + \frac{1}{2}(\partial_\mu \bar{c})[c, c] + (\partial_\mu b)c \right) \right) \quad (5.3.162)$$

or, integrating by parts in the term containing λ ,

$$Z = \int \mathcal{D}\phi \exp \left(-S_{eff} - \int d^4x \lambda \partial_\mu \left(bD_\mu c + \frac{1}{2}(\partial_\mu \bar{c})[c, c] + (\partial_\mu b)c \right) \right) \quad (5.3.163)$$

Since Z cannot depend on λ , we obtain after differentiation,

$$\frac{\delta \log Z}{\delta \lambda} = 0 = \langle \partial_\mu \left(bD_\mu c + \frac{1}{2}(\partial_\mu \bar{c})[c, c] + (\partial_\mu b)c \right) \rangle \quad (5.3.164)$$

so that we can identify the BRST Noether current as

$$J_\mu = bD_\mu c + \frac{1}{2}(\partial_\mu \bar{c})[c, c] + (\partial_\mu b)c \quad (5.3.165)$$

so that the BRST charge is given by

$$Q_{BRST} = \int d^3x \left(bD_0 c + \frac{1}{2}(\partial_0 \bar{c})[c, c] + (\partial_0 b)c \right) \quad (5.3.166)$$

BRST symmetry and Ward identities

How does one use this BRST invariance in order to prove renormalizability of gauge theories? We shall only sketch the procedure here. The interested reader should look, for example, the Itzykson-Zuber book for details [?]. The strategy is to obtain Ward identities exploiting precisely the properties of BRST transformations. One then uses these Ward identities to establish relations between Green functions and this in turn ensures cancellations of infinities order by order in perturbation theory.

As a simple example, consider the following trivial identity,

$$0 = \int \mathcal{D}b \mathcal{D}\bar{c} \mathcal{D}c \mathcal{D}A_\mu \bar{c} \exp \left(-S_{eff} + \int d^4x j_\mu A^\mu \right) \quad (5.3.167)$$

where

$$S_{eff} = \int d^4x L_{eff} \quad (5.3.168)$$

and j^μ is an external source. Identity (5.3.167) is valid because

$$\int \mathcal{D}\bar{c} \mathcal{D}c \bar{c} \exp(-\text{tr} \int d^4x \bar{c} \mathcal{M} c) = 0 \quad (5.3.169)$$

Now, we perform an infinitesimal BRST transformations as a change of variables, assuming that the measure is BRST invariant⁸. After the transformation we have

$$0 = \int \mathcal{D}b \mathcal{D}\bar{c} \mathcal{D}c \mathcal{D}A_\mu \bar{c} \exp \left(-S_{eff} + \int d^4x j_\mu (A^\mu + D_\mu c \lambda) \right) + \int \mathcal{D}b \mathcal{D}\bar{c} \mathcal{D}c \mathcal{D}A_\mu b \lambda \exp \left(-S_{eff} + \int d^4x j_\mu A^\mu \right) \quad (5.3.170)$$

so that, to order λ one has

$$0 = \int \mathcal{D}b \mathcal{D}\bar{c} \mathcal{D}c \mathcal{D}A_\mu \left(b(x) + \int d^4y \bar{c}(x) j_\mu(y) D_\mu c(y) \right) \exp \left(-S_{eff} + \int d^4x j_\mu A^\mu \right) \quad (5.3.171)$$

Now, the second term in (5.3.171) can be easily integrated over the ghost variables. Indeed, from the identity

$$I = \int \mathcal{D}\bar{c} \mathcal{D}c \exp \left(-\int d^4x (\bar{c} \mathcal{M} c + \bar{c} \theta + \bar{\theta} c) \right) = \det \mathcal{M} \exp \left(-\int d^4x d^4y \bar{\theta} \mathcal{M}^{-1} \theta \right) \quad (5.3.172)$$

one gets, by differentiation with respect to the anticommuting sources θ and $\bar{\theta}$

$$-\left. \frac{\delta^2 I}{\delta \theta \delta \bar{\theta}} \right|_{\theta=\bar{\theta}=0} = \det \mathcal{M} \mathcal{M}^{-1} \quad (5.3.173)$$

Using this result in (5.3.171) one gets

$$0 = \int \mathcal{D}b \mathcal{D}A_\mu \left(\det \mathcal{M} b(x) + \int d^4y j_\mu(y) D_\mu^y (\det \mathcal{M} \mathcal{M}^{-1}(x, y)) \right) \exp \left(-S_{eff} + \int d^4x j_\mu A^\mu \right) \quad (5.3.174)$$

We shall now integrate out the auxiliary field b . for the integrand which is linear in b we use

$$\begin{aligned} \int \mathcal{D}b b \exp \left(-\int d^4x b F[A] \right) &= \left. \frac{\delta}{\delta s} \int \mathcal{D}b \exp \left(-\int d^4x b (F[A] + s) \right) \right|_{s=0} \\ &= \left. \frac{\delta}{\delta s} \int \delta(F[A] + s) \right|_{s=0} \end{aligned} \quad (5.3.175)$$

⁸ In fact, once BRST invariance is discovered at the level of the classical action, it is consistent to define a quantum measure respecting this invariance, as it should respect gauge invariance, Lorentz invariance, etc).

Now, using the alternative representation for the delta function

$$\delta(F[A] + s) = \lim_{\alpha \rightarrow 0} \exp \left(-\frac{1}{2\alpha} \int d^4x (F[A] + s)^2 \right) \quad (5.3.176)$$

we get (we omit the explicit indication that the $\alpha \rightarrow 0$ limit should be taken at the end of calculations)

$$\int \mathcal{D}b \exp \left(-\int d^4x b F[A] \right) = \frac{F[A]}{2\alpha} \exp \left(-\frac{1}{2\alpha} \int d^4x F[A]^2 \right) \quad (5.3.177)$$

Putting all this together we end with

$$\int \mathcal{D}A_\mu \left(\det \mathcal{M}[\mathcal{A}] \frac{1}{2\alpha} F[A] + \int d^4y j_\mu(y) D_\mu^y[A] \left(\det \mathcal{M}[\mathcal{A}] \mathcal{M}^{-1}[A; x, y] \right) \right) \exp(-S_{eff}[j]) = 0 \quad (5.3.178)$$

or

$$\left(\det \mathcal{M}[\frac{\delta}{\delta j}] \frac{1}{2\alpha} F[\frac{\delta}{\delta j}] + \int d^4y j_\mu(y) D_\mu^y[\frac{\delta}{\delta j}] \left(\det \mathcal{M}[\frac{\delta}{\delta j}] \mathcal{M}^{-1}[\frac{\delta}{\delta j}; x, y] \right) \right) Z[j] = 0 \quad (5.3.179)$$

Here we called

$$S_{eff}[j] = S_{eff} + \int d^4x j_\mu A_\mu$$

and

$$Z[j] = \int \mathcal{D}fields \exp(-S[j])$$

This identity, obtained for the first time by Lee and Zinn-Justin [?] allows to obtain relations between n point functions. Indeed, differentiation of $Z[j]$ with respect to j n times gives the n -point function Γ^n . Now, if one differentiates identity (5.3.179) one gets relations among certain functionals containing Γ^n . This in turn allows to obtain relations between the renormalization constants which show that infinities cancel out, a result that can be then regarded as a consequence of gauge invariance (which, in this approach *à l'ancienne*, is behind BRST invariance)